Exam December 2020 - Results

Attempt 1 of Unlimited

Written 26 November, 2024 6:09 PM - 26 November, 2024 6:09 PM

Your quiz has been submitted successfully.

Attempt Score **0 %**

Overall Grade (Highest Attempt) **0 %**

Question 1 **0 / 5 points**

(W20_1) Methanol, also known as methyl alcohol amongst other names, is a chemical with the formula CH_3OH. Methanol is occasionally used to fuel internal combustion engines. It burns forming carbon dioxide and water according to the following reaction. Note that the reaction is not balanced.

$$CH_3\,OH \;+\; O_2 \;\rightarrow\; CO_2 \;+\; H_2O$$

If we burn 20 gram CH_3OH with 10 gram O_2, how many grams of CO_2 do we produce?

- 5.2
- 7.2
- ➡ 9.2
- 11.2
- 13.2

▼ Hide question 1 feedback

Feedback

Answer C

First, balance the reaction:

$$CH_3\,OH \;+\; 1.5\,O_2 \;\rightarrow\; CO_2 \;+\; 2\,H_2O$$

The molar mass of CH_3OH is 12.01 + 1.008*4 + 16 = 32.042 g/mol

$n(CH_3OH)$ = 20/32.042 = 0.62 mol

The molar mass of O_2 is 32.00 g/mol.

$n(O_2)$ = 10/32.00 = 0.31 mol

$n(O_2)/n(CH_3OH)$ = 0.31/0.62 = 0.5 < 1.5. Hence O_2 is the limiting reactant.

The amount of produced CO_2 is proportional to $n(O_2)$.

$n(CO_2) = n(O_2)/1.5$ = 0.31/1.5 = 0.21 mol.

The molar mass of CO_2 is 12.01 + 16*2 = 44.01 g/mol

The mass of produced CO_2 is 44.01*0.21 = 9.24 g

Question 2 — 0 / 2 points

(W20_2) Assign oxidation number to chromium (Cr) in CrO_2Cl_2:

- +2
- +3
- +4
- +5
- ⇨ +6

▼ Hide question 2 feedback

Feedback

Answer E

Cl has a valence of -1 and O has a valence of -2, so the valence of Cr is +6. Remember that the sum of the valences must equal the outer charge.

Question 3 — 0 / 5 points

(W20_3) Acid rain is mainly caused by the sulfur dioxide gas (SO_2) present in air. The concentration of SO_2 can be determined by titrating against a standard permanganate solution (such as a $KMnO_4$ solution) as follows:

$$SO_2 \;+\; \left[MnO_4\right]^- \;+\; H_2O \;\rightarrow\; \left[SO_4\right]^{2-} \;+\; Mn^{2+} \;+\; H^+$$

What is the molar ratio between SO_2 and H^+ in the balanced reaction? The reaction environment is acidic.

- 5:2 (5 SO_2 used for every 2 H^+)
- 2:5 (2 SO_2 used for every 5 H^+)

5:4 (5 SO_2 used for every 4 H^+)

4:5 (4 SO_2 used for every 5 H^+)

3:5 (3 SO_2 used for every 5 H^+)

Hide question 3 feedback

Feedback

Answer C

The reaction is a redox reaction:

S changes from +4 in SO_2 to +6 in $[SO_4]^{2-}$ (The oxidation number increases by 2)

Mn changes from +7 in $[MnO_4]^-$ to +2 (The oxidation number decreases by 5)

5 SO_2 are therefore needed for every 2 $[MnO_4]^-$ in order to keep the increase and decrease balanced
Then balance charges → 4 protons needed as product
Then balance oxygen atoms → 2 water needed as reactant
Check: 4 hydrogen atoms and 20 oxygen atoms on each side

$$5\,SO_2 \;+\; 2\left[MnO_4\right]^- \;+\; 2H_2O \;\rightarrow\; 5\left[SO_4\right]^{2-} \;+\; 2\,Mn^{2+} \;+\; 4H^+$$

The molar ratio between SO_2 and H^+ in the balanced reaction is 5:4, 5 SO_2 used for every 4 H^+.

Question 4	**0 / 3 points**

(W20_4) Estimate if the following unbalanced reaction is favorable at standard conditions.

$$Zn\ (s) + Cd^{2+}\ (aq) \rightarrow Zn^{2+}\ (aq) + Cd\ (s)$$

Favorable

Not favorable

Neither

Insufficient information

Hide question 4 feedback

Feedback

Answer A

List the standard reduction potential (Table 19.1) for the anode and cathode half-reactions:

Cathode: $Cd^{2+}\ (aq) + 2e^- \rightarrow Cd\ (s)$ E^o = -0.40 V

Anode: $Zn\ (s) \rightarrow Zn^{2+}\ (aq) + 2e^-$

$Zn^{2+}\ (aq) + 2e^- \rightarrow Zn\ (s)$ E^o = -0.76 V

$E_{cell}^{o} = E_{Cathode}^{o} - E_{Anode}^{o}$ = -0.40 – (-0.76) = 0.36 V >0
Hence, $\Delta G = -nFE < 0$
The reaction is favorable.

Question 5 — 0 / 2 points

(W20_5) Due to its highly negative redox potential, aluminium (Al) reacts with water to produce hydrogen gas according to the following:

$$2\ \mathrm{Al} + 3\,\mathrm{H_2O} \rightarrow 3\,\mathrm{H_2} + \mathrm{Al_2\,O_3}$$

By putting 10 gram aluminium powder into water, how many litres of hydrogen will it produce at 1 atmosphere and 25 ^{o}C? Assume all the aluminium reacts.

- 13.6 litre
- 15.6 litre
- 17.6 litre
- 19.6 litre
- 21.6 litre

Hide question 5 feedback

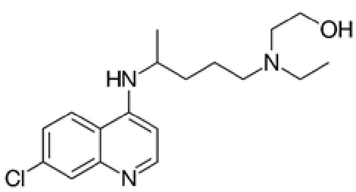
Feedback

Answer A

The atomic mass of Al is 26.98 g/mol.
10 g Al corresponds to 10/26.98 = 0.37 mol Al
If all the aluminum reacts, it will produce 0.37*3/2 = 0.555 mol H_2.
Then use the ideal gas law to calculate the volume of H_2 gas.
PV = nRT
V = nRT/P = 0.555 mol * 0.082057 (l atm)/(K mol) * 298.15 K / 1 atm = 13.6 l

Question 6 — 0 / 5 points

(W20_6) Hydroxychloroquine (HCQ) is a medication used to prevent and treat malaria in areas where malaria remains sensitive to chloroquine. It is also being studied as a treatment for coronavirus disease 2019 (COVID-19). Its structure is shown below:



Which one of the following listed functional groups are not found in HCQ's structure?

- Methyl
- Halogen
- Hydroxyl
- Carboxyl
- Amine

Hide question 6 feedback

Feedback

Answer D

Question 7 — 0 / 5 points

(W20_7) Pure benzene freezes at 5.5 °C. A solution prepared by dissolving 1 gram of an unknown substance in 60 g of benzene is found to freeze at 4.2 °C. Determine the molecular weight of the unknown substance. The freezing point constant for benzene is 5.12 °C/m. Note that the unknown substance does not ionize in benzene.

- ~36 g/mol
- ~46 g/mol
- ~56 g/mol
- ~66 g/mol
- ~76 g/mol

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Feedback

Answer D

Use the equation $\Delta Tf = iKfm$
(5.5-4.2) ^{o}C = 1 * 5.12 ^{o}C/m * (x mol/0.060 kg)
x = 0.01523 mol
The molecular weight is then calculated as 1 gram/0.01523 mol = 65.6 g/mol

Question 8 — 0 / 5 points

(W20_8) A bottle of an unknown aqueous solution is found in the lab. It may contain salts of Ag^{+}, or Ba^{2+}, or Pb^{2+} dissolved in water. Adding H_2SO_4 to the solution gives precipitation, but no precipitation is observed when adding NaOH. Which cation can this unknown solution contain?

- Ag^{+}
- Ba^{2+}

- Pb^{2+}
- Not enough information is given.

Hide question 8 feedback

Feedback

Answer B

Sulfates of Ag^+, Ba^{2+}, Pb^{2+} are all insoluble.
Hydroxides of Ag^+, Pb^{2+} are insoluble, but not with Ba^{2+}.
So the solution contains Ba^{2+} only.

Question 9	**0 / 5 points**

(W20_9) Vinegar is an aqueous solution consisting of about 5-20 % acetic acid (CH_3COOH). Take a vinegar with 5 % (weight percentage) CH_3COOH as an example. The vinegar has a density of 1.05 g/mL. CH_3COOH has a Ka of $1.8 * 10^{-5}$. What is the pH value of this vinegar with 5 % acetic acid?

- 1.6
- 2.0
- 2.4
- 2.8
- 3.2

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Feedback

Answer C

Assume 1 litre vinegar, the mass is equal to 1.05 * 1000 = 1050 g
The mass of CH_3COOH is 1050 * 0.05 = 52.5 g
The molecular mass of CH_3COOH is 12.01*2 + 1.008*4 + 16*2 = 60.052 g/mol
The molar concentration of CH_3COOH is then 52.5/60.052/1 = 0.87 mol/l = 0.87 M

$CH_3COOH = CH_3COO^- + H^+$
0.87
0.87-x   x   x

$x^2/(0.87-x) = 1.8*10^{-5}$
x = 0.004 M = $[H^+]$
pH = $-\log[H^+]$ = 2.4

Question 10	**0 / 5 points**

(W20_10) Carbon monoxide (CO) is a poisonous, colorless, odorless and tasteless gas. Carbon monoxide is harmful when breathed because it displaces oxygen in the blood and deprives the heart, brain and other vital organs of oxygen. Most of the safety standards prohibit worker exposure to more than 50 parts of CO gas per million parts of air (50 ppm) averaged during an 8-hour time period. In a closed room of 10 m * 12 m * 3 m at 1 atmosphere and 25 ^{o}C, what is maximum allowable amount of CO (in terms of gram) in the atmosphere in order to stay below 50 ppm?

- ~16 gram
- ~21 gram
- ~26 gram
- ~31 gram
- ~36 gram

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Feedback

Answer B

Use the ideal gas law equation:
PV = nRT
n = PV/RT = 101325 Pa * 360 m^3 / 8.314 J/(mol K) /298.15 K = 14715 mol
50 ppm corresponds to 14715 * 0.00005 = 0.74 mol
The molecular mass of CO is 28.01 g/mol
Then 0.74 * 28.01 = 20.7 gram

Question 11	**0 / 5 points**

(W20_11) Methanol (CH_3OH) has a boiling point of 64.6 ^{o}C at 1 atmosphere pressure. Its heat of vaporization is around 37.6 kJ/mol. If we increase the pressure to 50 atm, what will be the new boiling temperature for methanol then?

- 14 ^{o}C
- 104 ^{o}C
- 154 ^{o}C
- 204 ^{o}C
- 254 ^{o}C

Hide question 11 feedback

Feedback

Answer D

Use the following equation:

$$Ln\left(P_2/P_1\right) = \frac{\Delta H_{vap}}{R} * \frac{T_2 - T_1}{T_2 * T_1}$$

ΔH_{vap} = 37600 J/mol

P_1 = 1 atm
T_1 = 64.6+273.15 = 337.75 K
P_2 = 50 atm
Ln(50/1) = 37600/8.314 * (1/T1-1/T2)
T2 = 477.15 K → 204 ^{o}C

Question 12	0 / 2 points

(W20_12) Using the following reaction enthalpies:
$C\ (graphite)\ +\ O_2\ (g)\ \rightarrow\ CO_2\ (g)$ ΔH^o = -393.5 kJ/mol
$H_2\ (g)\ +\ 0.5\ O_2\ (g)\ \rightarrow\ H_2O\ (l)$ ΔH^o = -285.8 kJ/mol
$C\ (graphite)\ +\ 2\ H_2\ (g)\ +\ 0.5\ O_2\ (g)\ \rightarrow\ CH_3OH\ (l)$ ΔH^o = -238.7 kJ/mol

Calculate the reaction enthalpy for combustion of methanol:
$CH_3OH\ (l)\ +\ 1.5\ O_2\ (g)\ \rightarrow\ CO_2\ (g)\ +\ 2\ H_2O\ (l)$

All the reactions take place at 1 atmosphere and 25 ^{o}C.

- -526.4 kJ/mol
- -626.4 kJ/mol
- +626.4 kJ/mol
- ➡ -726.4 kJ/mol
- +726.4 kJ/mol

▼ Hide question 12 feedback

Feedback

Answer D

$C\ (graphite)\ +\ O_2\ (g)\ \rightarrow\ CO_2\ (g)$ ΔH^o = -393.5 kJ/mol
$2\ H_2\ (g)\ +\ O_2\ (g)\ \rightarrow\ 2\ H_2O\ (l)$ ΔH^o = -285.8*2 kJ/mol
$CH_3OH\ (l)\ \rightarrow\ C\ (graphite)\ +\ 2\ H_2\ (g)\ +\ 0.5\ O_2\ (g)$ ΔH^o = +238.7 kJ/mol

Sum the above 3 reactions, we get:
$CH_3OH\ (l)\ +\ 1.5\ O_2\ (g)\ \rightarrow\ CO_2\ (g)\ +\ 2\ H_2O\ (l)$
ΔH^o = 238.7 – 393.5 - 285.8*2 = -726.4 kJ/mol

Question 13	0 / 3 points

(W20_13) The Haber-Bosch Process combines nitrogen from the air with hydrogen derived mainly from natural gas (methane) into ammonia, as described by the reaction below. The reaction is reversible and the production of ammonia is exothermic.

$$N_2\ (g)\ +\ 3\ H_2\ (g)\ \rightleftarrows\ 2\ NH_3\ (g)$$

Which of the following changes will shift the equilibrium towards the right?

- Decrease temperature.
- Increase temperature.
- A catalyst is added.
- Decrease pressure.
- None of the above.

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Feedback

Answer A

The reaction is exothermic, i.e. releasing heat. So decreasing temperature will move the reaction towards the right. Adding catalyst will not shift the equilibrium. Decreasing pressure will move the reaction towards the left.

Question 14 — 0 / 5 points

(W20_14) Which of the following is true about a 1.0 M solution of a weak acid, HX?

- pH = 0
- $[X^-]$ = 1 M
- $[HX] > [H^+]$
- $[H^+]$ = 1 M
- both b and d

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Feedback

Answer C

As HX is a weak acid, only a small part of HX will dissociate. Only Answer C is correct.

Question 15 — 0 / 5 points

(W20_15) $Mg(OH)_2$ is insoluble in water. Its solubility product K_{sp} is 5.61 x 10^{-12}. What is the pH of the water solution saturated with $Mg(OH)_2$?

- ~8.4
- ~9.4
- ~10.4
- ~11.4
- ~12.4

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Feedback

Answer C

$Mg(OH)_2$ = Mg^{2+} + $2[OH]^-$

x 2x

K_{sp} = 5.61 x 10^{-12} = x*$(2x)^2$

x = 1.12 x 10^{-4} M

$[OH]^-$ = 2.24 x 10^{-4} M

pOH = 3.65

pH = 14 – 3.65 = 10.35

Question 16 **0 / 3 points**

(W20_16) The decomposition of dimethylether at 504 ^oC is a first order reaction with a half-life of 1570 seconds. What fraction of an initial amount of dimethylether remains after 3140 seconds?

- 1/2
- 1/3
- 1/4
- 1/6
- 1/8

Hide question 16 feedback

Feedback

Answer C

For first-order reactions,

$\ln([A]/[A]_0) = -kt$

From half-life, we get $k = 0.693/1570\ s^{-1}$

Then we calculate [A] at 3140 seconds
$\ln([A]/[A]_0) = -0.693/1570*3140$
$[A]/[A]_0 = 0.25$

Question 17	**0 / 2 points**

(W20_17) Dimethyl ether (DME, also known as methoxymethane) is the organic compound with the formula CH_3OCH_3, simplified to C_2H_6O. DME is produced by dehydration of methanol:

$$2\ CH_3\,OH \rightarrow (CH_3)_2O + H_2O$$

A potentially major use of DME is as fuel in household and industry. Below is a reaction for complete combustion of DME:

$$C_2H_6O + O_2 \rightarrow CO_2 + H_2O$$

In which of the following answers is the reaction correctly balanced?

- ◯ $C_2H_6O + 1.5\,O_2 \rightarrow 2\ CO_2 + 3\,H_2O$
- ➡ ◯ $C_2H_6O + 3\,O_2 \rightarrow 2\ CO_2 + 3\,H_2O$
- ◯ $2\,C_2H_6O + 5\,O_2 \rightarrow 4\ CO_2 + 6\,H_2O$
- ◯ $2\,C_2H_6O + 3\,O_2 \rightarrow 4\ CO_2 + 6\,H_2O$

▼ Hide question 17 feedback

Feedback

Answer B

Question 18	**0 / 3 points**

(W20_18) Which of the following statements is true?

- ◯ A catalyst changes the equilibrium concentration of the products.
- ◯ A catalyst does not affect the reaction energy path.

- Reactions with more negative values of ΔG^o are spontaneous and proceed at a higher rate than those with less negative values of ΔG^o.
- If a reaction is thermodynamically spontaneous, it also occurs rapidly.
- ⇨ None of the above.

▼ Hide question 18 feedback

Feedback

Answer E

Question 19 — 0 / 5 points

(W20_19) Which of the following molecules has a liner geometry?

- NH_3
- $[NH_2]^-$
- $[SO_3]^{2-}$
- H_2O
- ⇨ CO_2

▼ Hide question 19 feedback

Feedback

Answer E

NH_3 – trigonal pyramidal
$[NH_2]^-$ - bent
$[SO_3]^{2-}$ - trigonal pyramidal
H_2O - bent
CO_2 - linear

Question 20 — 0 / 2 points

(W20_20) How many unpaired electrons in total does a chromium (Cr) atom have?

- 2
- 3
- 4
- 5
- ⇨ 6

▼ Hide question 20 feedback

Feedback

Answer E

Cr: $[Ar]4s^1 3d^5$, one unpaired electron in the 4s orbit and five in the 3d orbital

Question 21 **0 / 3 points**

(W20_21) How many electrons in total are located in p-orbitals of a chromium (Cr) atom?

- 2
- 6
- 8
- 10
- ⇨ 12

Hide question 21 feedback

Feedback

Answer E

6 electrons in 2p orbital and 6 in 3p orbital.

Question 22 **0 / 5 points**

(W20_22) Rank the compounds according to increasing degree of ionic character of the bond: NaCl, K_2O, H_2, SO_2, CH_4

- $H_2 < CH_4 < K_2O < SO_2 < NaCl$
- $K_2O < H_2 < CH_4 < SO_2 < NaCl$
- ⇨ $H_2 < CH_4 < SO_2 < NaCl < K_2O$
- $H_2 < CH_4 < K_2O < NaCl < SO_2$
- $H_2 < CH_4 < SO_2 < K_2O < NaCl$

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Feedback

Answer C

Calculate the electronegativity difference in each molecule:

NaCl: 3-0.9=2.1
K_2O: 3.5-0.8=2.7
H_2: 0

SO_2: 3.5-2.5=1

CH_4: 2.5-2.1=0.4

The higher the difference, the more ionic.

Question 23 — 0 / 5 points

(W20_23) Which of the following ions will oxidize Br^- ions to Br_2?

- Pb^{2+}
- Fe^{3+}
- H^+
- ➡ Ce^{4+}
- Cu^{2+}

Hide question 23 feedback

Feedback

Answer D

List and compare standard reduction potentials

$Br_2 + 2e^- \rightarrow 2Br^-$ 1.07 V

$Pb^{2+} + 2e^- \rightarrow Pb$ -0.13 V

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ 0.77 V

$2\ H^+ + 2e^- \rightarrow H_2$ 0 V

$Ce^{4+} + e^- \rightarrow Ce^{3+}$ 1.61 V

$Cu^{2+} + e^- \rightarrow Cu^+$ 0.15 V

1.61 V > 1.07 V, only Ce^{4+} can oxidize Br^-.

Question 24 — 0 / 5 points

(W20_24) The activation energy for the decomposition of hydrogen peroxide is 42 kJ/mol.

$$2\ H_2O_2\ (aq)\ \rightarrow\ 2\ H_2O\ (l)\ +\ O_2\ (g)$$

The activation energy is reduced to only 7 kJ/mol when an enzyme is used. Assume the rate constant without and with using enzyme is k_1 and k_2, respectively. How much faster is the reaction with enzyme, i.e. the ratio of k_2/k_1? Assume that the reaction takes place at 25 ^oC.

- 1.4×10^7
- ➡ 1.4×10^6

- 1.4×10^5
- 1.4×10^4
- 1.4×10^3

Hide question 24 feedback

Feedback

Answer B

Use the following equation

$$Ln(k_2/k_1) = (E_{a1} - E_{a2})/RT$$

E_{a1} = 42000 J/mol

E_{a2} = 7000 J/mol

T = 298.15 K

$k_2/k_1 = 1.4*10^6$

Question 25	**0 / 5 points**

(W20_25) Gold has a molar heat capacity of 25.6 J/(mol·K), while the molar heat capacity for water is 75.2 J/(mol K). If 10 g of Au at 20.0 °C is placed in an insulated mug with 50 g of water at 97.0 °C, what will be the final temperature of the water after the system has equilibrated? Assume that no heat is transferred to the mug or to the surroundings.

- 94.5 ^oC
- 95.1 ^oC
- 95.7 ^oC
- 96.2 ^oC
- ➡ 96.5 ^oC

Hide question 25 feedback

Feedback

Answer E

The atomic mass of Au is 197 g/mol. 10 g Au corresponds to 10/197 = 0.05 mol
50 gram of water corresponds to 50/18.016 = 2.78 mol
The temperature of Au will increase, and that of water will decrease. They will have the same temperature in the end.
Assume the final temperature is x.
The amount of heat Au absorbed should equal to that of heat water released.
0.05*25.6*(x-20) = 2.78*75.2*(97-x)

x = 96.5 ^oC

Done

Exam May 2021 - Results

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Attempt 1 of Unlimited

Written 26 November, 2024 6:12 PM - 26 November, 2024 6:12 PM

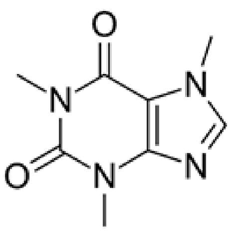
Your quiz has been submitted successfully.

Attempt Score **0 %**

Overall Grade (Highest Attempt) **0 %**

Question 1	**0 / 5 points**

(S21_1) What is the molecular formula of this compound? What is the empirical formula of this molecule?



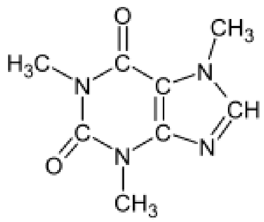
- ◯ $C_6H_4N_4O_2$; $C_6H_4N_2O$
- ➡ ◯ $C_8H_{10}N_4O_2$; $C_4H_5N_2O$
- ◯ $C_6H_4N_4O_2$; $C_5H_4N_4O_2$
- ◯ $C_8H_{12}N_4O_2$; $C_8H_{12}N_4O_2$
- ◯ None of the above

▼ Hide question 1 feedback

Feedback

Answer B

The molecular structure belongs to a compound known as 1,3,7 –Trimethylxanthine, also commonly known as Caffeine. Re-writing the molecular structure with all elements, shows the structure as shown below. The methyl groups, carbon atoms and hydrogen atoms are not always shown. So, there are 8 carbon atoms, 10 hydrogen atoms, 4 nitrogen atoms and 2 oxygen atoms.



The second part of the question is writing the empirical formula for this molecule. Empirical formula is the smallest possible whole number ratio of the atoms in the given molecular formula. Thus, the empirical formula of Caffeine is $C_4H_5N_2O$.

Question 2 **0 / 3 points**

(S21_2) Isopropanol has a strong odor and is a colourless flammable compound. It is used as a hand sanitizer when mixed with water. How many hydrogen atoms are present in 36.25 g of isopropanol? (Hint: Avogadro's number: 6.022×10^{23}).

- 2.90×10^{12} H atoms
- 2.40×10^{12} H atoms
- 2.90×10^{24} H atoms
- 5.80×10^{24} H atoms
- 4.80×10^{12} H atoms

Hide question 2 feedback

Feedback

Answer C

The molecular formula for isopropanol is C_3H_8O

The molecular weight of isopropanol is: 60 g/mol

36.25 g of isopropanol corresponds to 36.25/60 moles = 0.60 moles

1 mole of isopropanol contains 8 moles of hydrogen.

0.6 moles of isopropanol contains 0.6 x 8 mole of hydrogen = 4.8 moles of hydrogen.

1 mole of hydrogen contains 6.022×10^{23} hydrogen atoms

4.8 moles of hydrogen contains 4.8 x 6.022 x 10^{23} hydrogen atoms = 2.89 x 10^{24} hydrogen atoms

Question 3 — 0 / 5 points

(S21_3) Will the following reaction occur spontaneously at 25˚C, if the concentration of [Fe^{2+}] and [Cd^{2+}] are 0.20 M and 0.020 M respectively? At what ratio of [Cd^{2+}] to [Fe^{2+}], the reaction would become spontaneous?

$$\mathrm{Cd(s) + Fe^{2+}(aq) \rightarrow Cd^{2+}(aq) + Fe(s)}$$

The standard reduction potentials E° for the half-cell reactions are given below:

$Cd^{2+} + 2e^- \rightarrow Cd$ E° = -0.40 V

$Fe^{2+} + 2e^- \rightarrow Fe$ E° = -0.44 V

- Yes and the ratio should be less than 0.0246
- No and the ratio should be less than 0.0446
- Yes and the ratio should be greater than 0.0846
- No and the ratio should be greater than 0.0446
- Yes and the ratio should be equal to 0.0146

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Feedback

Answer B

$$E_0 = E_0^{red} - E_0^{ox} = (-0.44 + 0.40)V = -0.04V$$

$$E = E^0 - \frac{R \cdot T}{n \cdot F} \cdot \ln(Q)$$

Up on substituting R and T (298 K)

$$E = E^0 - \frac{0.0257V}{n} \ln(Q)$$

$$E = E^0 - \frac{0.0257V}{n} \ln \frac{[Cd^{2+}]}{[Fe^{2+}]}$$

n = 2, $[Cd^{2+}]$ = 0.020 M and $[Fe^{2+}]$ = 0.20 M and E^0 = -0.04 V

E = -0.01 V

Because E is negative, the reaction is not spontaneous.

To find out the ratio of $[Cd^{2+}]$ to $[Fe^{2+}]$ for spontaneous reaction, consider the equilibrium situation.
At equilibrium E = 0, rewrite the above equation

Thus

$$E = E^0 - \frac{0.0257\ \mathrm{V}}{n} \ln \frac{[Cd^{2+}]}{[Fe^{2+}]}$$

becomes

$$0 = -0.04 - \frac{0.0257\ \mathrm{V}}{2} \ln \frac{[\mathrm{Cd}^{2+}]}{[\mathrm{Fe}^{2+}]}$$

$$\ln \frac{[Cd^{2+}]}{[Fe^{2+}]} = -3.11$$

$$\frac{[\mathrm{Cd}^{2+}]}{[\mathrm{Fe}^{2+}]} = e^{-3.11} = 0.0446$$

The ratio must be smaller than 0.0446

Question 4 **0 / 2 points**

(S21_4) Which of the following statements is true under normal circumstances?

- Metallic elements may have either positive or negative oxidation states
- Nonmetallic elements may have only negative oxidation states
- Nonmetallic elements may have only positive oxidation states
- ➡ Nonmetallic elements may have either positive or negative oxidation states
- Metallic elements may have only negative oxidation states

▼ Hide question 4 feedback

Feedback

Answer D

Metals have only positive oxidation states and thus statements (a) and (e) are not true. Nonmetallic element may have both negative or positive oxidation states and thus statements (b) and (c) are not true.

Question 5	**0 / 3 points**

(S21_5) Aluminium nitrate ($Al(NO_3)_3$) is a strong oxidizing agent and widely used in leather tanning industry. When $Al(NO_3)_3$ solution is mixed with KOH solution, which of the following can precipitate?

- KNO_3
- $K2(NO_3)_2$
- ➡ $Al(OH)_3$
- $Al_2(OH)_4$
- $Al_2(NO_3)_2(OH)_4$

▼ Hide question 5 feedback

Feedback

Answer C

The net ionic equation for the reaction will be

$$\mathrm{Al^{3+}\ (aq) + 3\,OH^-\ (aq) \rightarrow\ Al(OH)_3\ (s)}$$

The KNO_3 is soluble and others are not feasible compounds.

Question 6 — 0 / 2 points

(S21_6) Automobile antifreeze solutions with different concentrations were prepared by mixing ethylene glycol, $CH_2(OH)CH_2(OH)$ and water in different ratios. Which of the following solutions will have highest freezing point depression?

- 2.6 m solution
- 3.3 m solution
- 1.1 m solution
- ➡ 5.7 m solution
- 4.4 m solution

▼ Hide question 6 feedback

Feedback

Answer D

$$\Delta T_f = K_f m$$

Freezing point depression is proportional to the molar concentration of the solution and thus higher concentration solution will have higher freezing point depression.

Question 7 — 0 / 5 points

(S21_7) During the reactions between sulfur and oxygen, different sulfur oxide gas compounds were synthesized. One of these compounds has a density of 2.61 g/L at 100˚C and 1 atm. pressure. Which of the following sulfur oxide gas compound was this compound?

- SO_2
- ➡ SO_3
- SO
- S_6O
- S_2O_2

Hide question 7 feedback

Feedback

Answer B

pV=nRT

n = m/M and density ρ = m/V
Up on substituting these in above equation and adjusting the terms

M= (ρRT)/p

Up on substituting all the values from the question, M = 79.96 g/mol.

Among the given options, molecular weight of SO_3 comes close to this and thus the answer is B.

Question 8 **0 / 5 points**

(S21_8) Which of the following molecules has a tetrahedral geometry?

- XeF_4
- SF_4
- $[AlCl_4]^-$
- $[NO_3]^-$
- $[IF_4]^-$

Hide question 8 feedback

Feedback

Answer C

XeF_4 is squre planar
SF_4 is trigonal bipyramidal
$[AlCl_4]^-$ is tetrahedral
$[NO_3]^-$ is trigonal planar
$[IF_4]^-$ is square planar

Question 9 — 0 / 3 points

(S21_9) Potassium hydroxide (KOH) is used for making the liquid soaps. During this process, 1.7 g of KOH is dissolved in 30 L of water. What is the pH of the resultant solution?

- ≈ 14
- ≈ 13
- ≈ 12
- ≈ 11
- ≈ 10

Hide question 9 feedback

Feedback

Answer D

Molecular weight of KOH = M = 56.10 g/mol.

m = 1.7 g

V = 30 L

c (KOH) = n/V = m/V.M = 0.001 M
KOH is a strong base and is fully dissociated.

Therefore $[OH^-]$ = 0.001

pH = 14 – pOH = 14 + log $[OH^-]$ = 14 + log [0.001] = 11

Question 10 — 0 / 2 points

(S21_10) When hydrogen is reacted with alkyne and ketone respectively, which of the following compounds are formed?

- A carboxylic acid and an alcohol
- An alcohol and an aldehyde
- An alcohol and an alkane
- An alkene and an alcohol
- An alkane and an alkyne

Hide question 10 feedback

Feedback

Answer D

The reaction between hydrogen and alkyne breaks one bond in triple bond and forms an alkene. And reaction between ketone and hydrogen forms an alcohol.

Question 11	**0 / 5 points**

(S21_11) The low solubility of AgCl is attractive to make the pottery glazes. During dissolution of AgCl in water, with the addition of 2 g AgCl, particles were observed even after long dissolution time. What is the solubility of the AgCl? K_{sp} of AgCl is 1.6 x 10^{-10}.

- ➡ ◯ 1.82 x 10^{-3} g/L
- ◯ 3.82 x 10^{-3} g/L
- ◯ 1.82 x 10^{-5} g/L
- ◯ 3.82 x 10^{-5} g/L
- ◯ 1.82 x 10^{-7} g/L

Hide question 11 feedback

Feedback

Answer A

Consider the dissociation of AgCl in water

$$\mathrm{AgCl\ (s)} \rightleftharpoons \mathrm{Ag^+\ (aq)} + \mathrm{Cl^-\ (aq)}$$

	AgCl (s)	⇌	Ag+ (aq)	+	Cl- (aq)
Initial (M)			0		0
Change (M)	-s		+ s		+ s
Equilibrium (M)			s		s

Solubility product, Ksp = $[Ag^+][Cl^-]$ = (s)(s) = s^2

Ksp (AgCl) = 1.6 x 10^{-10}

Therefore $1.6 \times 10^{-10} = s^2$

Molar solubility of AgCl; $s = 1.27 \times 10^{-5}$ M

Molecular weight of AgCl, M(AgCl) = 143.32 g/mol.

Solubility of AgCl = Molar solubility of AgCl x M(AgCl) = $1.27 \times 10^{-5} \times 143.32$

$= 1.82 \times 10^{-3}$ g/L

Question 12 0 / 5 points

(S21_12) More than half of benzene, C_6H_6, produced is used as a precursor for the production of polymers and plastics based on polystyrene. The vapor pressure of C_6H_6 is 100 mmHg at 26˚C. If the temperature is decreased to -37˚C, what will be the vapor pressure? The heat of vaporization for benzene is 31 kJ/mol.

- 1.9 mmHg
- 1.1 mmHg
- 4.6 mmHg
- 2.1 mmHg
- 3.6 mmHg

Hide question 12 feedback

Feedback

Answer E

ΔHvap = 31 kJ/mol.

T1 = 26˚C = 299 K   P1 = 100 mmHg
T2 = -37˚C = 236 K   P2 = ?

Use the Clausius-Clapeyron equation:

$$\ln \frac{P_1}{P_2} = \frac{\Delta H_{\text{vap}}}{R}\left(\frac{T_1 - T_2}{T_1 T_2}\right)$$

$$\ln \frac{100}{P_2} = \frac{31000}{8.314}\left[\frac{299 - 236}{(299)(236)}\right] = 3.33$$

$$\frac{100}{P_2} = e^{3.33} = 27.94$$

P2 = 3.58 mmHg

Question 13 **0 / 5 points**

(S21_13) The combustion of benzene, C_6H_6 in the presence of air produces carbon dioxide and water. What is the amount of heat released in kilojoules if 23 g of C_6H_6 is combusted in the presence of air? The standard enthalpy of formation of C_6H_6, CO_2 and water are 49.04 kJ/mol, -393.5 kJ/mol and -285.8 kJ/mol, respectively.

- ◯ -1021 kJ
- ➡ ◯ -962 kJ
- ◯ 1073 kJ
- ◯ 962 kJ
- ◯ -1073 kJ

▼ Hide question 13 feedback

Feedback

Answer B

Write the combustion equation and balance it:

$$2C_6H_6\ (g)\ +\ 15O_2\ (g)\ \rightarrow\ 12\,CO_2\ (g)\ +\ 6H_2O\ (l)$$

$\Delta H(C_6H_6)$ = 49.04 kJ/mol.; $\Delta H(CO_2)$ = −393.5 kJ/mol.; $\Delta H(H_2O)$ = −285.8 kJ/mol.

ΔH_{rxn} = [12 $\Delta H(CO_2)$ + 6 $\Delta H(H_2O)$] − [2 $\Delta H(C_6H_6)$ + 15 $\Delta H(O_2)$]

= [12 (−393.5) + 6 (−285.8)] − [2 (49.04) + 15 (0)]
= [− 4722 − 1714.8 − 98.08]
= − 6534.88 kJ/mol.

This is the amount of heat released for 2 moles of C_6H_6.

So, per 1 mole : – 3267.44 kJ/mol.

Molecular weight of C_6H_6, M(C_6H_6) = 78.11 g/mol.; mass of C_6H_6 used, m(C_6H_6) = 23 g

Number of moles used: n(C_6H_6) = m(C_6H_6) / M(C_6H_6) = 23/78.11 = 0.2945 mol.

So, the amount of heat released for 0.2945 mol. of C_6H_6 = 0.2945 x (– 3267.44) = – 962.26 kJ

Question 14 **0 / 5 points**

(S21_14) From the decomposition of steam, hydrogen and oxygen can be produced according to the reaction below

$$H_2O\ (g)\ \rightarrow\ H_2\ (g)\ +\ ½\ O_2\ (g)$$ ΔH = 241.8 kJ/mol.

Calculate the internal energy change, when 4 moles of H_2O (g) are úsed to produce the hydrogen and oxygen at a pressure of 1 atm. at 130˚C

- 980 kJ/mol
- ➡ 960 kJ/mol
- 990 kJ/mol
- 970 kJ/mol
- 1000 kJ/mol

▼ Hide question 14 feedback

Feedback

Answer B

$$H_2O\ (g)\ \rightarrow\ H_2\ (g)\ +\ ½\ O_2\ (g)$$ ΔH = 241.8 kJ/mol.

The internal energy change to be calculated when 4 moles of H_2O (g) is used. Re-write the above equation.

$$4H_2O\ (g)\ \rightarrow\ 4H_2\ (g)\ +\ 2O_2\ (g)$$ ΔH = 967.2 kJ/mol.

Change in the number of moles of gases, Δn

Δn = number of moles of product gases – number of moles of reactant gases

= 6 – 4
= 2

R = 8.314 J/K.mol = (8.314×10^{-3} kJ/K.mol); T = 403 K

Internal energy change ΔU

$\Delta U = \Delta H - RT\Delta n$

= 967.2 – ($8.314 \times 10^{-3} \times 403 \times 2$)
= 967.2 – 6.701
= 960.499 kJ/mol.

Question 15	**0 / 5 points**

(S21_15) Kerosene ($C_{12}H_{26}$) is popular fuel with oxygen as an oxidant for rocket engines.

$$C_{12}H2_6\ (g)\ +\ O_2\ (g)\ \rightarrow\ CO_2\ (g)\ +\ H_2O\ (g)$$

If 200 g of kerosene is burned with 4000 L oxygen at 1 atm. pressure and 1273 K, which of the following are limiting reactants? Assume all the gases are ideal gases

- ◯ Neither kerosene nor oxygen
- ◯ Both kerosene and oxygen
- ➡ ◯ Kerosene
- ◯ Oxygen
- ◯ Insufficient information

▼ Hide question 15 feedback

Feedback

Answer C

Write the balanced combustion equation

$$2C_{12}H_{26}\ (l)\ +\ 37O_2\ (g)\ \rightarrow\ 24\,CO_2\ (g)\ +\ 26H_2O\ (g)$$

Amount of $C_{12}H_{26}$ used:

Molecular weight of $C_{12}H_{26}$, $M(C_{12}H_{26})$ = 170.33 g/mol

Number of moles of $C_{12}H_{26}$, $n(C_{12}H_{26}) = m/M(C_{12}H_{26})$ = 200/170.33 = 1.17 mol.

Amount of O_2 used:

$n(O_2)$= (PV)/(RT)= (1 x 4000)/(0.0821 x 1273)=38 mol.

As $2n(O_2) > 37n(C_{12}H_{26})$, kerosene is the limiting reactant.

Question 16	**0 / 5 points**

(S21_16) Consider this reaction:

$X + Y \rightarrow Z$

Determine the order of the reaction and calculate the rate constant for the above reaction based on the data at 500˚C given below

[X] (M)	[Y] (M)	Rate (M/s)
1.50	1.50	3.20×10^{-1}
1.50	2.50	3.20×10^{-1}
3.00	1.50	6.40×10^{-1}

- 1 order reaction and 0.2133 M^2.s
- 2 order reaction and 0.4266 M^2.s
- 0 order reaction and 0.3166 M^2.s
- 2 order reaction and 0.4233 M^2.s
- 1 order reaction and 0.3133 M^2.s

Hide question 16 feedback

Feedback

Answer A

Experiments 1 and 3 show that the concentration of X doubled at constant Y concentration and the rate doubles.

From the rate law:

Rate for experiment 1, (R1) = k[X]^X x[Y]^Y = k (1.50)^X x (1.50)^Y

Rate for experiment 2, (R2) = k[X]^X x[Y]^Y = k (1.50)^X x (2.50)^Y

Rate for experiment 3, (R3) = k[X]^X x[Y]^Y = k (3.00)^X x (1.50)^Y

The ratio of rates from experiments 1 and 3

$$\frac{R3}{R1} = \frac{6.40 \times 10^{-1}}{3.20 \times 10^{-1}} = 2 = \frac{k(3.00)^x x(1.50)^y}{k(1.50)^x x(1.50)^y}$$

$$2^x = 2$$

Therefore x = 1 indicating the reaction is 1st order in X

The ratio of rates from experiments 1 and 2

$$\frac{R2}{R1} = \frac{3.20 \times 10^{-1}}{3.20 \times 10^{-1}} = 1 = \frac{k(1.50)^x x(2.50)^y}{k(1.50)^x x(1.50)^y}$$

$$\left(\frac{5}{3}\right)^y = 1 \Rightarrow y = 0$$

This indicates that the reaction is zero order in Y

$$rate = k[X][Y]^0 = k[X]$$

Thus the reaction is 1st order overall.

Rate constant calculation:

By re-arranging the rate law

$$k = \frac{\text{rate}}{[X]}$$

Data from any experiment can be used to calculate k, (here data from experiment 1 is used)

$$k = \frac{3.20 \times 10^{-1}}{1.50} = 0.2133\mathrm{M}^2.\mathrm{s}$$

Question 17 — 0 / 5 points

(S21_17) Which of the following list of compounds does not contain base?

- CH_3CH_2COOH, NaOH, CH_4
- HCN, NaH, H_3PO_4
- NH_3, C_2H_5OH, H_2SO_4
- NO_2, CH_3CH_2COOH, C_2H_2
- C_4H_9OH, C_5H_5N, H_2S

Hide question 17 feedback

Feedback

Answer D

NaOH in option (a), NaH in option (b), NH_3 in option (c) and C_5H_5N in option (e) are all either strong or weak bases and there are none in option (d)

Question 18 — 0 / 5 points

(S21_18) The reaction $2K_3PO_4 + 3Ca(NO_3)_2 \rightleftharpoons Ca_3(PO_4)_2 + 6KNO_3$ takes place in aqueous solution. The equilibrium concentrations of the four compounds are as given below.

$[K_3PO_4]$ = 0.8 M, $[Ca(NO_3)_2]$ = 0.6 M, $[Ca_3(PO_4)_2]$ = 0.8 M and $[KNO_3]$ = 0.6 M

Calculate the reaction free energy at 80^{o}C.

- 3.1 kJ/mol
- 3.8 kJ/mol

4.1 kJ/mol

4.5 kJ/mol

2.9 kJ/mol

Hide question 18 feedback

Feedback

Answer B

First step is to calculate the equilibrium constant

$$K = \frac{[\mathrm{Ca_3(PO_4)_2}] \cdot [\mathrm{KNO_3}]^6}{[K_3PO_4]^2 \cdot [\mathrm{Ca(NO_3)_2}]^3}$$

$[K_3PO_4]$ = 0.8 M, $[Ca(NO_3)_2]$ = 0.6 M, $[Ca_3(PO_4)_2]$ = 0.8 M and $[KNO_3]$ = 0.6 M

$$K = \frac{(0.8) \cdot (0.6)^6}{(0.8)^2 \cdot (0.6)^3} = 0.27$$

K = 0.27

ΔG^o_r = - RT ln (K) = - 8.3144 J/K/mol. x 353 K x ln (0.27) = 3842.87 J/mol = 3.843 kJ/mol

Question 19	**0 / 5 points**

(S21_19) An example of the disproportionate reaction is given below:

$$\mathrm{NO_2 + OH^- \rightarrow NO_2^- + NO_3^- + H_2O}$$

Identify the right combination of the coefficients below.

4, 3, 2, 2, 2

4, 3, 2, 2, 1

- 2, 1, 1, 2, 1
- 2, 1, 2, 2, 1
- ➡ 2, 2, 1, 1, 1

▼ Hide question 19 feedback

Feedback

Answer E

First assign the oxidation states:

NO_2	+	OH^-	→	NO_2^-	+	NO_3^-	+	H_2O
+4 -2		-2 +1		+3 -2		+5 -2		+1 -2

Balance the N on both sides

$$2\,NO_2 + OH^- \rightarrow NO_2^- + NO_3^- + H_2O$$

Balance the H on both sides

$$2\,NO_2 + 2\,OH^- \rightarrow NO_2^- + NO_3^- + H_2O$$

Verify that the O amount is same on both sides.

Question 20 — 0 / 5 points

(S21_20) Hydrofluric acid, HF is a weak acid. pH was measured to be 4.20 in a 1.00 M HF solution. Calculate the ratio of [conjugate base]/[acid] and the strength of the acid in this solution. Disregard the autoionization of the water in the calculations.

- The ratio is 6.31×10^{-4} and the acid strength is 1%
- The ratio is 5.03×10^{-4} and the acid strength is 0.008%
- ➡ The ratio is 6.31×10^{-5} and the acid strength is 0.006%
- The ratio is 3.02×10^{-5} and the acid strength is 0.5%
- The ratio is 4.03×10^{-5} and the acid strength is 0.2%

▼ Hide question 20 feedback

Feedback

Answer C

pH of HF is given, so calculate the $[H^+]$ from pH

$pH = -\log [H^+]$

$4.20 = -\log [H^+]$

$[H^+] = 10^{-4.2} = 6.31 \times 10^{-5}$ M

HF	+	H_2O	$\rightleftharpoons$	H_3O^+	+	F^-
1				0		0
1-X				X		X

Therefore : $[H_3O^+ +] = [H^+] = [F^-] = 6.31 \times 10^{-5}$ M

The ratio of [conjugate base]/[acid] = $[F^-] / [HF] = 6.31 \times 10^{-5}$ M

The strength of the acid = percent ionization = $([H^+])/([HF]) \times 100 = (6.31 \times 10^{-5})/1 \times 100 = 6.31 \times 10^{-3}\%$

Question 21 **0 / 5 points**

(S21_21) Ammonia (NH_3) is widely used as a precursor for the preparation of fertilizer and can be synthesized from nitrogen (N_2) and hydrogen (H_2). By using 7 kg of N_2 and 1 kg of H_2, how much NH_3 can be produced?

- 3.65 kg
- 3.56 kg
- 6.35 kg
- ➡ 5.63 kg
- 6.53 kg

Hide question 21 feedback

Feedback

Answer D

Write the balanced chemical reaction:

$N_2 + 3H_2 \rightarrow 2NH_3$

Molecular weight of N_2 = 28.02 g/mol.
Molecular weight of H_2 = 2.016 g/mol.

$n(N_2)$ = mass of N_2/molecular weight of N_2 = 7000 g/28.02 g = 249.82 mol.
$n(H_2)$ = mass of H_2/molecular weight of H_2 = 1000 g/2.016 g = 496.03 mol.

1 mole of N_2 requires 3 moles of H_2
Number of moles of H_2 is the limiting factor as the $n(H_2) < 3 \times n(N_2)$

Number of moles of NH_3 that can be produced, n(NH3) = 2/3 x $n(H_2)$ = 2/3 x 496.03 mol. = 330.68 mol.

Molecular weight of NH_3 = 17.03 g/mol.

Mass of NH_3 produced = $n(NH_3)$ x Molecular weight of NH_3 = 330.68 mol. x 17.03 g/mol.

= 5631.61 g = 5.63 kg

Question 22 **0 / 2 points**

(S21_22) What is the correct order of the coordination numbers for the compounds listed below?

$[Co(NH_3)_4Cl_2]^+$, $[Au(CN_2)]^-$, $[Ce(NO_3)_6]^{2-}$, $[Pt(NH_3)_4]^{2+}$

- 8, 4, 2, 6
- 6, 2, 12, 4
- 12, 8, 6, 4
- 2, 4, 6, 8
- 6, 6, 4, 4

Hide question 22 feedback

Feedback

Answer B

The coordination number of $[Co(NH_3)_4Cl_2]^+$ ion is 6

The coordination number of $[Au(CN_2)]^-$ ion is 2

The coordination number of $[Ce(NO_3)_6]^{2-}$ ion is 12

The coordination number of $[Pt(NH_3)_4]^{2+}$ ion is 4

Question 23 — 0 / 2 points

(S21_23) How many unpaired electrons in total does the ruthenium (Ru) atom has in the ground state configuration and how many of these total electrons are located in d-orbital?

- 4 in total and 3 in d-orbital
- 8 in total and 7 in d-orbital
- 4 in total and 7 in d-orbital
- 8 in total and 3 in d-orbital
- 7 in total and 7 in d-orbital

Hide question 23 feedback

Feedback

Answer A

The electron configuration of Ru is [Kr] $5s^1\ 4d^7$
Among the 5 sub-shells in d-orbital, 2 are paired, leaving 3 unpaired electrons in d-orbital and 1 unpaired electron in s-orbital.

Question 24 — 0 / 3 points

(S21_24) Which of the following compounds contain a polyatomic cation and polyatomic anion?

- NH_4ClO_3
- $K_2Cr_2O_7$
- NaOCN
- $NaC_2H_3O_2$

None of the above

Hide question 24 feedback

Feedback

Answer A

Only NH_4ClO_3 contains the polyatomic cation, $[NH_4]^+$ and polyatomic anion, $[ClO_3]^-$
All other compounds contain monoatomic cations.

Question 25 — 0 / 3 points

(S21_25) Identify the list of compounds that is arranged in increasing order of ionic character.

- IBr, LiBr, KBr, HBr
- Cl_2, CsCl, KCl, HCl
- LiF, CsF, KF, HF
- HI, KI, LiI, CsI
- None of the above

Hide question 25 feedback

Feedback

Answer E

In option (a) HBr has lower ionic character than LiBr and KBr
In option (b) HCl has lower ionic character than KCl and CsCl
In option (c) HF has the lowest ionic character.
In option (d) KI has higher ionic character than LiI

Done

Exam 26030 E16

Question 1:

Below is given a list of pairs of acids and bases:

CCl_3COOH	CH_3COOH
NH_4^+	NH_3
$[Co(H_2O)_6]^{2+}$	$[Co(H_2O)_6]^{3+}$
$[Al(H_2O)_6]^{3+}$ ($pk_a = 4.89$)	$[Fe(H_2O)_6]^{3+}$ ($pK_a = 2.22$)
$[Fe(H_2O)_6]^{2+}$ ($pk_a = 8.30$)	$[Zn(H_2O)_6]^{2+}$ ($pk_b = 5.00$)

In the below scheme tick the box where only the strongest acid of each pair is present. For some pairs this can be decided by logical arguments while you will need to use the given acid (pk_a) and base (pk_b) constants for other pairs:

A	CCl_3COOH;	NH_4^+;	$[Co(H_2O)_6]^{2+}$;	$[Al(H_2O)_6]^{3+}$;	$[Fe(H_2O)_6]^{2+}$
B	CH_3COOH;	NH_3;	$[Co(H_2O)_6]^{3+}$;	$[Fe(H_2O)_6]^{3+}$;	$[Zn(H_2O)_6]^{2+}$
C	CCl_3COOH;	NH_4^+;	$[Co(H_2O)_6]^{2+}$;	$[Fe(H_2O)_6]^{3+}$;	$[Zn(H_2O)_6]^{2+}$
D	CH_3COOH;	NH_3;	$[Co(H_2O)_6]^{3+}$;	$[Al(H_2O)_6]^{3+}$;	$[Fe(H_2O)_6]^{2+}$
E	CCl_3COOH;	NH_4^+;	$[Co(H_2O)_6]^{3+}$;	$[Fe(H_2O)_6]^{3+}$;	$[Fe(H_2O)_6]^{2+}$

Question 2:

A thermite is a pyrotechnic composition of metal powder and metal oxide which, when ignited by heat, will undergo an exothermic redox reaction. Most varieties are not explosive but can create brief bursts of high temperatures in a small area.

One classic example is:

$$2\ Al\ (s) + Fe_2O_3\ (s) \rightarrow Al_2O_3\ (s) + 2\ Fe\ (s)$$

Due to the large amount of energy released the reaction enthalpy cannot be determined directly but rather through a slow inlet of oxygen to minute amounts of the separate metals. The following results were obtained:

(1) $4\ Al\ (s) + 3\ O_2\ (g) \rightarrow 2\ Al_2O_3\ (s)$ $\quad \Delta H_r = -3339.6\ ^{kJ}/_{mol}$
(2) $4\ Fe\ (s) + 3\ O_2\ (g) \rightarrow 2\ Fe_2O_3\ (s)$ $\quad \Delta H_r = -1644.4\ ^{kJ}/_{mol}$

Use the above half reactions to determine ΔH_r for the reaction of aluminium with Fe_2O_3:

A $1695.2\ ^{kJ}/_{mol}$
B $-1695.2\ ^{kJ}/_{mol}$
C $847.6\ ^{kJ}/_{mol}$
D $-847.6\ ^{kJ}/_{mol}$
E $4984\ ^{kJ}/_{mol}$

Question 3:

Ethanol (CH_3CH_2OH) has a standard boiling point of 78.3 °C. At 25 °C the vapour pressure is measured to be 0.1 atm. Calculate the molar heat of evaporation based on these data:

A Insufficient information is given to calculate the molar heat of evaporation
B 0.7 kJ/mol
C 39.3 kJ/mol
D 1.4 kJ/mol
E 37.6 kJ/mol

Question 4:

In the book a spectrochemical series of ligands is presented. A similar series is known for metals:

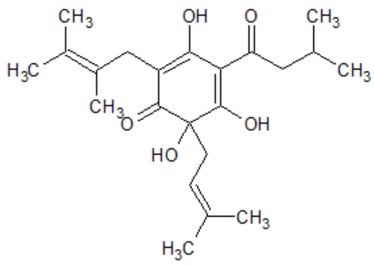
$Mn^{2+} < Ni^{2+} < Co^{2+} < Fe^{2+} < V^{2+} < Fe^{3+} < Cr^{3+} < V^{3+} < Co^{3+}$

The metals found to the right of this series are in other words more likely to have a large splitting between the d-orbital energy levels. Assuming perfect octahedral geometry mark the right d-orbital diagrams for $[Co(NH_3)_6]^{2+}$ and $[Co(NH_3)_6]^{3+}$:

	$[Co(NH_3)_6]^{2+}$	**$[Co(NH_3)_6]^{3+}$**
		[ ] [ ]
A	[↑] [↑]	
	[↑↓] [↑↓] [↑]	[↑↓] [↑↓] [↑↓]
		[↑] [ ]
B	[↑] [↑]	
	[↑↓] [↑] [↑]	[↑↓] [↑↓] [↑↓]
		[↑] [↑]
C	[ ] [ ]	
	[↑↓] [↑↓] [↑↓]	[↑↓] [↑↓] [↑]
		[↑] [↑]
D	[↑] [ ]	
	[↑↓] [↑↓] [↑↓]	[↑↓] [↑] [↑]
		[ ] [ ]
E	[↑] [↑]	
	[↑] [↑] [↑]	[↑↓] [↑] [↑]

Question 5:

Humulone is a bitter tasting compound found in mature hops:

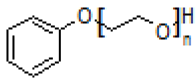


It is commonly used in the production of beer due to its food preserving properties. Which functional groups are present?

A Alkyne, ketone and alcohol
B Alkene, ketone and alcohol
C Alkyne, aldehyde and alcohol
D Alkene, ketone and an aromatic ring
E Alkene, aldehyde and ketone

Question 6:

Pycal 94 is a commercial plasticiser envisaged to replace the commonly used but toxic phthalates. The general structure can be seen here:



To determine the average n the hydroxyl value was measured to 208 $^{mg\ KOH}/_{g\ Pycal\ 94}$. The hydroxyl value gives the content of free OH groups in a chemical substance as the mass of potassium hydroxide equivalent to the OH content. From this the inverse molar weight of Pycal 94 can be determined.
Use this information to calculate n:

A $n \approx 1$
B $n \approx 2$
C $n \approx 3$
D $n \approx 4$
E $n \approx 5$

Question 7:

The production of water free HCl can be achieved through the treatment of kitchen salt (NaCl) with concentrated sulphuric acid (H_2SO_4) as shown in reaction (1):

(1) $NaCl\ (s) + H_2SO_4\ (aq) \rightarrow NaHSO_4\ (s) + HCl\ (g)$

A similar approach might not work for the production of HI as the following side reaction, reaction (2), is a possibility:

(2) $4\ H^+\ (aq) + 2\ I^-\ (aq) + SO_4^{2-}\ (aq) \rightarrow SO_2\ (g) + 2\ H_2O\ (l) + I_2\ (s)$

Using values from the book, calculate the standard potential for the side reaction, given by reaction 2, and evaluate if this side reaction is a real possibility:

A E = 0.33 V; Side reaction will happen
B E = - 0.33 V; Side reaction will not happen
C E = 0.73 V; Side reaction will not happen
D E = 0.73 V; Side reaction will happen
E E = -0.73 V; Side reaction will happen

Question 8:

In the laboratory nitrogen monoxide (NO) can be made through the reduction of dilute nitric acid (HNO_3) by elemental copper:

H^+ (aq) + Cu (s) + NO_3^- (aq) $\rightarrow$ Cu^{2+} (aq) + NO (g) + H_2O (l)

Balance the equation and mark the right set of coefficients below:

	H^+	Cu	NO_3^-	Cu^{2+}	NO	H_2O
A	2	3	2	3	2	1
B	8	2	3	2	3	4
C	16	2	3	2	3	8
D	16	3	2	3	2	8
E	8	3	2	3	2	4

Question 9:

As a consequence of the increase of global CO_2 levels in the atmosphere a change of pH has also been observed throughout the world oceans:

H_2O (l) + CO_2 (g) $\rightarrow$ H_2CO_3 (aq) $\rightarrow$ H^+ (aq) + HCO_3^- (aq)

It has recently been reported that the pH level of the oceans has changed by approx. 0.1 from the preindustrial average of 8. How much has the [H^+] changed?

A ≈ 50 %
B ≈ 25 %
C ≈ 20 %
D ≈ 15 %
E ≈ 1.25 %

Question 10:

Use the VSEPR model to predict the molecular geometry for XeF_2, SOF_4 and ClO_3^-:

A XeF_2: AB_2E_2, bent SOF_4: AB_5E, square pyramidal ClO_3^-: AB_3E, trigonal pyramidal
B XeF_2: AB_2E_2, bent SOF_4: AB_5E, square pyramidal ClO_3^-: AB_3E_2, T-Shaped
C XeF_2: AB_2E_2, bent SOF_4: AB_5, trigonal bipyramidal ClO_3^-: AB_3E_2, T-shaped
D XeF_2: AB_2E_3, linear SOF_4: AB_5, trigonal bipyramidal ClO_3^-: AB_3E_2, T-shaped
E XeF_2: AB_2E_3, linear SOF_4: AB_5, trigonal bipyramidal ClO_3^-: AB_3E, trigonal pyramidal

Question 11:

A bottle containing either Ag^+, Ba^{2+}, Ca^{2+}, Hg^{2+} or Pb^{2+} gives a precipitate when adding SO_4^{2-} but not when adding Cl^- or OH^-. Which of the cations can it contain?

A Ca^{2+} or Ba^{2+}
B Only Ca^{2+}
C Ca^{2+} or Pb^{2+}
D Only Ba^{2+}
E Ba^{2+} or Pb^{2+}

Question 12:

For the gas reaction 2 ClO (g) $\rightarrow$ Cl_2 (g) + O_2 (g) the following sets of rate constants and temperatures have been measured:

T (°C)	32.5	55.0	81.5	110
k (M^{-1} s^{-1})	$2.7 \cdot 10^6$	$4.20 \cdot 10^6$	$6.0 \cdot 10^6$	$8.9 \cdot 10^6$

What is the activation energy of the reaction?

A ≈ 1.5 kJ/mol
B ≈ 15 kJ/mol
C ≈ 0.15 kJ/mol
D ≈ 150 kJ/mol
E ≈ 50 kJ/mol

Question 13:

Consider the unbalanced reaction between ammonia (NH_4^+) and nitrite (NO_2^-):

NH_4^+ (aq) + NO_2^- (aq) $\rightarrow$ N_2 (g) + H_2O (l)

During an experiment the following sets of concentrations and initial reaction rates were measured:

[NH_4^+] (M)	**[NO_2^-] (M)**	**Rate (M/s)**
0.01	0.2	$5.4 \cdot 10^{-7}$
0.02	0.2	$10.8 \cdot 10^{-7}$
0.2	0.02	$10.8 \cdot 10^{-7}$
0.04	0.2	$21.5 \cdot 10^{-7}$
0.2	0.04	$21.7 \cdot 10^{-7}$
0.2	0.08	$43.3 \cdot 10^{-7}$

What is the rate law for the reaction and the reaction constant?

A rate = k · [NH_4^+] · [NO_2^-] ; $2.7 \cdot 10^{-4}$ s^{-1} M^{-1}
B rate = k · [NH_4^+] · [NO_2^-] ; $2.7 \cdot 10^{-2}$ s^{-1} M^{-1}
C rate = k · [NH_4^+] · [NO_2^-]^2 ; $2.7 \cdot 10^{-4}$ s^{-1} M^{-2}
D rate = k · [NH_4^+] · [NO_2^-]^2 ; $2.7 \cdot 10^{-2}$ s^{-1} M^{-2}
E rate = k · [NH_4^+]^2 · [NO_2^-] ; $2.7 \cdot 10^{-2}$ s^{-1} M^{-1}

Question 14:

Using values from the book calculate ΔH_r° for the above reaction between NH_4^+ and NO_2^- and determine if it is exothermic:

$\Delta H_f^\circ (NO_2^-) = -105\ ^{kJ}/_{mol}$

A 333.8 $^{kJ}/_{mol}$ endothermic
B 333.8 $^{kJ}/_{mol}$ exothermic
C -333.8 $^{kJ}/_{mol}$ exothermic
D 48 $^{kJ}/_{mol}$ endothermic
E -48 $^{kJ}/_{mol}$ exothermic

Question 15:

An accurate way of determining solubility products is through the measurement of standard reduction potentials. For the determination of the solubility of AgCl the following potentials were measured:

(1) $Ag^+ (aq) + e^- \rightarrow Ag (s)$ $E^0 = 0.8$ V
(2) $AgCl (aq) + e^- \rightarrow Ag (s) + Cl^- (aq)$ $E^0 = 0.215$ V

Determine the solubility product at 25 °C for $K_{sp} = [Ag^+] \cdot [Cl^-]$ based on the above measurements:

A $6.4 \cdot 10^9\ M^2$
B $1.3 \cdot 10^{-10}\ M^2$
C $7.8 \cdot 10^9\ M^2$
D $1.9 \cdot 10^{-10}\ M^2$
E $1.6 \cdot 10^{-10}\ M^2$

Question 16:

Lead chloride ($PbCl_2$) is generally regarded as an insoluble salt but the solubility product in water is as high as $1.6 \cdot 10^{-5}\ M^3$ making it somewhat soluble in water. Assuming that $PbCl_2$ is added to a 0.4 M solution of NaCl what will the concentration of Pb^{2+} then be at equilibrium?

A $1.0 \cdot 10^{-5}$ M
B $1.0 \cdot 10^{-4}$ M
C $1.0 \cdot 10^{-6}$ M
D $5.0 \cdot 10^{-6}$ M
E $5.0 \cdot 10^{-5}$ M

Question 17:

How many unpaired electrons does a free sulphur atom have?

A 0
B 1
C 2
D 3
E 4

Question 18:

Predict, by using qualitative arguments, if ΔS will be positive, nearly zero or negative for the following reactions:

1 H_2CO_3 (aq) $\rightarrow$ H_2O (g) + CO_2 (g)
2 N_2 (g) + 3 H_2 (g) $\rightarrow$ $2NH_3$ (g)
3 F_2 (g) + H_2 (g) $\rightarrow$ 2 HF (g)

A 1: nearly zero	2: nearly zero	3: positive
B 1: nearly zero	2: negative	3: positive
C 1: nearly zero	2: negative	3: nearly zero
D 1: positive	2: negative	3: nearly zero
E 1: positive	2: negative	3: positive

Question 19:

For mountain bikes a tyre pressure as high as 3 atm is often recommended. A tyre was pumped to this pressure on a hot summer day at 30 °C and then left until a cold winter day with -10 °C. Assuming that no gas has leaked and that the volume is approximately constant what will the pressure be on the cold winter day?

A 2.5 atm
B 2.6 atm
C 2.7 atm
D 2.8 atm
E 2.9 atm

Question 20: Equilibrium

The hypothetical gas reaction A (g) + 2 B (g) $\rightarrow$ C (g) + D (g) is known to be very exothermic. In what direction will the reaction shift upon 1) an increase in temperature, 2) an increase in pressure, 3) an addition of extra B and 4) a removal of the product D?

A 1) left	2) left	3) right	4)	right
B 1) left	2) right	3) right	4)	right
C 1) right	2) left	3) right	4)	right
D 1) right	2) right	3) left	4)	left
E 1) left	2) right	3) left	4)	left

Suggested solution for exam 26030 E16

Question 1:

Answer E

CCl_3COOH as Cl is more electronegative than H and thus "pulls" more in the electrons
NH_4^+ with NH_3 being the corresponding base
$[Co(H_2O)_6]^{3+}$ due to higher charge on the central atom
$[Fe(H_2O)_6]^{3+}$ as pK_a is lower
$[Fe(H_2O)_6]^{2+}$ One value is given as a pK_a the other a pK_b (thus very weakly acidic)

Question 2:

Answer D

The reaction scheme can be written as scheme 2 – scheme 1 divided by 2 $\Rightarrow$

$$\Delta H_r = \frac{\Delta H_{r1} - \Delta H_{r2}}{2} = -847.6\frac{kJ}{mol}$$

Question 3:

Answer E

P_1 = 1.0 atm ; T_1 = (78.3 + 273.15)K (P is given by the term standard)
P_2 = 0.1 atm ; T_2 = (25.0 + 273.15)K
R = 8.3145 $^{J}/_{mol \cdot K}$

$$ln\left(\frac{P_1}{P_2}\right) = \frac{\Delta H_{VAP}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \Rightarrow \Delta H_{VAP} = \frac{ln\left(\frac{P_1}{P_2}\right) \cdot R}{\left(\frac{1}{T_2} - \frac{1}{T_1}\right)} = 37.6\frac{kJ}{mol}$$

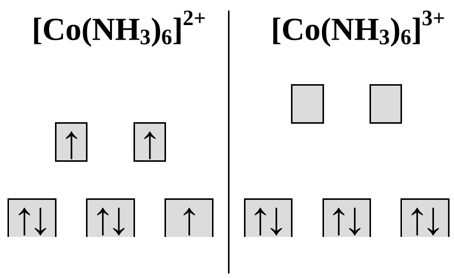
Question 4:

Answer A

From table 20.1:

Co^{2+}: $3d^7$ configuration
Co^{3+}: $3d^6$ configuration

Then use figure 20.1 and the information that Co^{3+} has the larger splitting:



Question 5:

Answer B

Use table 11.4 and information regarding the difference between aldehydes and ketones on p.388

Question 6:

Answer D

$m_{Pycal\ 94}$ = 1.0 g (from 208 $^{mg\ KOH}/_{g\ Pycal\ 94}$)
m_{KOH} = 0.208 g (from 208 $^{mg\ KOH}/_{g\ Pycal\ 94}$)
M_{KOH} = 56.1073 $^{g}/_{mol}$

$$n_{Pycal\ 94} = n_{KOH} = \frac{m_{KOH}}{M_{KOH}} = \frac{0.208g}{56.1073\frac{g}{mol}} 0.003707mol$$

$$M_{Pycal\ 94} = \frac{m_{Pycal\ 94}}{n_{Pycal\ 94}} = \frac{1.0g}{0.003707mol} = 269.7466\frac{g}{mol}$$

$$M_{Pycal\ 94} = M(C_6H_6O) + n \cdot M(C_2H_4O) \Rightarrow n = \frac{M_{Pycal\ 94} - M(C_6H_6O)}{M(C_2H_4O)}$$

$$= \frac{269.7466\frac{g}{mol} - 94.1128\frac{g}{mol}}{44.053\frac{g}{mol}} \approx 4$$

Question 7:

Answer B

From table 19.1:

E^0 (SO_4^{2-} + 4 H^+ + 2 e^- → SO_2 + 2 H_2O) = 0.2 V
E^0 (I_2 + 2 e^- → 2 I^-) = 0.53 V

E^0 = E^0 (SO_4^{2-} + 4 H^+ + 2 e^- → SO_2 + 2 H_2O) - E^0 (I_2 + 2 e^- → 2 I^-) = 0.53 V = -0.33 V
E^0 < 0 ⇒ Side reaction will not happen

Question 8:

Answer E

Identify species changing oxidation state first

Oxidation state of Cu increases by 2 from 0 to 2
Oxidation state of N decreases by 3 from 5 to 2

That is 3 Cu for every 2 N:

3 Cu + 2 NO_3^- → 3 Cu^{2+} + 2 NO

Adjust charges by adding 8 H^+ on left side (acidic environment as reaction happens in dilute nitric acid):

$8\ H^+ + 3\ Cu + 2\ NO_3^- \rightarrow 3\ Cu^{2+} + 2\ NO$

Balance hydrogen atoms by adding 4 water on right side:

$8\ H^+ + 3\ Cu + 2\ NO_3^- \rightarrow 3\ Cu^{2+} + 2\ NO + 4\ H_2O$

Control : 6 oxygen on each side!

Question 9:

Answer B

The pH scale is logarithmic and it is therefore necessary to convert from pH to $[H^+]$ to find the change in $[H^+]$:

$$pH = -\log[H^+] \Rightarrow [H^+] = 10^{-pH}$$

Change in $[H^+]$: $\frac{\left[[H^+]_{before} - [H^+]_{after}\right]}{[H^+]_{before}} \cdot 100\% = \frac{\left[10^{-8.0} - 10^{-7.9}\right]}{10^{-8.0}} \cdot 100\% \approx 25\%$

Question 10:

Answer E

F has 7 electrons, Xe has 8 electrons $\Rightarrow$ 1 electron from Xe to bond with each F, 6 electrons remaining on Xe to be located in 3 lone pairs $\Rightarrow$ XeF_2 AB_2E_3 linear
S has 6 electrons, F has 7 electrons, O has 6 electrons $\Rightarrow$ 2 electrons from S to bond with O, 1 electron from S to bond with each F, no electrons left on S for lone pairs $\Rightarrow$ SOF_4 AB_5 trigonal bipyramidal
Cl has 7 electrons, O has 6 electrons, 1 outer charge $\Rightarrow$ 2 electrons from Cl to bond with each O, 1 electron remaining on Cl to be located in 1 lone pair together with outer charge $\Rightarrow$ ClO_3^- AB_3E trigonal pyramidal

Question 11:

Answer D

Use table 4.2 $\Rightarrow$ Only Ba^{2+} can be in the bottle

Question 12:

Answer B

Plot ln K vs 1/T with T in K and find the slope α
E_a is then $-\ \alpha \cdot 8.3145\ {}^{J}/_{mol \cdot K} \approx 15\ {}^{kJ}/_{mol}$

Question 13:

Answer A

Doubling $[NH_4^+]$ at constant $[NO_2^-]$ doubles rate $\Rightarrow$ 1st order in respect to $[NH_4^+]$
Doubling $[NO_2^-]$ at constant $[NH_4^+]$ doubles rate $\Rightarrow$ 1st order in respect to $[NO_2^-]$

$$k = \frac{rate}{[NH_4^+] \cdot [NH_2^-]}$$

Choose any set of data and calculate k $\Rightarrow$ k $\approx 2.7 \cdot 10^{-4}$ $s^{-1} \cdot M^{-1}$

Question 14:

Answer C

Balance reaction: NH_4^+ (aq) + NO_2^- (aq) $\rightarrow$ N_2 (g) + 2 H_2O (l)

$\Delta H_r = \Delta H_{products} - \Delta H_{reactants}$
$\Delta H_r = 2 \cdot \Delta H(H_2O\ (l)) + \Delta H(N_2\ (g)) - \Delta H(NH_4^+\ (aq)) - \Delta H(NO_2^-\ (aq))$
$\Delta H_r = (2 \cdot - 280.5 - (- 105 - 132.8))$ $^{kJ}/_{mol}$ = - 333.8 $^{kJ}/_{mol}$

Question 15:

Answer B

$E^0 = E^0_{r2} - E^0_{r1} = (0.215 – 0.8)$ V = - 0.585 V

$$E = E^0 - \frac{R \cdot T}{n \cdot F} \cdot \ln(Q)$$

Equilibrium $\Rightarrow$ Q = K_{sp} $\Rightarrow$ E = 0 $\Rightarrow$

$$0 = E^0 - \frac{R \cdot T}{n \cdot F} \cdot \ln(K_{sp}) \Rightarrow K_{sp} = e^{\frac{E^0 \cdot n \cdot F}{R \cdot T}} = 1.29 \cdot 10^{-10} M^2$$

Question 16:

Answer B

$K_{sp} = [Pb^{2+}] \cdot [Cl^-]^2 = x \cdot (0.4+x)^2$

0.4 + x ≈ 0.4 as salt is only slightly soluble (alternatively solve 3. order equation in e.g. Maple)

$K_{sp} \approx x \cdot 0.4^2 \Rightarrow x \approx 10^{-4}$ M

Question 17:

Answer C

Overall configuration from table 7.3. Then apply Hund’s rule giving as many electrons as possible with parallel spin $\Rightarrow$ 2 unpaired electrons

Question 18:

Answer D

1: high entropy gas molecules formed at the expense of low entropy molecules in solution $\Rightarrow$ positive
2: high entropy gas molecules are removed $\Rightarrow$ negative
3: bonds rearranged but total amount of high entropy gas molecules maintained $\Rightarrow$ nearly zero

Question 19:

Answer B

Ideal gas law with n, R and V all constant $\Rightarrow \frac{P_1}{T_1} = \frac{n \cdot R}{V} = \frac{P_2}{T_2} \Longrightarrow P_1 = \frac{P_2 \cdot T_1}{T_2}$
Remember temperatures in K $\Rightarrow$ 2.6 atm

Question 20:

Answer B

Use Le Chatelier's principle:

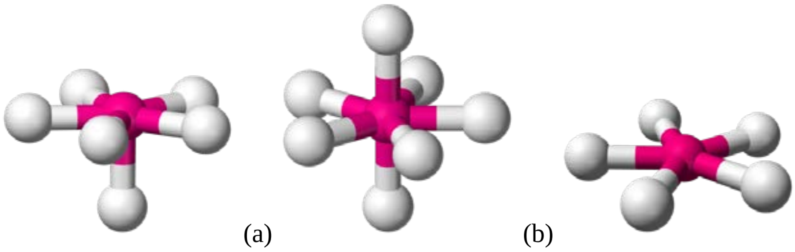
1: left – heat is removed when going left (endothermic) thus lowering temperature as a response to the change
2: right – amount of gas molecules are being reduced when going right lowering the pressure as a response to the change
3: right – some of the added b is thereby being removed as a response to the change
4: right – some of the removed d is thereby being generated as a response to the change

Exam 26030 E17

<u>Question 1</u>:

(5 points)

The VSEPR model describes optimal molecular geometries arising from repulsions between electron pairs. The text book deals with up to 6 electron pairs but the theory can easily be extended to geometries containing more electron pairs. Rank the below geometries according to an increasing amount of lone pairs:



(a) (b) (c)

Please note that only bonding pairs are shown.

A: A > B > C
B: C > B > A
C: C > A > B
D: B > A > C
E: C = A > B

<u>Question 2</u>:

(5 points)

In a reaction between A and B the compounds C and D are formed. Below are given 3 different reaction schemes for the same reaction:

1) $2\,A + B \rightleftarrows 2\,C + D$
2) $4\,A + 2\,B \rightleftarrows 4\,C + 2\,D$
3) $6\,A + 3\,B \rightleftarrows 6\,C + 3\,D$

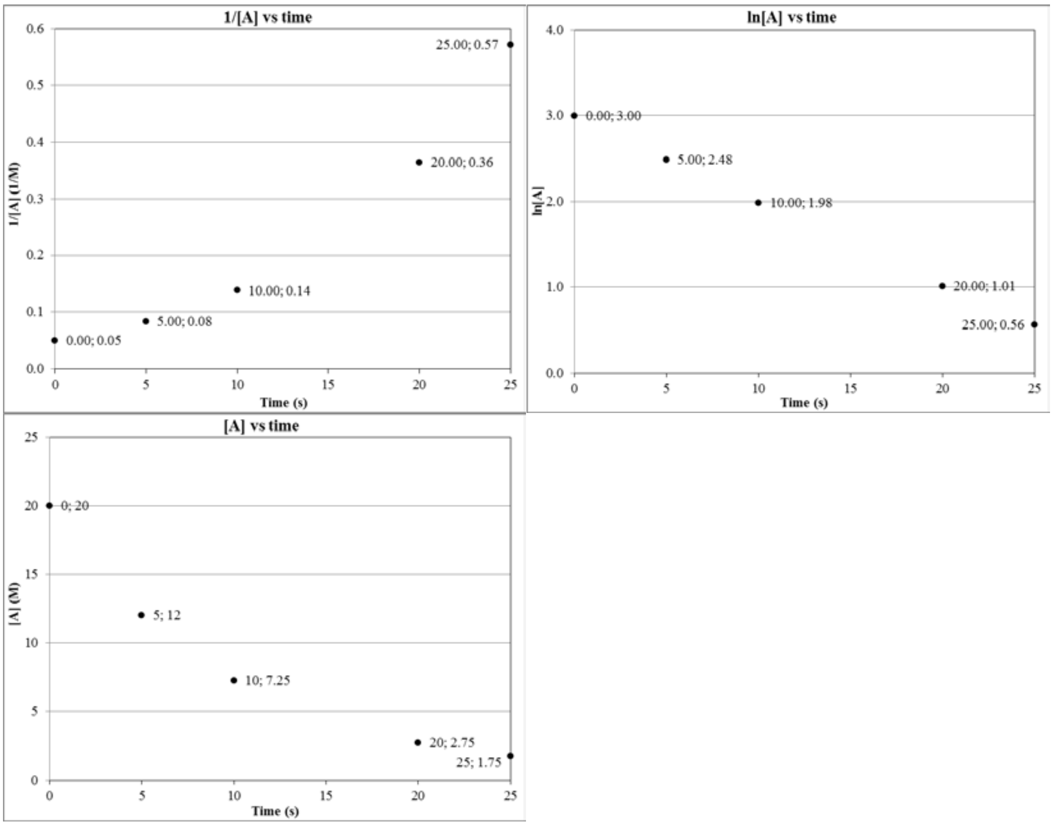
Given that the concentrations at equilibrium are [A] = 0.1M, [B] = 0.2 M, [C] = 0.3 M and [D] = 0.4 M how are the equilibrium constants of the 3 reactions then related?

A: No calculations are needed: $K_1 = K_2 = K_3$ as it is the same reaction just multiplied with different coefficients
B: $K_3 > K_2 > K_1$
C: $K_3 < K_2 < K_1$
D: No calculations are needed: $K_1 = K_2 = K_3$ as it is the same reaction just multiplied with different coefficients. Additionally, the value is 1.0 as the coefficient of A equals that of C and the coefficient of B equals that of C
E: No calculations are needed: The value is 1.0 as the coefficient of A equals that of C and the coefficient of B equals that of C

Question 3:

(3 points)

For the decomposition of a compound A the concentration of A has been measured as a function of time. Below the data has been recorded in 3 different ways:



Coordinates are given for easier data treatment.

Use the above plots to determine the reaction order with respect to A:

A 1st order
B 2nd order
C 0th order
D Plots are insufficient to determine the reaction order
E Plots are insufficient to determine the reaction order but indicate a catalysed reaction

Question 4:

(2 points)

Determine the rate constant k from the data given in question 3:

A $\approx 0.5\ s^{-1}$

B ≈ 0.4 s^{-1}
C ≈ 0.3 s^{-1}
D ≈ 0.2 s^{-1}
E ≈ 0.1 s^{-1}

Question 5:

(2 points)

For the reaction 2 NO (g) + 2 H_2 (g) → N_2 (g) + 2 H_2O (g) the rate law has been determined experimentally:

$$Rate = k \cdot [NO]^2 \cdot [H_2]$$

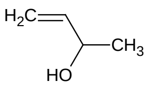
What is the reaction order in respect to [NO], [H_2] and the overall reaction order?

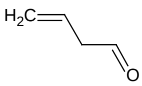
A 2, 1, 1½
B 2, 1, 3
C 2, 1, 2
D 1, 1, 3
E 1, 1, 2

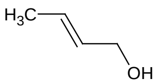
Question 6:

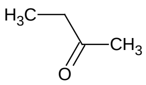
(2 points)

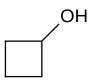
A molecule has the molecular formula C_4H_8O. Which structure is not consistent with the molecular formula?

A 

B 

C 

D 

E 

Question 7:

(4 points)

Assuming that the molecule from question 6 is an aldehyde how many isomers are then possible? Please note that there might be more isomers with the molecular formula C_4H_8O than those shown in question 6.

A 1
B 2
C 3
D 4
E 5

Question 8:

(2 points)

A salt has been investigated and both the anion and the cation are found to be oxidised by oxygen. Which salt is it?

A CaS
B $CaSO_4$
C FeS
D $FeSO_4$
E $KMnO_4$

Question 9:

(3 points)

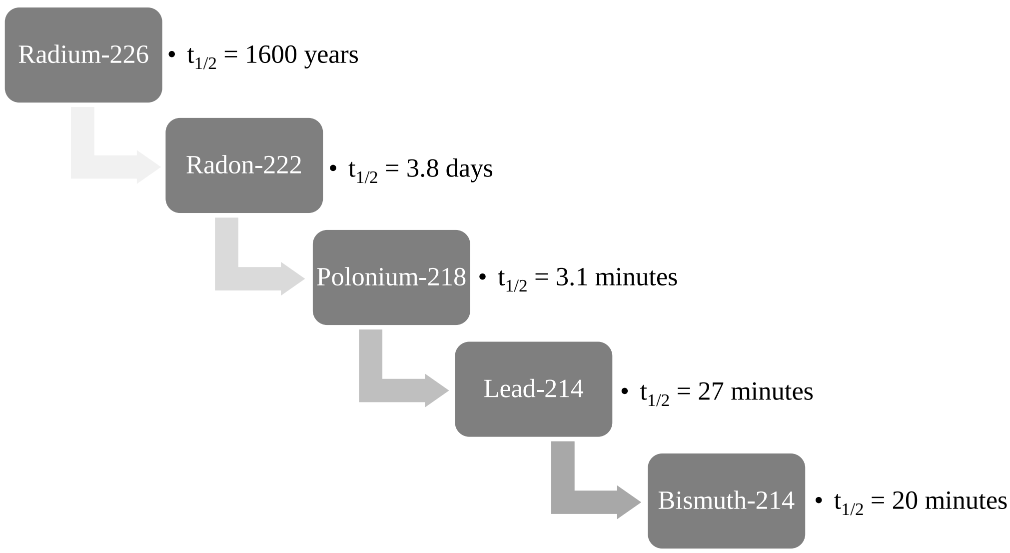
$FeSO_4$ is reacted with $KMnO_4$ in an acidic aqueous solution. What is $FeSO_4$ converted into?

A Fe^{3+} and SO_4^{2-}
B Fe^{2+} and S
C Fe^{3+} and S
D Fe^{2+} and SO_4^{2-}
E Fe^{3+} and S^{2-}

Question 10:

(3 points)

Being a noble gas radon is from a chemical point of view considered harmless but from a physical point of view e.g. the isotope radon-222 is considered unstable with a half-life of 3.8 days. Radon-222 is therefore considered a significant health hazard when evaluating the indoor environment in buildings. Below you can find a part of the decay chain for radon-222:



Given that the rate law is 1st order for all the decay processes which process will then have the higher rate constant?

A Decay of radium-226
B Decay of radon-222
C Decay of polonium-218
D Decay of lead-214
E Decay of bismuth-214

<u>Question 11</u>:

(3 points)

Given the half-life constants from question 10 which element will then be present at the highest concentration after 1 half-life period of radium-226 (1600 years) assuming that everything is radium-226 at the beginning?

A Radium-226
B Radon-222
C Polonium-218
D Lead-214
E Bismuth-214

<u>Question 12</u>:

(5 points)

Copper can be produced by roasting a copper sulphide using controlled amounts of oxygen. E.g.:

$CuS\ (s) + O_2\ (g) \rightarrow Cu\ (s) + SO_2\ (g)$

Calculate the reaction enthalpy using the following reactions:

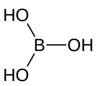
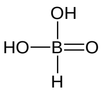
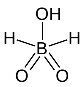
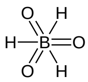
1 $2\ SO_2\ (g) + O_2\ (g) \rightarrow 2\ SO_3\ (g)$ $\Delta H_1° = -198\ ^{kJ}/_{mol}$
2 $S\ (s) + O_2\ (g) \rightarrow SO_2\ (g)$ $\Delta H_2° = -296\ ^{kJ}/_{mol}$
3 $2\ S\ (s) + 3\ O_2\ (g) \rightarrow 2\ SO_3\ (g)$ $\Delta H_3° = -790\ ^{kJ}/_{mol}$
4 $CuS\ (s) + 2\ O_2\ (g) \rightarrow CuSO_4\ (s)$ $\Delta H_4° = -721\ ^{kJ}/_{mol}$
5 $2\ Cu\ (s) + O_2\ (g) \rightarrow 2\ CuO\ (s)$ $\Delta H_5° = -310\ ^{kJ}/_{mol}$
6 $2\ CuO\ (s) + 2\ SO_2\ (g) + O_2\ (g) \rightarrow 2\ CuSO_4\ (s)$ $\Delta H_6° = -638\ ^{kJ}/_{mol}$

A $\approx 250\ ^{kJ}/_{mol}$
B $\approx 500\ ^{kJ}/_{mol}$
C $\approx -500\ ^{kJ}/_{mol}$
D $\approx 0\ ^{kJ}/_{mol}$
E $\approx -250\ ^{kJ}/_{mol}$

<u>Question 13</u>:

(5 points)

The molecular formula of boric acid is H_3BO_3 but this gives no information about the actual structure. Four possible examples are given below:


a),

b),

c) and

d)

Besides a violation of the octet rule for the structures b), c) and d) the formal charges can also be used to indicate the likelihood of the individual structures. Rank the likelihood of the above structures based on an evaluation of formal charges only:

A A > B > C > D
B D > C > B > A
C A > B = C = D
D A < B = C = D
E A = B = C = D

<u>Question 14</u>:

(*5 points*)

Mark the option where all the compounds are acids:

A HCl, H_2SO_4, CH_3CH_2COOH
B HCl, H_2SO_4, NaH
C HCl, H_2SO_4, CH_4
D CH_3CH_2COOH, H_2SO_4, KOH
E HCl, CH_3CH_2COOH, CH_3CH_2OH

<u>Question 15</u>:

(1 point)

The reaction $2\ A\ (s) + B\ (g) \rightarrow 3\ C\ (g)$ is known to be endothermic towards the right. What will happen upon addition of B?

A The reaction will shift to the right
B The reaction will shift to the left
C Nothing

Question 16:

(1 point)

The reaction 2 A (s) + B (g) → 3 C (g) is known to be endothermic towards the right. What will happen upon the addition of catalyst (volume change can be neglected)?

A The reaction will shift to the right
B The reaction will shift to the left
C Nothing

Question 17:

(1 point)

The reaction 2 A (s) + B (g) → 3 C (g) is known to be endothermic towards the right. What will happen upon an increase in pressure?

A The reaction will shift to the right
B The reaction will shift to the left
C Nothing

Question 18:

(1 point)

The reaction 2 A (s) + B (g) → 3 C (g) is known to be endothermic towards the right. What will happen upon lowering the temperature?

A The reaction will shift to the right
B The reaction will shift to the left
C Nothing

Question 19:

(2 points)

Estimate if the following unbalanced reaction is entropically favourable at standard conditions?

SO_2 (g) + O_2 (g) → SO_3 (g)

If necessary use thermodynamic properties from the book.

A Favourable
B Not favourable
C Neither

Question 20:

(2 points)

Estimate if the following unbalanced reaction is favourable at standard conditions?

SO_3^{2-} (aq) + Mn^{2+} (aq) → SO_4^{2-} (aq) + MnO_4^- (aq)

If necessary use thermodynamic properties from the book.

A Favourable
B Not favourable
C Neither

Question 21:

(2 points)

Estimate if the following unbalanced reaction is favourable at standard conditions?

Cu^{2+} (aq) + Fe (s) → Cu (s) + Fe^{2+} (aq)

If necessary use thermodynamic properties or similar from the book.

A Favourable
B Not favourable
C Neither

Question 22:

(2 points)

Estimate if a reaction with $\Delta_rH° = 134.63\ ^{kJ}/_{mol}$ and $\Delta_rS° = 200\ ^{J}/_{mol\ K}$ is favourable at 400 °C?

A Favourable
B Not favourable
C Neither

Question 23:

(5 points)

Which of the following mixtures will have the largest freezing-point depression?

A 3.0 m CH_3CH_2OH
B 2.5 m CH_3CH_2COOH
C 2.0 m NaCl
D 1.5 m Na_3PO_4
E 1.0 m H_2SO_4

Question 24:

(5 points)

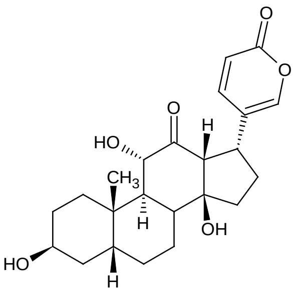
Which of the following compounds will exert the largest pressure at 25 °C assuming equal amounts of moles from the start?

A Acetone (Boiling point = 56 °C, ΔH_{vap} = 31.3 $^{kJ}/_{mol}$)
B Ammonia (Boiling point = -33 °C, ΔH_{vap} = 23.4 $^{kJ}/_{mol}$)
C Methanol (Boiling point = 65 °C, ΔH_{vap} = 35.2 $^{kJ}/_{mol}$)
D Ethanol (Boiling point = 78 °C, ΔH_{vap} = 38.6 $^{kJ}/_{mol}$)
E Water (Boiling point = 100 °C, ΔH_{vap} = 40.7 $^{kJ}/_{mol}$)

Question 25:

(4 points)

Arenobufagin is a cardiotoxic bufanolide steroid secreted by the Argentine toad *Bufo Arenarum*. It has effects similar to digitalis, blocking the Na+/K+ pump in heart tissue. It is a rather complex heterocyclic compound:



Besides a number of alcohol groups which functional groups are present?

A Ether, ketone, alkene
B Ester, aldehyde, alkyne
C Ester, ketone, alkene
D Ether, aldehyde, alkene
E Ester, ketone, alkyne

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Question 26:

(5 points)

Hydrogen cyanide (HCN) is a weak acid with pK_a = 9.21. What is the ratio between HCN and the corresponding base (CN^-) at equilibrium when the formal concentration of HCN is 1.0 M?

A ≈ 2.5/1000000
B ≈ 2.5/100000
C ≈ 2.5/10000
D ≈ 2.5/1000
E ≈ 2.5/100

Question 27:

(5 points)

Calcium (Ca), potassium (K), argon (Ar), chlorine (Cl) and sulphur (S) can occur as isoelectronic species with the $1s^2 2s^2 2p^6 3s^2 3p^6$ electronic structure. Below mark the correct combination of charges with the isoelectronic species arranged according to an increased radius:

A $Ca^{2+} < K^+ < Ar < Cl^- < S^{2-}$
B $Ca^{2-} < K^- < Ar < Cl^+ < S^{2+}$
C $Ca^{2+} > K^+ > Ar > Cl^- > S^{2-}$
D $Ca^{2-} > K^- > Ar > Cl^+ > S^{2+}$
E K > Ca > S > Cl > Ar

<u>Question 28</u>:

(5 points)

A disproportionation is a specific type of redox reaction in which an element from a reaction undergoes both oxidation and reduction to form two different products. An example is the reaction of chlorine (Cl_2) with hydroxide (OH^-):

Cl_2 (g) + OH^- (aq) → Cl^- (aq) + ClO_3^- (aq) + H_2O (l)

Mark the right combination of coefficients below:

A 1, 2, 1, 1, 1
B 3, 6, 5, 1, 3
C 1, 2, 1, 1, 2
D 3, 6, 5, 1, 6
E 2, 4, 2, 2, 3

<u>Question 29</u>:

(5 points)

An adult male needs to consume food corresponding to an energy intake of approx. 10.000 kJ each day. Most recommendations suggest that about 50% of that should come from carbohydrates which corresponds to about 400 g. For simplicity, we can assume it all to be glucose ($C_6H_{12}O_6$). How many grams of oxygen (O_2) is necessary for a stoichiometric consumption?

A ≈ 50 g
B ≈ 175 g
C ≈ 300 g
D ≈ 425 g
E ≈ 550 g

<u>Question 30</u>:

(5 points)

Assuming that oxygen (O_2) is used at 100 % efficiency how large a volume of air should then be inhaled to cover the approximately 25 moles of oxygen needed to convert a balanced diet? The partial pressure of oxygen is 0.2 atm and the temperature and the total pressure are 25 °C and 1 atm respectively.

A ≈ 3 L
B ≈ 30 L
C ≈ 300 L
D ≈ 3000 L
E ≈ 30000 L

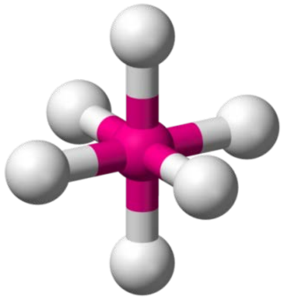
Suggested solution for exam 26030 E17

Question 1:

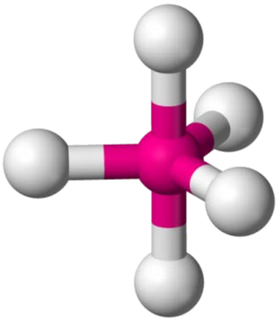
Answer C

VSEPR theory basically states that electrons will stay as far as part as possible. For the structures in A) and C) this is not the case indicating the presence of one or more lone pairs. For A) the free space above the central atom indicates the presence of a lone pair here. For C) the free space both above and below the central atom indicates the presence of a lone pair at both locations

Without the presence of lone pairs different geometries would have been expected:



Optimal geometry of bonding pairs for (a) without lone pairs



Optimal geometry of bonding pairs for (c) without lone pairs:

Question 2:

Answer B

As the value of [C] > [A] and [D] > [B] the value of K must increase from K_1 to K_3.

Alternatively do the calculations

$$K_1 = \frac{[C]^2 \cdot [D]}{[A]^2 \cdot [B]} = 18$$
$$K_2 = \frac{[C]^4 \cdot [D]^2}{[A]^4 \cdot [B]^2} = 324$$
$$K_3 = \frac{[C]^6 \cdot [D]^3}{[A]^6 \cdot [B]^3} = 5832$$

From a thermodynamic point of view the reaction Gibbs free energy also depends on the specific coefficients: $\Delta G_r^0 = \sum n\Delta G_f^0(products) - m\Delta G_f^0(reactants)$ with m and n being stoichiometric coefficients

It is related to the equilibrium constant through the following equation: $\Delta G^0 = -R \cdot T \cdot ln(K)$. As $\Delta G°$ changes with the coefficients so must the equilibrium constant.

Question 3:

Answer A

Since a plot of ln [A] as a function of time gives a straight line the reaction order is 1 with respect to A

Question 4:

Answer E

The reaction rate is given as the negative of the slope (α):

$$k = -\alpha = -\frac{\Delta Y}{\Delta X} = -\frac{0.56 - 3}{25s - 0s} \approx 0.1s^{-1}$$

Question 5:

Answer B

The reaction order of the individual components is the power to which they are raised so 2 for [NO] and 1 for $[H_2]$
The overall reaction order is the sum of the individual reaction orders so 2 + 1 = 3.

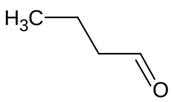
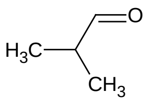
Question 6:

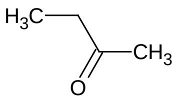
Answer B

Each double bond or ring corresponds to the removal of 2 hydrogen atoms. Since structure B has 2 double bonds this will not be consistent with the molecular formula. The molecular formula for this compound is C_4H_6O.

Question 7:

Answer B

Only  and  are valid options given that it has to be an aldehyde

The structure  is a ketone and not an aldehyde

Question 8:

Answer C

Ca and K are already in their highest oxidation states (+II and +I) and cannot be oxidised any further).
Fe is both known as Fe(II) and Fe(III) and can be further oxidised to Fe(III)
S is fully reduced in S^{2-} but fully oxidised in SO_4^{2-}. Mn in MnO_4^- is fully oxidised. Since both ions can be oxidised it must be FeS.

Question 9:

Answer A

Fe^{2+} is not fully oxidised and will be oxidised by the strong oxidant MnO_4^-. While oxidising Fe the Mn is itself being reduced. The products are Fe^{3+} and SO_4^{2-}

Question 10:

Answer C

For a 1st order reaction the half-life ($t_{1/2}$) and the rate constant are inversely related:

$$t_{\frac{1}{2}} = \frac{ln(2)}{k}$$

A small $t_{1/2}$ thus corresponds to a high k

Question 11:

Answer A

Since $t_{1/2}$ is much larger for radium-226 than for the other elements the other elements are converted almost instantly when compared to radium-226. After 1600 years the concentration of radium-226 will be half of the original value while the remaining concentrations will be close to zero as they have already almost completely converted further to the stable isotope lead-206 (They have completed many more half-life periods).

Question 12:

Answer E

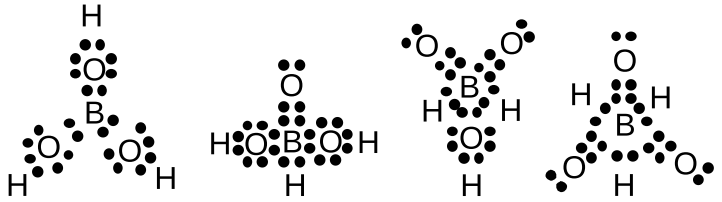
The reaction enthalpy can be obtained using reactions 4, 5 and 6 only:

$$\Delta H^0 = \frac{2 \cdot \Delta H_4^0 - \Delta H_5^0 - \Delta H_6^0}{2} \approx -250 \frac{kJ}{mol}$$

Question 13:

Answer A

Draw the Lewis dot structures:



Bonding electrons are equally distributed between the two bonding atoms.
Lone pair electrons belong to the atom.

Hydrogen gets the charge 0 in all the structures.
Oxygen gets the charge 0 when single bonded to hydrogen or boron.

Oxygen gets the charge +2 when double bonded to boron.
Boron gets the charge 0 when no oxygen is double bonded to it.
Boron gets the charge -2 when 1 oxygen is double bonded to it.
Boron gets the charge -4 when 2 oxygen are double bonded to it.
Boron gets the charge -6 when 3 oxygen are double bonded to it.

Separation of charges should be avoided when writing Lewis structures. Structures B), C) and D) are thus increasingly unlikely both when considering the octet rule and the formal charges.

Question 14:

Answer A

HCl is a stong acid
NaH is ionic but here H will have the formal charge -1 as it is bonded to an alkali metal. It will thus not release H^+.
CH_4 is covalently bonded and will not release H^+.
H_2SO_4 is a strong acid.
KOH is a strong base.
CH_3CH_2OH is neither a base nor an acid.
CH_3CH_2COOH is an organic acid.

Question 15:

Answer A

Use Le Chateliers principle. Reaction will shift to the right in order to remove the added B.

Question 16:

Answer C

Nothing. A catalysator only changes the reaction speed and not the position of the equilibrium.

Question 17:

Answer B

Use Le Chateliers principle. Reaction will shift to the left in order to reduce the amount of gas and thus the pressure.

Question 18:

Answer B

Use Le Chateliers principle. Reaction will shift to the left as it will then be exothermic and produce heat. This will compensate the reduction in temperature.

Question 19:

Answer B

Balance equation:

$2\ SO_2\ (g) + O_2\ (g) \rightarrow 2\ SO_3\ (g)$

Notice the decrease in number of gas molecules which is highly disfavoured.

<u>Question 20</u>:

Answer B

Both SO_3^{2-} and Mn^{2+} are being oxidised
S from IV to VI
Mn from II to VII

An oxidising agent is needed

<u>Question 21</u>:

Answer A

E.g. look at the electrochemical series or the standard reduction potentials

<u>Question 22</u>:

Answer C

Convert $\Delta_r H°$ = to $^{J}/_{mol}$ and temperature to Kelvin

$\Delta_r G° = \Delta_r H° - T \cdot \Delta_r S° = 0$

Neither favourable nor disfavourable

<u>Question 23</u>:

Answer D

$\Delta T_f = I \cdot K_f \cdot m$

K_f is constant so just find the larger $i \cdot m$

$i \cdot m = 3$ for CH_3CH_2OH ($i = 1$)
$i \cdot m \approx 2.5$ for CH_3CH_2COOH (weak acid $\Rightarrow$ very little dissociation $\Rightarrow I \approx 1$)
$i \cdot m = 4$ for NaCl ($i = 2$)
$i \cdot m = 6$ for Na_3PO_4 ($i = 4$)
$i \cdot m = 3$ for H_2SO_4 (strong acid $\Rightarrow$ almost full dissociation $\Rightarrow I \approx 3$)

<u>Question 24</u>:

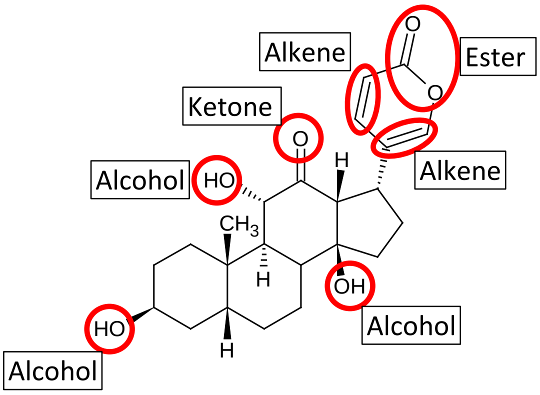
Answer B

Ammonia as it is already fully converted to a gas at 25 °C and will have a pressure determined by the ideal gas equation. The remaining compounds will have a pressure determined by the Clausius-

Clapeyron equation and of these the one with the lower enthalpy of vaporization will have the higher pressure.

Question 25:

Answer C



Question 26:

Answer B

	HCN	+	H_2O	→	H_3O^+	+	CN^-
Start	1.0				0		0
Equilibrium	1.0-X				X		X

$$K_a = \frac{[H_3O^+]\cdot[CN^-]}{[HCN]} = \frac{X \cdot X}{1.0 - X} \approx X^2 \text{ as HCN is a weak acid X<<<1}$$

$$K_a = 10^{-pK_a}$$

$$X = \sqrt{10^{-pK_a}} \approx 2.48 \cdot 10^{-5} M$$

$$\frac{[CN^-]}{[HCN]} = \frac{2.48 \cdot 10^{-5}}{1.0} \approx \frac{2.5}{100000}$$

Question 27:

Answer A

The given electronic structure matches that of argon with 18 electrons
Sulphur is element 16 and thus needs to gain 2 electrons
Chlorine is element 17 and thus needs to gain 1 electron
Potassium is element 19 and thus needs to lose 1 electron
Calcium is element 20 and thus needs to lose 2 electrons
Going from a double charged cation to a double charged anion the size will increase due to changes in electron-electron repulsion and screening of the core charge.

Question 28:

Answer B

Assign oxidation states:

0				-I		+V		

Cl_2 (g)	+	OH^- (aq)	→	Cl^- (aq)	+	ClO_3^- (aq)	+	+ H_2O (l)

Cl↓1
Cl↑5

For every ClO_3^- 5 Cl^- are needed:

3 Cl_2 (g) + OH^- (aq) → 5 Cl^- (aq) + ClO_3^- (aq) + H_2O (l)

1 negative charge on left side, 6 negative on right side. As environment is alkaline balance charges with extra OH^- on left side:

3 Cl_2 (g) + 6 OH^- (aq) → 5 Cl^- (aq) + ClO_3^- (aq) + H_2O (l)

Balance H by adding water on right side:

3 Cl_2 (g) + 6 OH^- (aq) → 5 Cl^- (aq) + ClO_3^- (aq) + 3 H_2O (l)

Verify that the amount on oxygen is equal on both sides

Question 29:

Answer D

First balance the equation:

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$$

Find the amount of glucose used:

$$M(C) = 12.01\frac{g}{mol}, M(O) = 16.00\frac{g}{mol} \text{ and } M(H) = 1.008\frac{g}{mol}$$

$$M(C_6H_{12}O_6) = 6 \cdot M(C) + 12 \cdot M(H) + 6 \cdot M(O) = 180.56\frac{g}{mol}$$

$$m = n \cdot M \Rightarrow n(C_6H_{12}O_6) = \frac{m(C_6H_{12}O_6)}{M(C_6H_{12}O_6)} = \frac{400g}{180.56\frac{g}{mol}} = 2.215mol$$

6 oxygen consumed for each glucose:

$$M(O_2) = 2 \cdot M(O) = 32\frac{g}{mol}$$
$$m(O_2) = n(O_2) \cdot M(O_2) = 6 \cdot n(C_6H_{12}O_6) \cdot M(O_2) = 425.32g$$

Question 30:

Answer D

$$P(O_2) = X(O_2) \cdot P(total) = 0.2\text{atm}$$
$$P(total) = 1.0\text{atm}$$
$$X(O_2) = \frac{P(O_2)}{P(total)} = 0.2 = \frac{n(O_2)}{n(total)} \Longrightarrow n(total) = \frac{n(O_2)}{0.2} = 125mol$$

$$P \cdot V = n(total) \cdot R \cdot T \Longrightarrow V = \frac{n(total) \cdot R \cdot T}{P} = 3060L$$

Remember T in Kelvin

Re-exam 26030 F17

Question 1:

The solubility of silver(I) sulphate (Ag_2SO_4) is known to be 12.4 g/L in pure water at 80 °C. What is the solubility product at this elevated temperature?

A K_{sp} (80 °C) ≈ $2.5 \cdot 10^{-2}$ M^3
B K_{sp} (80 °C) ≈ $2.5 \cdot 10^{-3}$ M^3
C K_{sp} (80 °C) ≈ $2.5 \cdot 10^{-4}$ M^3
D K_{sp} (80 °C) ≈ $2.5 \cdot 10^{-5}$ M^3
E K_{sp} (80 °C) ≈ $2.5 \cdot 10^{-6}$ M^3

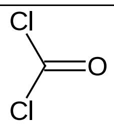
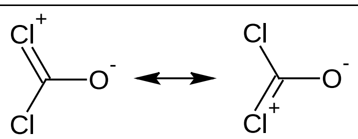
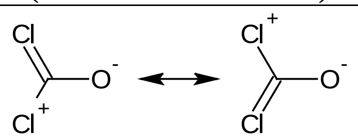
Question 2:

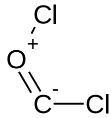
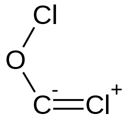
Both silver(I) sulphate (Ag_2SO_4) and calcium sulphate ($CaSO_4$) are rather insoluble in water at low temperatures. In spite of the rather similar solubility products in pure water at 25 °C ($K_{sp}(Ag_2SO_4) = 1.6 \cdot 10^{-5}$ M^3, $K_{sp}(CaSO_4) = 2.3 \cdot 10^{-5}$ M^2) only silver(I) sulphate will dissolve upon the addition of concentrated ammonia (NH_3). What is the origin of this difference?

A The acidic properties of ammonia are sufficient for dissolving silver(I) sulphate but not calcium sulphate
B Ammonia can oxidise Ag(I) to Ag(II) under the formation of NO and/or NO_2. This oxidation is not possible for Ca which is already at its maximum oxidation state
C Actually they will both initially dissolve but Ca will rapidly re-precipitate as $Ca(OH)_2$ and it will appear like nothing happens in this case
D Being a transition metal silver has a much larger tendency to form coordination compounds with a strong ligand such as ammonia
E They will both dissolve over time but the kinetics are much slower in the case of calcium sulphate and it only appears like nothing happens

Question 3:

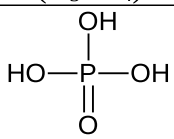
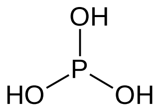
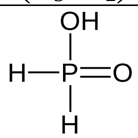
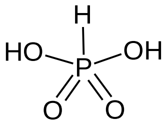
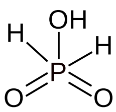
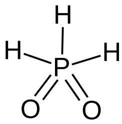
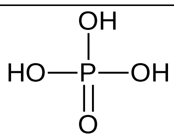
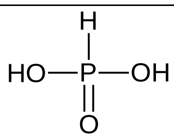
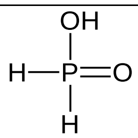
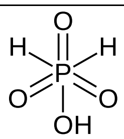
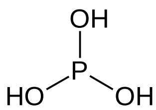
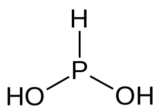
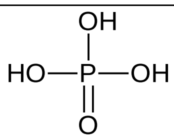
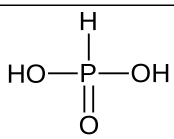
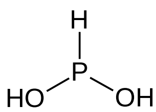
Phosgene ($COCl_2$) was used as a chemical weapon during World War I. It forms hydrochloric acid (HCl) upon reaction with water and crosslinks proteins through reactions with amines altogether causing damages to the lungs and subsequent suffocation.
Which is the more plausible configuration of this reactive molecule?

A	
B	 (A resonance structure)
C	 (A resonance structure)

D	
E	

<u>Question 4</u>:

Mark the correct structure of (ortho)phosphoric acid (H_3PO_4), phosphorous acid (H_3PO_3) and hypophosphorous acid (H_3PO_2) knowing that they have 3, 2 and 1 acidic protons respectively:

	(H_3PO_4)	(H_3PO_3)	(H_3PO_2)
A			
B			
C			
D			
E			

<u>Question 5</u>:

What is the formal oxidation number of phosphorus (P) in (ortho)phosphoric acid (H_3PO_4), phosphorous acid (H_3PO_3) and hypophosphorous acid (H_3PO_2) and which is the stronger reducing agent?

A + V, + III, + I, H_3PO_2
B - V, - III, - I, H_3PO_2
C + V, + III, + I, H_3PO_4
D - V, - III, - I, H_3PO_4
E - V, - III, - I, H_3PO_3

<u>Question 6</u>:

Borax ($Na_2B_4O_7 \cdot 10\ H_2O$) is commonly used to prepare buffer solutions due to the formation of equal amounts of boric acid ($B(OH)_3$) and its corresponding base tetrahydroxyborate ($B(OH)_4^-$) when reacting with water:

$$B_4O_7^{2-} + 7\ H_2O \rightarrow 2\ B(OH)_3 + 2\ B(OH)_4^-$$

Calculate the pH of a 1.0 M solution of borax in water assuming that the K_a of boric acid is $6 \cdot 10^{-10}$ M?

A pH ≈ 7.2
B pH ≈ 8.2
C pH ≈ 9.2
D pH ≈ 10.2
E pH ≈ 11.2

Question 7:

Both alkali metals and alkaline earth metals are reactive towards air and water and with water they will form bases and hydrogen gas (H_2) as illustrated with barium (Ba):

$Ba (s) + 2 H_2O (l) \rightarrow Ba(OH)_2 (aq) + H_2 (g)$

The reactivity towards water varies greatly and systematically. Rank cesium (Cs), rubidium (Rb), potassium (K), calcium (Ca) and magnesium (Mg) according to their reactivity towards water:

A Cs > Rb > K > Ca > Mg
B Mg > Ca > K > Rb > Cs
C K > Rb > Cs > Mg > Ca
D K < Rb < Cs < Mg < Ca
E Cs > Ca > Rb > Mg > K

Question 8:

Rank the following species according to a decreasing degree of ionic character of the bond:

Sodium fluoride (NaF), barium sulphide (BaS), barium oxide (BaO), difluorine (F_2) or nitrogen monoxide (NO)

A NaF > BaO > BaS > NO > F_2
B NaF > F_2 > BaO > NO > BaS
C NaF > BaS > BaO > NO > F_2
D NaF > BaO > BaS > F_2 > NO
E NaF < F_2 < BaO < NO < BaS

Question 9:

During an experiment a nitrogen oxide was synthesized and collected in a bag and the following properties were measured:

T = 100 °C
P = 1.0 atm
ρ = 3.0 g/L

Which nitrogen oxide was most likely synthesised?

A NO
B NO_2

C N_2O
D N_2O_4
E N_2O_5

Question 10:

Ammonia (NH_3) can be synthesised from nitrogen (N_2) and hydrogen (H_2) through the following reaction:

N_2 (g) + 3 H_2 (g) $\rightarrow$ 2 NH_3 (g)

The following values have been measured:

	ΔH° (kJ/mol)	ΔS° (J/(mol · K))
N_2 (g)	0	191.61
H_2 (g)	0	130.79
NH_3 (g)	-46.11	192.4

Calculate ΔG° and K at 25 °C using the above values:

A $\Delta G° = -\ 32.8$ kJ/mol and $K = 5.65 \cdot 10^{5}$
B $\Delta G° = -\ 32.8$ kJ/mol and $K = 5.65 \cdot 10^{-5}$
C $\Delta G° = +\ 32.8$ kJ/mol and $K = 5.65 \cdot 10^{5}$
D $\Delta G° = +\ 32.8$ kJ/mol and $K = 1.75 \cdot 10^{-6}$
E $\Delta G° = -\ 32.8$ kJ/mol and $K = 1.75 \cdot 10^{-6}$

Question 11:

The synthesis of ammonia, through the above reaction, becomes less favourable with increasing temperatures and more favourable with increasing pressures. What is the reason for that?

A The reaction pathway changes with temperature and pressure changing the reaction kinetics and thus the equilibrium constant.
B As the reaction is exothermic energy is released in the reaction disfavouring an increase in temperature.
Additionally the reaction results in an overall decrease in the number of molecules which, for entropic reasons, is disfavoured at high temperatures but favoured by high pressures.
C As the reaction is endothermic energy is used in the reaction disfavouring an increase in temperature.
Additionally the reaction results in an overall decrease in the number of molecules which, for entropic reasons, is disfavoured at high temperatures but favoured by high pressures.
D The only effect is the overall decrease in the number of molecules which, for entropic reasons, is disfavoured at high temperatures but favoured by high pressures.
E It is correct that changes in temperature and pressure will have the effects described but in reality the impact is so small that they can be completely ignored.

Question 12:

Even though the production of ammonia is disfavoured by high temperatures such conditions are still favoured for the industrial production of ammonia. Typical conditions are 150 – 250 bar and 400 – 500 °C. Additionally a catalyst is used. Why are high temperatures used even though they are unfavourable and why is the catalyst used?

A The effect of the high temperature is more than counteracted by applying the high pressure and is actually just a minor side effect caused by applying the pressure. The catalyst further shifts the equilibrium in favour of product.
B The catalyst needs high temperatures to be active and the change in equilibrium caused by the catalyst more than counteracts the disadvantages of the higher temperature.
C At high temperatures the use of the catalyst allows for a new reaction pathway thus altering the overall reaction enthalpy while keeping the entropy approximately constant.
D The catalyst increases the forward reaction rate so much that the reaction becomes rather uncontrollable thus raising the pressure and temperature significantly. Therefor the production is always combined with other processes that can use the pressure and temperature. An example could be a steam turbine for power production.
E The higher temperature is needed to break the strong triple bond of nitrogen which requires a significant input of energy. The catalyst helps by reducing the activation energy.

Question 13:

For the following reaction the rate constant were obtained at different temperatures:

$2\ N_2O_5\ (g) \rightarrow 4\ NO_2\ (g) + O_2\ (g)$,

Temperature (°C)	45	50	55	60
Rate constant k (s^{-1})	$4.8 \cdot 10^{-4}$	$8.8 \cdot 10^{-4}$	$1.6 \cdot 10^{-3}$	$2.8 \cdot 10^{-3}$

Calculate the activation energy (E_a) for this reaction:

A $E_a \approx 1$ kJ/mol
B $E_a \approx 50$ kJ/mol
C $E_a \approx 10$ kJ/mol
D $E_a \approx 5$ kJ/mol
E $E_a \approx 100$ kJ/mol

Question 14:

Consider the following kinetic data for the reaction:

$S_2O_8^{2-}\ (aq) + 3\ I^-\ (aq) \rightarrow 2\ SO_4^{2-}\ (aq) + I_3^-\ (aq)$

$[S_2O_8^{2-}]$ (M)	$[I^-]$ (M)	Initial rate (M/s)
0.038	0.60	$1.5 \cdot 10^{-5}$
0.076	0.60	$2.8 \cdot 10^{-5}$
0.038	1.20	$2.9 \cdot 10^{-5}$
0.076	1.20	$5.8 \cdot 10^{-5}$

What is the rate law for the reaction?

A Rate = $k \cdot [S_2O_8^{2-}]^2 \cdot [I^-]^2$
B Rate = $k \cdot [S_2O_8^{2-}]^2 \cdot [I^-]$
C Rate = $k \cdot [S_2O_8^{2-}] \cdot [I^-]^2$
D Rate = $k \cdot [S_2O_8^{2-}] \cdot [I^-]$
E Rate = $k \cdot [S_2O_8^{2-}]$

Question 15:

A student wants to change the pH of a dilute solution of a strong acid from 4 to 8. What is the better approach?

A Since the pH is doubled the $[H^+]$ needs to be half of the original. A dilution with a factor of two is thus need
B By definition the pH scale is logarithmic and a dilution with a factor of 10^4 is needed
C More acid needs to be added as a dilution with a factor of more than ten is impractical
D A change from acidic to alkaline conditions cannot be achieved by simple dilution. A titration with a base is needed
E The acid anion should be precipitated as an insoluble salt, through a titration, thus shifting the equilibrium constant

Question 16:

Hess's law states that for some thermodynamic properties, such as Gibbs free energy, the changes are independent of the path and the properties are thus additive.
This is however not always true for standard reduction potentials. E.g.:

(1) $E°(Fe^{3+}/Fe^{2+}) = 0.77$ V
(2) $E°(Fe^{2+}/Fe) = - 0.44$ V
(3) $E°(Fe^{3+}/Fe) = - 0.04$ V $\neq$ (1) + (2)

What is the origin of this difference between Gibbs free energy and standard reduction potentials?

A The amount of transferred electrons are different in the above case. The standard reduction potentials therefore needs to be converted to Gibbs free energy which is directly additive
B The standard reduction potentials depend on concentration as stated by the Nernst Equation
C The standard reduction potentials are only additive when each half-cell reaction has the same sign. That is favourable and unfavourable reactions are not additive with each other
D Being an intensive property changing the stoichiometric coefficient has no effect in the case of standard reduction potentials. In other words standard reduction potentials are unaffected by e.g. size of the electrode or the amount of solutions present and are thus not directly additive
E The reference states are defined differently for Gibbs free energy and standard reduction potentials

Question 17:

Free metals can often be produced, at high temperatures, by the reduction of a metal oxide with coal. E.g.:

$2\ Fe_2O_3\ (s) + 3\ C\ (s) \rightarrow 4\ Fe\ (s) + 3\ CO_2\ (g)$

Calculate $\Delta H°$ by using the following reactions:

$C\ (s) + O_2\ (g) \rightarrow CO_2\ (g)$	$\Delta H° = -393.5$ kJ/mol
$C\ (s) + CO_2\ (g) \rightarrow 2\ CO\ (g)$	$\Delta H° = 172.5$ kJ/mol
$Fe_2O_3\ (s) + C\ (s) \rightarrow 2\ FeO\ (s) + CO\ (g)$	$\Delta H° = 167.7$ kJ/mol
$FeO\ (s) + CO\ (g) \rightarrow Fe\ (s) + CO_2\ (g)$	$\Delta H° = - 11.0$ kJ/mol

Further state if the reaction is favourable for enthalpic reasons only:

A $\Delta H° = 463.9$ kJ/mol – not favourable
B $\Delta H° = 463.9$ kJ/mol – favourable
C $\Delta H° = - 463.9$ kJ/mol – not favourable
D $\Delta H° = - 329.2$ kJ/mol – favourable
E $\Delta H° = 329.2$ kJ/mol – not favourable

Question 18:

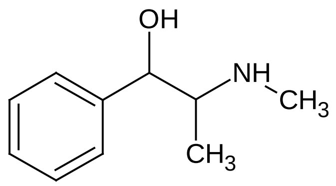
Phosphorous (P) has the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^3$

How are the 3p electrons distributed?

A	↑↓	↓	
B	↑↓	↑	
C	↑	↑	↑
D	↑	↓	↑
E	↓	↑	↓
	p_x	p_y	p_z

Question 19:

Ephedrine has been a popular supplement for bodybuilders as it can induce short-term weight loss before a competition:



What will happen to the alcohol upon oxidation and what are the names of the two other functional groups?

A A ketone is formed upon oxidation. The two other functional groups are an amine and a double bond
B An aldehyde is formed upon oxidation. The two other functional groups are an amine and a double bond
C A ketone is formed upon oxidation. The two other functional groups are an amine and an aromatic ring
D An aldehyde is formed upon oxidation. The two other functional groups are an amine and an aromatic ring
E A carboxylic acid is formed upon oxidation. The two other functional groups are an amine and an aromatic ring

Question 20:

Which of the following salts are soluble in water?

Na_2CO_3, $MgCO_3$, $BaSO_4$, $CuSO_4$, $PbCl_2$ or $NiCl_2$

A Chlorides and sulphates are generally soluble in water so the last four salts
B All except $MgCO_3$ as alkali salts, chlorides and sulphates are generally soluble in water
C Only Na_2CO_3 as the others are salts of heavy metals
D Only Na_2CO_3, $CuSO_4$ and $NiCl_2$ as $BaSO_4$ and $PbCl_2$ are insoluble exceptions
E Being salts they are all soluble

Suggested solution for re-exam 26030 F17

Question 1:

Answer C

	Ag_2SO_4 (s)	$\rightleftarrows$	$2\ Ag^+$ (aq)	+	SO_4^{2-}(aq)
Initial			0		0
Change	-s		2 s		s

$$K_{sp,80\ C} = [Ag^+]^2 \cdot [SO_4^{2-}] = (2s)^2 \cdot s = 4s^3$$
$$M(Ag_2SO_4) = 2 \cdot M(Ag) + M(S) + 4 \cdot M(O) = (2 \cdot 107.87 + 32.06 + 4 \cdot 16.00)\frac{g}{mol} = 311.80\frac{g}{mol}$$
$$S(Ag_2SO_4)_{80\ C} = 12.4\frac{g}{L}$$
$$[Ag_2SO_4] = s = \frac{S(Ag_2SO_4)_{80\ C}}{M(Ag_2SO_4)} = \frac{12.4\frac{g}{L}}{311.80\frac{g}{mol}} = 0.0398\frac{mol}{L}$$
$$K_{sp,80\ C} = 4s^3 = 4 \cdot \left(0.0398\frac{mol}{L}\right)^3 = 2.52 \cdot 10^{-4} M^3$$

Question 2:

Answer D

From section 17.7 and 20.2 it is given that transition metals have a particular tendency to form coordination compounds.

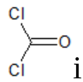
For silver(I) ions with ammonia (table 17.4):

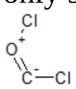
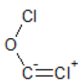
$$Ag^+ + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+, K_f = 1.5 \cdot 10^7$$

The concentration of free silver(I) ions is thus very small in the presence of ammonia

Question 3:

Answer A

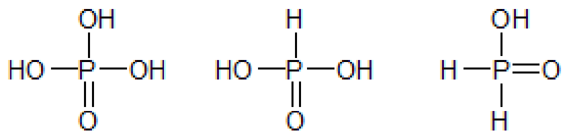
From section 9.7 p. 304 a Lewis structure with no formal charge is preferable and as  is the only structure without formal charges it is the preferred structure

 and  are both very unlikely as the negative formal charges are additionally not placed on the more electronegative atoms.

Question 4:

Answer C

From section 16.8 in the subsection on oxo acids on p. 574 it is given that the acidity of oxo acids is due to the polarization of the OH bond. The right structures thus needs to have a number of OH groups matching the number of acidic protons:



(H_3PO_4) (H_3PO_3) (H_3PO_2)

Alternatively Lewis structures can be drawn for all the molecules. For most structures this will result in formal charges making those unlikely.

The structure of (ortho)phosphoric acid (H_3PO_4) and phosphorous acid (H_3PO_3) are additionally given directly in figure 16.5.

<u>Question 5</u>:

Answer A

H is +1
O is -2
No outer charges on any of the structures and sum of oxidation numbers needs to equal outer charge
$\Rightarrow$ P=5 (H_3PO_4), P=3 (H_3PO_3) and P=1 (H_3PO_2)
P in H_3PO_2 has the lowest oxidation number and is thus the one most likely to be oxidised. H_3PO_2 will thus be a reducing agent

<u>Question 6</u>:

Answer C

For buffers the Henderson Hasselbalch equation applies:

$$pH = pK_a + log\frac{[Conjugate\ base\ ]}{[Acid]}$$

From the text and the reaction equation given in the question it is evident that [conjugate base] = [acid] $\Rightarrow log\frac{[Conjugate\ base\ ]}{[Acid]} = 0 \Rightarrow pH = pK_a = -logK_a = 9.22$

<u>Question 7</u>:

Answer A

The reactivity can be estimated from the ionization energies with a small ionization energy corresponding to a more favourable reaction. From p.265 it is given that the smallest ionization energies can be found at the lower left in the periodic table and that they will increase either going right or up.

The order is therefore Cs>Rb>K>Ca>Mg

The text on group 2A p. 272-273 can be used to draw the same conclusions

<u>Question 8</u>:

Answer A

Figure 9.5 p. 296 shows the electronegativity of most elements. A larger difference in electronegativity between the two atoms corresponds to a more ionic bond.

Na: 0.9
Ba: 0.9
S: 2.5
N: 3.0
O: 3.5
F: 4.0

NaF: 4.0 – 0.9 = 3.1
BaO: 3.5 – 0.9 = 2.6
BaS: 2.5 – 0.9 = 1.6
NO: 3.5 – 3.0 = 0.5
F_2: 4.0 – 4.0 = 0.0

NaF>BaO>BaS>NO>F_2

Question 9:

Answer D

$p \cdot V = n \cdot R \cdot T, n = \frac{m}{M}$, and $\rho = \frac{m}{V} \Rightarrow p \cdot V = \frac{m}{M} \cdot R \cdot T \Rightarrow p \cdot M = \frac{m}{V} \cdot R \cdot T \Rightarrow p \cdot M = \rho \cdot R \cdot T$
$\Rightarrow M = \frac{\rho \cdot R \cdot T}{p}$

T = 100 °C = 373.15K
P = 1.0 atm
ρ = 3.0 g/L
R=0.0821 $^{L \cdot atm}/_{mol \cdot K}$

$$M = \frac{3.0\,\frac{g}{L} \cdot 0.0821\frac{L \cdot atm}{mol \cdot K} \cdot 373.15K}{1.0atm} = 91.91\frac{g}{mol}$$
$$M(N_2O_4) = 2 \cdot M(N) + 4 \cdot M(O) = (2 \cdot 14.01 + 4 \cdot 16.00)\frac{g}{mol} = 92.02\frac{g}{mol}$$

The gas is likely N_2O_4

Question 10:

Answer A

$$\Delta_r H^\circ = \sum n \cdot H_f^\circ\,(products) - \sum m \cdot H_f^\circ(reactants)$$
$$= \left(2 \cdot -46.11 - (3 \cdot 0 + 0)\right)\frac{kJ}{mol} = -92.22\frac{kJ}{mol} = -92220\frac{J}{mol}$$
$$\Delta_r S^\circ = \sum n \cdot S^\circ\,(products) - \sum m \cdot S^\circ(reactants)$$
$$= \left(2 \cdot 192.4 - (3 \cdot 130.79 + 191.61)\right)\frac{J}{mol \cdot K} = -199.18\frac{J}{mol \cdot K}$$
$$\Delta_r G^\circ = \Delta_r H^\circ - T\Delta_r S^\circ = -92220\frac{J}{mol} - 298.15K \cdot -199.18\frac{J}{mol \cdot K} = -32835\frac{J}{mol}$$
$$\Delta_r G^\circ = -R \cdot T \cdot ln(K) \Rightarrow K = e^{\frac{-\Delta_r G^\circ}{R \cdot T}} = e^{\frac{32835\frac{J}{mol}}{8.314\frac{J}{mol \cdot K} \cdot 298.15K}} = 5.65 \cdot 10^5$$

Question 11:

Answer B

$\Delta_r H^\circ < 0 \equiv exothermic\ reaction$

A thorough description of the effects of changes in pressure and temperature can be found in section 15.4 p. 532-534

As the reaction is exothermic energy is released in the reaction disfavouring an increase in temperature.
Additionally the reaction results in an overall decrease in the number of molecules which, for entropic reasons, is disfavoured at high temperatures but favoured by high pressures.

Question 12:

Answer E

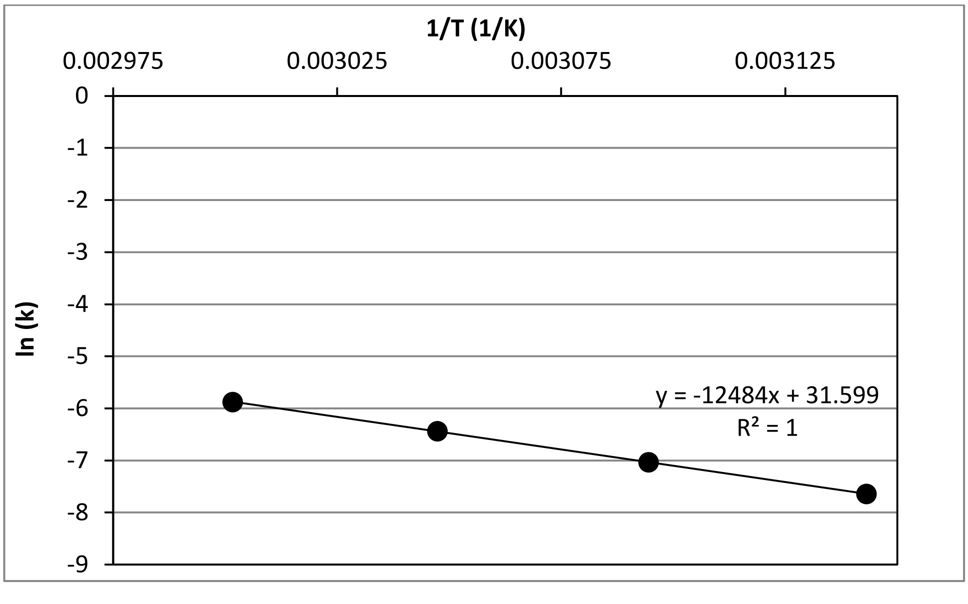
The triple bond in N_2 is one of the strongest bonds known and requires a high energy input to be broken. This can e.g. be supplied as heat. The catalyst reduces the activation energy making it easier to break the bond. The catalyst affects the rate of the reaction but not the location of the equilibrium.

E.g. consult figure 14.2 and section 15.4 p. 535

Question 13:

Answer E

Convert temperatures from celcius to Kelvin
Plot ln k as a function of 1/T



$\alpha = -\frac{E_a}{R} \Rightarrow E_a = -\alpha \cdot R = 12484\frac{1}{K} \cdot 8.314\frac{J}{mol \cdot K} = 103792\frac{J}{mol} \approx 100\frac{kJ}{mol}$

Question 14:

Answer D

At constant [$S_2O_8^{2-}$] the rate doubles when [I^-] doubles ⇒ 1st order regarding [I^-]
At constant [I^-] the rate doubles when [$S_2O_8^{2-}$] doubles ⇒ 1st order regarding [$S_2O_8^{2-}$]

Rate = k · [$S_2O_8^{2-}$] · [I^-]

Question 15:

Answer D

In order to reach alkaline conditions [OH^-] needs to be added. A simple dilution will cause an asymptomatic approach to neutral pH (pH 7 – the pH of pure water).

Question 16:

Answer A

When the amount of transferred electrons is different the electrode potentials need to be converted to Gibbs free energy through the relation $\Delta G° = -n \cdot F \cdot E°$

Question 17:

Answer A

(1) 2 Fe_2O_3 (s) + 3 C (s) → 4 Fe (s) + 3 CO_2 (g)

(2) C (s) + O_2 (g) → CO_2 (g) (not used!)
(3) C (s) + CO_2 (g) → 2 CO (g)
(4) Fe_2O_3 (s) + C (s) → 2 FeO (s) + CO (g)
(5) FeO (s) + CO (g) → Fe (s) + CO_2 (g)

(1) = 2 · (4) + (3) + 4 · (5) Hess's law

$$\Delta H°(1) = 2 \cdot \Delta H°(4) + \Delta H°(3) + 4 \cdot \Delta H°(5) = (2 \cdot 167.7 + 172.5 + 4 \cdot -11.0)\frac{kJ}{mol} = 463.9\frac{kJ}{mol}$$
$$\Delta_r H° > 0 \equiv endothermic\ reaction/not\ favourable$$

Question 18:

Answer C

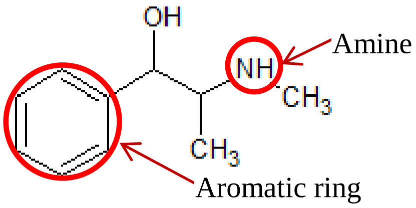
All configurations in agreement with the Pauli exclusion principle – maximum two electrons in each orbital and electrons in same orbital have opposite spins

Hund's rule: *The most stable arrangement of electrons in a subshell is the one with the greatest number of parallel spins:*

↑	↑	↑
p_x	p_y	p_z

Question 19:

Answer C



The alcohol is oxidised to a ketone. Only an aldehyde can be further oxidised to a carboxylic acid

Question 20:

Answer D

Referring to table 4.2 p. 101 it can be seen that Na_2CO_3, $CuSO_4$ and $NiCl_2$ are soluble. $BaSO_4$ and $PbCl_2$ are insoluble exceptions.

Re-exam 26030 F18

Question 1:

(5 points)

Mark the option where all the compounds are bases:

A KOH, $Ba(OH_2)$, NH_4^+
B $Ba(OH_2)$, $NH_{3,}$ CH_3CH_2OH
C KOH, $NH_{3,}$ CH_3CH_2OH
D KOH, $Ba(OH_2)$, NH_3
E KOH, $Ba(OH_2)$, CH_3CH_2OH

Question 2:

(2 points)

Consider the following reaction:

2 A (s) + B (g) ⇄ 3 C (g)

A mixture of A, B and C is added to a container and allowed to reach equilibrium. What will happen if the volume of the container is then increased?

A The reaction will shift to the right (more of C is formed)
B The reaction will shift to the left (C is used)
C Nothing

Question 3:

(1 point)

Estimate if the following reaction is favourable at standard conditions?

A + B ⇄ C + D

At equilibrium the following concentrations have been found:
[A] = [B] = 0.1 M and [C] = [D] = 1.0 M

A Favourable
B Not favourable
C Neither

Question 4:

(5 points)

For people getting lost in arctic regions two main considerations are not to dehydrate and not to use too much energy. A general recommendation is not to use ice to cover the water intake. In the following we will examine if this is a relevant consideration.

How large a percentage of the daily energy intake will be used for heating 2 kg of ice from – 30 °C to water at a final temperature of 37 °C?

The heat capacity of ice is 2.1 J/(g · K), the heat capacity of water 4.2 J/(g · K), the heat of fusion 333.5 J/g and the average daily energy intake 10000 kJ.

A ≈ 0.1 %
B ≈ 1.0 %
C ≈ 10.0 %
D ≈ 100 %
E ≈ 1000 %

<u>Question 5</u>:

(5 points)

SO_3 can be formed through a reaction between SO_2 and O_2. Below are given 3 different reaction schemes for the same reaction:

1) $2\ SO_2\ (g) + O_2\ (g) \rightleftarrows 2\ SO_3\ (g)$
2) $4\ SO_2\ (g) + 2\ O_2\ (g) \rightleftarrows 4\ SO_3\ (g)$
3) $6\ SO_2\ (g) + 3\ O_2\ (g) \rightleftarrows 6\ SO_3\ (g)$

How is $|\Delta_r G|$ (the absolute value of $\Delta_r G$) related between the 3 different reaction schemes?

Note: The standard free energies of formation can be found in the book.

A $|\Delta_r G_1| > |\Delta_r G_2| > |\Delta_r G_3|$

B $|\Delta_r G_1| < |\Delta_r G_2| < |\Delta_r G_3|$

C $|\Delta_r G_1| = |\Delta_r G_2| = |\Delta_r G_3|$

D $|\Delta_r G_1| = |\Delta_r G_2| = |\Delta_r G_3| \equiv 0$ (all reactants and product are in their reference states)

<u>Question 6</u>:

(2 points)

Consider the formation of ammonia (NH_3) from dinitrogen (N_2) and dihydrogen (H_2):

$N_2\ (g) + 3\ H_2\ (g) \rightarrow 2\ NH_3\ (g)$

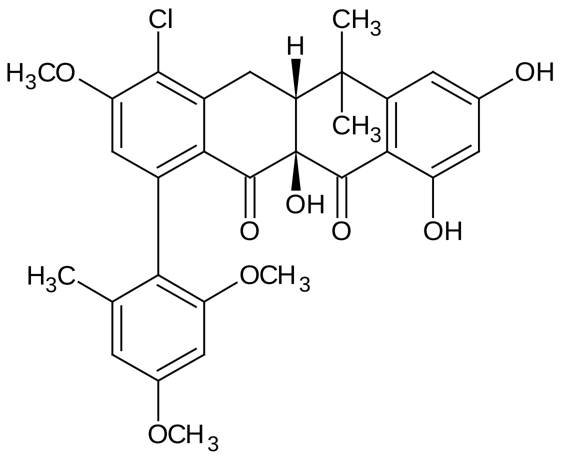
How many moles of dinitrogen are needed to synthesise six moles of ammonia?

A 3
B 6
C 12

<u>Question 7</u>:

(2 points)

Tetraponera penzigi is a fungus growing ant found in Africa. The ant is host to bacteria which produces a new class of antibiotics which is still effective against otherwise multidrug-resistant bacteria. Formicamycin A has been isolated as one of the active compounds:



How many ether groups are present in Formicamycin A?

A 0
B 1
C 2
D 3

<u>Question 8:</u>

(2 points)

How many aldehyde groups are present in Formicamycin A (structure can be found in question 7)?

A 0
B 1
C 2
D 3

<u>Question 9</u>:

(2 points)

Estimate if the following reaction is favourable at standard conditions?

Au^{3+} (aq) + Cr (s) → Au (s) + Cr^{3+} (aq)

Hint: Use standard reduction potentials or the electrochemical series which can all be found in the book

A Favourable
B Not favourable
C Neither

<u>Question 10</u>:

(5 points)

Consider the following reaction:

$2\ NO\ (g) + 2\ H_2\ (g) \rightarrow N_2\ (g) + 2\ H_2O\ (g)$

The rate law can in general be written as:

$\text{Rate} = k \cdot [NO]^x \cdot [H_2]^y$

Experiments have shown that the reaction is of 2^{nd} order with respect to NO and of 1^{st} order with respect to H_2.

Use the following set of experiments to determine k:

Experiment	[NO] (M)	$[H_2]$ (M)	Initial Rate (M/s)
1	0.050	0.100	0.0625
2	0.050	0.050	0.03125
3	0.100	0.050	0.1250

A $k = 12.5\ M^{-2} \cdot s^{-1}$
B $k = 125\ M^{-2} \cdot s^{-1}$
C $k = 250\ M^{-2} \cdot s^{-1}$
D $k = 2500\ M^{-2} \cdot s^{-1}$

<u>Question 11</u>:

(5 points)

Consider the following reaction:

$2\ KMnO_4 + 5\ Hg_2Cl_2 + 16\ HCl \rightarrow 10\ HgCl_2 + 2\ MnCl_2 + 2\ KCl + 8\ H_2O$

If 5.0 g of each reactant is mixed which will then be the limiting reactant?

A $KMnO_4$
B Hg_2Cl_2
C HCl
D $KMnO_4$ and HCl
E All three are limiting reactants

<u>Question 12</u>:

(5 points)

The enthalpies of the following two reactions are known:

Reaction 1:	$N_2\ (g) + 2\ O_2\ (g) \rightarrow 2\ NO_2\ (g)$	$\Delta H_{r1} = 66.4$ kJ/mol
Reaction 2:	$2\ NO\ (g) + O_2\ (g) \rightarrow 2\ NO_2\ (g)$	$\Delta H_{r2} = -114.2$ kJ/mol

From these, calculate the enthalpy of reaction (ΔH_{r3}) for reaction 3:

Reaction 3: N_2 (g) + O_2 (g) $\rightarrow$ 2 NO (g) ΔH_{r3} = ?

A ΔH_{r3} = -47.8 kJ/mol
B ΔH_{r3} = 47.8 kJ/mol
C ΔH_{r3} = - 180.6 kJ/mol
D ΔH_{r3} = 180.6 kJ/mol

Question 13:

(5 points)

Which compound is incorrectly matched with bond angles?

A $SiCl_4$: 109.5°
B PCl_3: 120°
C SF_6: 90° and 180°
D BF_3: 120°
E AsF_5: 90°, 120° and 180°

Question 14:

(5 points)

0.087 g of a sample is enclosed into a balloon and the sample is then converted into an ideal gas. At 273.15 K and 1.0 atm the final volume of the gas will be 33.6 mL. What is the molecular formula of the compound?

A CH_4
B C_2H_6
C C_3H_8
D C_4H_{10}
E C_5H_{12}

Question 15:

(2 points)

Consider the following reaction:

2 ClO_2 (aq) + 2 OH^- (aq) $\rightarrow$ ClO_3^- (aq) + ClO_2^- (aq) + H_2O (l)

The following starting concentrations and initial rates of reactions were observed:

Experiment	$[ClO_2]$ (M)	$[OH^-]$ (M)	Initial Rate (M/s)
1	0.060	0.030	0.02480
2	0.020	0.030	0.00276
3	0.020	0.090	0.00828

What is the order of the reaction with respect to OH^-?

A 0
B 1
C 2
D 3

<u>Question 16</u>:

(3 points)

Again consider the reaction:

$2\ ClO_2\ (aq) + 2\ OH^-\ (aq) \rightarrow ClO_3^-\ (aq) + ClO_2^-\ (aq) + H_2O\ (l)$

The following starting concentrations and initial rates of reactions were observed:

Experiment	$[ClO_2]$ (M)	$[OH^-]$ (M)	Initial Rate (M/s)
1	0.060	0.030	0.02480
2	0.020	0.030	0.00276
3	0.020	0.090	0.00828

What is the order of the reaction with respect to ClO_2?

A 0
B 1
C 2
D 3

<u>Question 17</u>:

(5 points)

What is the freezing point of an aqueous 1.00 m solution of $MgCl_2$? The freezing-point depression constant is 1.86 °C/m for water.

Hint: Assume complete dissociation of the salt

A 1.86 °C
B - 1.86 °C
C 3.72 °C
D 5.58 °C
E - 5.58 °C

<u>Question 18</u>:

(5 points)

Calculate the pH of a buffer solution with 0.35 M acetic acid ($K_a = 1.8 \cdot 10^{-5}$) and 0.65 M sodium acetate:

A pH ≈ 4
B pH ≈ 5
C pH ≈ 6

D pH ≈ 7
E pH ≈ 8

Question 19:

(3 points)

In which of the following compounds is manganese most oxidised?

A Mn_2O_3
B MnO_2
C Mn
D MnO_4^-
E Mn^{2+}

Question 20:

(2 points)

What is the oxidation state of oxygen in Na_2O_2?

A + I
B - I
C 0
D - ½
E - II

Question 21:

(2 points)

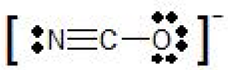
Which of the following elements is able to form an expanded octet?

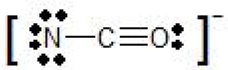
A C
B N
C O
D S

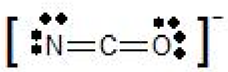
Question 22:

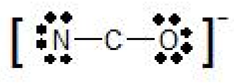
(3 points)

Considering formal charge, what is the preferred Levis structure of NCO^-?

A 

B 

C 

D 

Question 23:

(5 points)

What is the vapour pressure of water at 50 °C?

Hint: The boiling point and the heat of vaporisation can be found in the book.

A 0.0013 atm
B 0.013 atm
C 0.13 atm
D 1.3 atm

Question 24:

(1 point)

How many electrons are located in p-orbitals in the bromine atom (Br)? The electron configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$.

A 16
B 17
C 18

Question 25:

(2 points)

How many electrons are located in s-orbitals in the chlorine atom (Cl)?

Hint: The electron configuration can be found in the book

A 5
B 6
C 7

Question 26:

(2 points)

How many electrons are located in s-orbitals or p-orbitals in the sodium ion (Na^+)?

Hint: The electron configuration of the uncharged sodium atom can be found in the book. This needs to be modified for the charge

A 10
B 11
C 12

Question 27:

(5 points)

Determine the pH of a 0.015 M aqueous solution of HF ($K_a = 6.8 \cdot 10^{-4}$):

A pH ≈ 0.5
B pH ≈ 1.5
C pH ≈ 2.5
D pH ≈ 3.5
E pH ≈ 4.5

Question 28:

(2 points)

Consider the following reaction:

$CaBr_2$ (aq) + K_2SO_4 (aq) → $CaSO_4$ (s) + 2 KBr (aq)

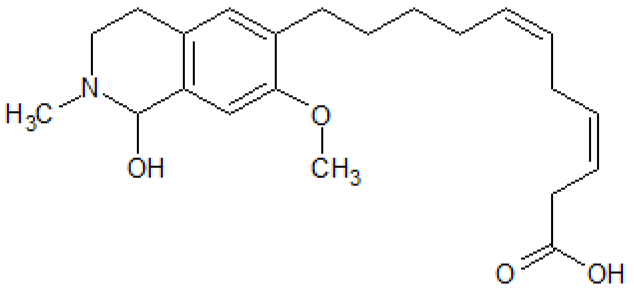
What kind of reaction is it?

A An acid-base reaction
B A redox reaction
C A precipitation reaction
D None of the above

Question 29:

(4 points)

The Polka-Dot Treefrog lives in trees in moist woodlands of South America. The skin fluoresces with blue to green colours when exposed to UV-light. One of the fluorescent compounds is a tetrahydroisoquinolin derivate:



Which functional groups are not present in this molecule?

A An alkyne and an ester
B An alkyne and an ether
C An alkene and an ester
D An alkene and an ether
E An alkyne and a carboxylic acid

Question 30:

(3 points)

Consider the combustion of oxalic acid ($(COOH)_2$). In which of the following cases is the reaction scheme balanced?

A $2\ (COOH)_2\ (s) + O_2\ (g) \rightarrow 4\ CO_2\ (g) + H_2O\ (g)$
B $(COOH)_2\ (s) + O_2\ (g) \rightarrow 2\ CO_2\ (g) + H_2O\ (g)$
C $2\ (COOH)_2\ (s) + O_2\ (g) \rightarrow 4\ CO_2\ (g) + 2\ H_2O\ (g)$
D $2\ (COOH)_2\ (s) + 2\ O_2\ (g) \rightarrow 2\ CO_2\ (g) + H_2O\ (g)$

Suggested solution for re-exam 26030 F18

Question 1:

Answer D

$Ba(OH)_2$ is a strong base.
NH_4^+ is a weak acid.
NH_3 is a weak base.
KOH is a strong base.
CH_3CH_2OH is neither a base nor an acid (it is an alcohol)

Question 2:

Answer A

Use Le Chateliers principle. The increased volume will cause a drop in the pressure. The reaction will therefor shift to the right in order to increase the amount of gas and thus reduce the drop in the pressure.

Question 3:

Answer A

$$K = \frac{[C] \cdot [D]}{[A] \cdot [B]} = \frac{1.0M \cdot 1.0M}{0.1M \cdot 0.1M} = 100$$

The reaction is favourable as K >> 1

Question 4:

Answer C

There are several contributions when converting ice at – 30 °C to water at 37 °C

Heat to 0 °C: $\Delta H = \Delta T \cdot m \cdot C_{ice} = 30K \cdot 2000g \cdot 2.1\frac{J}{g \cdot K} = 126kJ$
Melting: $\Delta H = m \cdot \Delta_{fus}H = 2000g \cdot 333.5\frac{kJ}{mol} = 667kJ$
Heat to 37 °C: $\Delta H = \Delta T \cdot m \cdot C_{water} = 37K \cdot 2000g \cdot 4.2\frac{J}{g \cdot K} = 311kJ$

Total 1104 kJ or approx. 10 %

Question 5:

Answer B

$$\Delta_r G = \sum \Delta_f G(products) - \sum \Delta_f G(reactants)$$

The larger the stoichiometric coefficients are the larger the $|\Delta_r G|$ will be $\Rightarrow$ $|\Delta_r G_1| < |\Delta_r G_2| < |\Delta_r G_3|$

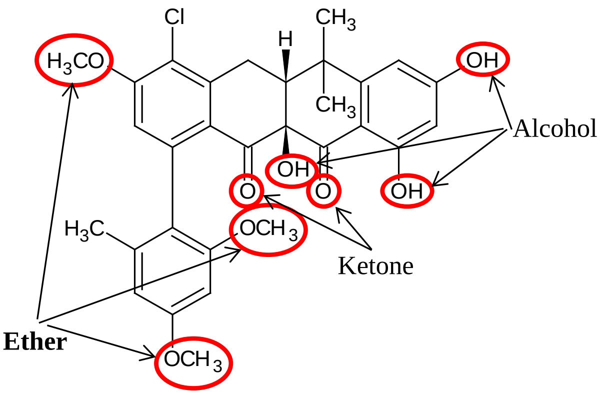
Question 6:

Answer A

From the reaction scheme it is seen that one nitrogen molecule is needed to form two ammonia molecules. Three moles of dinitrogen are thus needed to form six moles of ammonia.

Question 7:

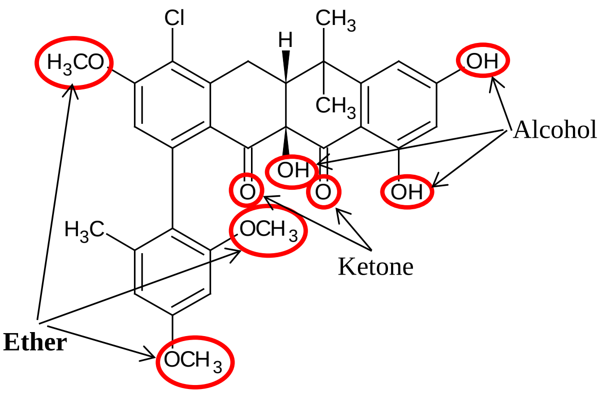
Answer D



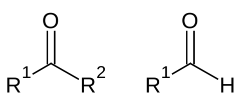
Three ether groups can be found as marked above

Question 8:

Answer A



There are 2 double bonded oxygens present as marked above. These are, however, ketones and not aldehydes:



Ketone Aldehyde

Question 9:

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Answer A

Either calculate the standard emf (E^o_{cell}):

$E^o_{cell} = E^o_{cathode} - E^o_{anode} = E^o_{Au^{3+}/_{Au}} - E^o_{Cr^{3+}/_{Cr}} = 1.5\,V - (-0.74V) = 2.24V >$
$0 \Longrightarrow$Favourable

Alternatively both the electrochemical series and the standard reduction potentials identifies Au^{3+} as the one most likely to be reduced (take up electrons) thus forcing Cr to be oxidised (release electrons).

<u>Question 10</u>:

Answer C

2^{nd} order with respect to NO $\Rightarrow$ x = 2
1^{st} order with respect to H_2 $\Rightarrow$ y = 1

$$rate = k \cdot [NO]^2 \cdot [H_2] \Longrightarrow k = \frac{rate}{[NO]^2 \cdot [H_2]}$$

Use any of the data points. With the 1^{st} datapoint:

$$k = \frac{0.0625\frac{M}{s}}{(0.050M)^2 \cdot 0.100M} = 250\frac{1}{M^2 \cdot s}$$

<u>Question 11</u>:

Answer B

$$M(KMnO_4) = M(K) + M(Mn) + 4 \cdot M(O) = 39.0983\frac{g}{mol} + 54.9380\frac{g}{mol} + 4 \cdot 15.9994\frac{g}{mol} = 158.0339\frac{g}{mol}$$
$$M(Hg_2Cl_2) = 2 \cdot M(Hg) + 2 \cdot M(Cl) = 2 \cdot 200.59\frac{g}{mol} + 2 \cdot 35.453\frac{g}{mol} = 472.086\frac{g}{mol}$$
$$M(HCl) = M(H) + M(Cl) = 1.0079\frac{g}{mol} + 35.453\frac{g}{mol} = 36.4609\frac{g}{mol}$$

$$n(KMnO_4) = \frac{m(KMnO_4)}{M(KMnO_4)} = \frac{5.0g}{158.0339\frac{g}{mol}} = 0.0316mol$$
$$n(Hg_2Cl_2) = \frac{m(Hg_2Cl_2)}{M(Hg_2Cl_2)} = \frac{5.0g}{472.086\frac{g}{mol}} = 0.0106mol$$
$$n(HCl) = \frac{m(HCl)}{M(HCl)} = \frac{5.0g}{36.4609\frac{g}{mol}} = 0.1371mol$$

When finding the limiting factor we need to divide with the stoichiometric coefficients to find the reactant which is used the fastest (The larger the coefficient the faster the amount is used)
As $\frac{n(Hg_2Cl_2)}{5} < \frac{n(HCl)}{16} < \frac{n(KMnO_4)}{2}$ Hg_2Cl_2 will be the limiting factor

<u>Question 12</u>:

Answer D

Use Hess's law which allows you to divide the reaction into steps and then adding the individual steps.
The enthalpy for reaction 3 can be obtained by reversing reaction 2 (thus also changing the sign) and then adding the reverse of reaction 2 to reaction 1:

$$\Delta H_{r3} = \Delta H_{r1} - \Delta H_{r2} = 180.6\frac{kJ}{mol}$$

<u>Question 13</u>:

Answer B

Use VSEPR theory to predict the structure (some are already given as examples in the text book).

$SiCl_4$:	AB_4-type	⇒	Tetrahedral	⇒	Bond angles = 109.5°
PCl_3:	AB_3E-type	⇒	Trigonal pyramidal	⇒	Bond angles ≈ 109.5° (≠ 120°)
SF_6:	AB_6-type	⇒	Octahedral	⇒	Bond angles = 90° and 180°
BF_3:	AB_3-type	⇒	Trigonal planar	⇒	Bond angles = 120°
AsF_5:	AB_5-type	⇒	Trigonal bipyramidal	⇒	Bond angles = 90°, 120° and 180°

<u>Question 14</u>:

Answer D

Use the ideal gas law to calculate the amount of substance:

$$p \cdot V = n \cdot R \cdot T \Longrightarrow n = \frac{p \cdot V}{R \cdot T} = \frac{1.0atm \cdot 0.0336L}{0.0821\frac{L \cdot atm}{mol \cdot K} \cdot 273.15K} = 0.001498mol$$

$$M(gas) = \frac{m}{n} = \frac{0.087g}{0.001498mol} = 58.077\frac{g}{mol} \approx M(C_4H_{10})$$

For comparison:

$$M(CH_4) = M(C) + 4 \cdot M(H) = 12.011\frac{g}{mol} + 4 \cdot 1.0079\frac{g}{mol} = 16.04\frac{g}{mol}$$
$$M(C_2H_6) = 2 \cdot M(C) + 6 \cdot M(H) = 2 \cdot 12.011\frac{g}{mol} + 6 \cdot 1.0079\frac{g}{mol} = 30.07\frac{g}{mol}$$
$$M(C_3H_8) = 3 \cdot M(C) + 8 \cdot M(H) = 3 \cdot 12.011\frac{g}{mol} + 8 \cdot 1.0079\frac{g}{mol} = 44.10\frac{g}{mol}$$
$$M(C_4H_{10}) = 4 \cdot M(C) + 10 \cdot M(H) = 4 \cdot 12.011\frac{g}{mol} + 10 \cdot 1.0079\frac{g}{mol} = 58.12\frac{g}{mol}$$
$$M(C_5H_{12}) = 5 \cdot M(C) + 12 \cdot M(H) = 5 \cdot 12.011\frac{g}{mol} + 12 \cdot 1.0079\frac{g}{mol} = 72.15\frac{g}{mol}$$

<u>Question 15</u>:

Answer B

For a fixed $[ClO_2]$ ($[ClO_2]$ = 0.020 M) a 3-fold increase in $[OH^-]$ (from 0.030 M to 0.090 M) will cause a 3-fold increase in the initial rate (from 0.00276 M/s to 0.00828 M/s). This corresponds to first order with respect to OH^-

Question 16:

Answer C

For a fixed $[OH^-]$ ($[OH^-]$ = 0.030 M) a 3-fold increase of $[ClO_2]$ (from 0.020 M to 0.060 M) will cause a 9-fold increase in the initial rate (from 0.00276 M/s to 0.02480 M/s). This corresponds to second order with respect to ClO_2

Question 17:

Answer E

Complete dissociation means that the van't Hoff factor is 3. The reduction in the temperature can then be calculated as following:

$$\Delta T_F = i \cdot K_F \cdot m = 3 \cdot 1.86\frac{℃}{m} \cdot 1.00m = 5.58℃$$

The final temperature is thus – 5.58 °C

Question 18:

Answer B

For a buffer solution use the Henderson-Hasselbalch equation:

$$pH = pK_a + log\frac{[conjugate\ base]}{[acid]} = -log(K_a) + log\frac{[acetate]}{[acetic\ acid]}$$
$$= -log(1.8 \cdot 10^{-5}) + log\frac{[0.65]}{[0.35]} \approx 5$$

Question 19:

Answer D

Oxygen is normally assigned the oxidation number –II
The sum of the oxidation numbers should equal the net charge of the ion

Mn is +III in Mn_2O_3
Mn is +IV in MnO_2
Mn is 0 in Mn
Mn is +VII in MnO_4^-
Mn is +II in Mn^{2+}

Mn has the higher oxidation number in MnO_4^- and is thus most oxidised in this case

Question 20:

Answer B

Peroxides is an exception to the general rules and the oxidation state is –I (vs. the normal –II)

Question 21:

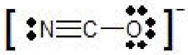
Answer D

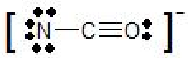
Second period elements cannot have an expended octet. Sulphur can, however, accommodate more bonding pairs through its more spatially expanded 3s and 3p orbitals.

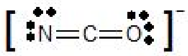
<u>Question 22</u>:

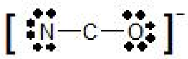
Answer A

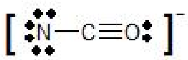
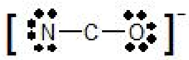
In general a charge separation should be avoided and if needed it should be as small as possible and the negative charges should be placed on the more electronegative atoms.

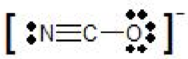
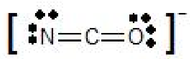
For  N formally has 5 electrons (0 formal charge), C 4 electrons (0 formal charge) and O 7 electrons (-1 formal charge)

 N formally has 7 electrons (-2 formal charge), C 4 electrons (0 formal charge) and O 5 electrons (+1 formal charge)

 N formally has 6 electrons (-1 formal charge), C 4 electrons (0 formal charge) and O 6 electrons (0 formal charge)

 N formally has 7 electrons (-2 formal charge), C 2 electrons (+2 formal charge) and O 7 electrons (-1 formal charge)

 and especially  are highly unlikely due to large charge separations. Furthermore the largest negative charge is not placed on the most electronegative atom (O: 3.5 vs N: 3.0)

 is favoured over  as the sole negative charge is placed on the more electronegative oxygen (O: 3.5 vs N: 3.0)

<u>Question 23</u>:

Answer C

Use the Clausius-Clapeyron equation:

$R = 8.31451 \frac{J}{mol \cdot K}$, T_2 = 100 °C = 373.15 K (from boiling point) P_2 = 1 atm (from boiling point), T_1 = 50 °C = 323.15 K, $\Delta_{vap}H = 40790 \frac{J}{mol}$

$$ln\left(\frac{P_1}{P_2}\right) = \frac{\Delta H_{vap}}{R} \cdot \left(\frac{1}{T_2} - \frac{1}{T_1}\right) \Rightarrow ln(P_1) - ln(P_2) = \frac{\Delta H_{vap}}{R} \cdot \left(\frac{1}{T_2} - \frac{1}{T_1}\right) \Rightarrow P_1 = e^{\frac{\Delta H_{vap}}{R} \cdot \left(\frac{1}{T_2} - \frac{1}{T_1}\right) + ln(P_2)}$$

$$= e^{\frac{40790\frac{J}{mol}}{8.31451\frac{J}{mol \cdot K}} \cdot \left(\frac{1}{373.15K} - \frac{1}{323.15K}\right) + ln(1atm)} = 0.131atm$$

Question 24:

Answer B

6 electrons are located in the 2p orbital
6 electrons are located in the 3p orbital
5 electrons are located in the 4p orbital.

Total: 17

Question 25:

Answer B

Look up the ground state electron configuration.
Count s-electrons only!

Cl	$1s^2 2s^2 2p^6 3s^2 3p^5$	6 s-electrons

Question 26:

Answer A

Look up the ground state electron configuration and correct for the removed electron. The electron configuration will resemble that of the nearest noble gas. In this case neon:

Na	$1s^2 2s^2 2p^6 3s^1$	
Na^+	$1s^2 2s^2 2p^6$	4 s-electrons and 6 p-electrons

Question 27:

Answer C

	HF	+	H_2O	$\rightleftharpoons$	H_3O^+	+	F^-
Start	0.015				0		0
Equilibrium	0.015-X				X		X

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$$K_a = \frac{[H_3O^+] \cdot [F^-]}{[HF]} = \frac{X \cdot X}{0.015 - X} \approx \frac{X^2}{0.015}$$

Note that K_a indicates a week acid and X is then small compared to 0.015 M

$$pH = -log[H_3O^+] = -log[X] = -log\left[\sqrt{0.015 \cdot K_a}\right] = 2.50$$

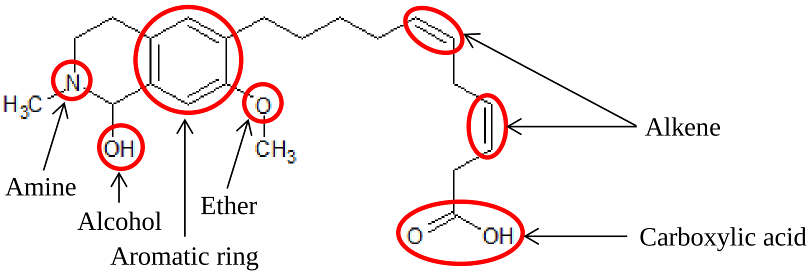
Question 28:

Answer C

The reaction is by definition a precipitation reaction. Additionally neither of the compounds are acids or bases and no change of redox states are observed.

<u>Question 29</u>:

Answer A



<u>Question 30</u>:

Answer C

E.g. count the atoms or alternatively solve as a redox reaction.

Exam exercises adapted from course 26027

Exercise 1:

Commercially 5 different buffer solutions of acetic acid (CH_3COOH) and sodium acetate ($NaCH_3COO$) are available. The pH of these can be described by the Henderson-Hasselbalch equation. For each of the buffers, 0.20 mol of HCl is added to one litre of the solutions. For which of the following solutions will the pH change the most?

A 1.0 M CH_3COOH and 0.25 M $NaCH_3COO$
B 0.5 M CH_3COOH and 0.5 M $NaCH_3COO$
C 0.75 M CH_3COOH and 0.5 M $NaCH_3COO$
D 0.25 M CH_3COOH and 0.5 M $NaCH_3COO$
E 0.25 M CH_3COOH and 1.0 M $NaCH_3COO$

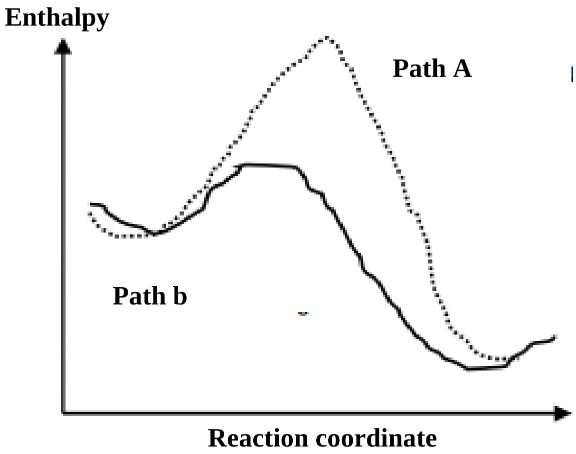
Exercise 2:

The volume of liquid water expands by 0.5% when the temperature is increased by 25 K. What will happen to the molarity and molality of a very dilute aqueous solution of glucose when the temperature is increased from 25 °C to 75 °C?

A Both the molarity and the molality will increase by 1%
B The molarity will increase by 1% while the molality will decrease by 1%
C The molarity remains unchanged while the molality decreases by 1%
D The molarity remains unchanged while the molality increases by 1%
E The molarity decreases by 1% while the molality remains unchanged

Exercise 3:

For a chemical reaction the enthalpy varies with the progress (Denoted the reaction coordinate) as shown in the figure below. Two possible reaction paths are shown:



What can be said about the reaction?

A	The reaction is endothermic towards the right and path A is catalysed
B	The reaction is endothermic towards the right and path B is catalysed
C	The reaction is exothermic towards the right and path A is catalysed
D	The reaction is exothermic towards the right and path B is catalysed
E	The reaction is neither exothermic or endothermic

Exercise 4:

Consider the below non-balanced oxidation of 10.0 g of glucose ($C_6H_{12}O_6$) by 4 L of oxygen at 1 atm and 298 K:

$C_6H_{12}O_6 + O_2 \rightarrow CO_2 + H_2O$

Determine if there is a limiting reactant. Oxygen is assumed to be an ideal gas.

A	Glucose
B	O_2
C	Neither glucose or O_2
D	Both glucose and O_2
E	None of the above answers

Exercise 5:

NaCl and $AgNO_3$ are two examples of salts which are highly soluble in water while their reaction product, AgCl, is not. For both NaCl and $AgNO_3$ two solutions with a concentration of $1.0 \cdot 10^{-5}$ M are prepared. One set of the reactants is kept at 25 °C and the other at 100 °C. The solubility of AgCl, in water, is 0.0020 g/L at 25 °C and 0.022 g/L at 100 °C. Which scenario can be observed when mixing the reactants at the two temperatures?

A AgCl is not precipitated at neither 25 °C or at 100 °C
B AgCl is precipitated at 25 °C but not at 100 °C
C AgCl is precipitated at 100 °C but not at 25 °C
D AgCl is precipitated at both 25 °C and at 100 °C
E Not AgCl but both NaCl and $AgNO_3$ will precipitate when mixing

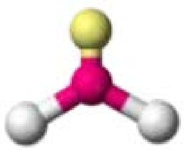
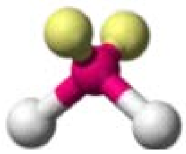
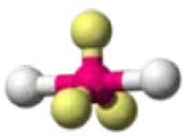
Exercise 6:

Which of the following atoms has the smallest atomic radii in the oxidation state 0?

A Al
B Si
C P
D S
E Cl

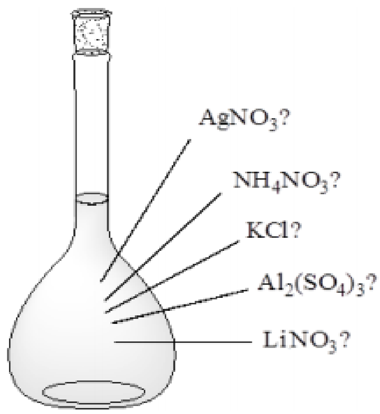
Exercise 7:

Choose the correct electronic geometrical representation of $[ICl_2^-]$ using VSEPR theory. The central iodine atom is shown in red, the chlorine atoms in white and possible lone pairs in yellow.

A	
B	
C	
D	

Exercise 8:

An unlabelled volumetric flask containing an aqueous solution of either $AgNO_3$, NH_4NO_3, KCl, $Al_2(SO_4)_3$ or $LiNO_3$ has been left in the laboratory. Different reagents are added to determine the content. With addition of NaCl nothing happens while the addition of $Ba(NO_3)_2$ results in an immediate precipitation. What is likely to be in the volumetric flask?



A $AgNO_3$
B NH_4NO_3
C KCl
D $Al_2(SO_4)_3$
E $LiNO_3$

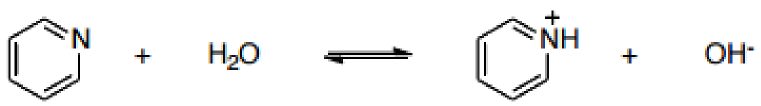
Exercise 9:

For the equilibrium $3\ O_2\ (g) \rightleftharpoons 2\ O_3\ (g)$ predict in which direction the reaction will shift upon a) addition of O_2 b) an increase of pressure c) an increase of temperature. The reaction is known to consume heat.

A a) Towards right b), towards right and c) towards left
B a) Towards right b), towards right and c) towards right
C a) Towards left b), towards left and c) towards left
D a) Towards left b), towards right and c) towards right
E a) Towards right b), towards left and c) towards left

Exercise 10:

Pyridine is a week, monovalent base:



The base dissociation constant, K_b, is $1.7 \cdot 10^{-9}$. Calculate pH for a 1.0 M solution.

A 8.62
B 9.62
C 10.62
D 11.62

E 12.62

Exercise 11:

For a 1.0 M solution of pyridine (Py) what is the $[PyH^+]$? The base dissociation constant, K_b, is $1.7 \cdot 10^{-9}$

If the solution is diluted by a factor of 10 will the fraction of protonated pyridine then increase, decrease or remain unchanged?

A	$[PyH^+] = 2.0 \cdot 10^{-5}$;	Fraction will increase upon dilution
B	$[PyH^+] = 2.0 \cdot 10^{-5}$;	Fraction will decrease upon dilution
C	$[PyH^+] = 2.0 \cdot 10^{-5}$;	Fraction will remain unchanged upon dilution
D	$[PyH^+] = 4.0 \cdot 10^{-5}$;	Fraction will increase upon dilution
E	$[PyH^+] = 4.0 \cdot 10^{-5}$;	Fraction will decrease upon dilution

Exercise 12:

Using the below reaction enthalpies:

1) $N_2\,(g) + O_2\,(g) \rightarrow 2\,NO\,(g)$ $\Delta H° = 180.8\ {}^{kJ}/_{mol}$
2) $H_2\,(g) + ½\,O_2\,(g) \rightarrow H_2O\,(g)$ $\Delta H° = -241.8\ {}^{kJ}/_{mol}$
3) $4\,NH_3\,(g) + 5\,O_2\,(g) \rightarrow 4\,NO\,(g) + 6\,H_2O\,(g)$ $\Delta H° = -904.0\ {}^{kJ}/_{mol}$

Calculate the reaction enthalpy for the synthesis of ammonia:

$N_2\,(g) + 3\,H_2\,(g) \rightarrow 2\,NH_3\,(g)$

A -90 ${}^{kJ}/_{mol}$
B -100 ${}^{kJ}/_{mol}$
C -110 ${}^{kJ}/_{mol}$
D -120 ${}^{kJ}/_{mol}$
E -130 ${}^{kJ}/_{mol}$

Exercise 13:

For the sulphate ion, SO_4^{2-}, the sulphur atom is placed centrally surrounded by the four oxygen atoms. Using VSEPR theory determine the geometry of the sulphate ion:

A Square planar
B Tetrahedral
C Trigonal bipyramidal
D Octahedral

E Distorted tetrahedral

Exercise 14:

Urea can be used for gently de-freezing concrete surfaces where most common salts will otherwise degrade the surface. Urea is composed of N, C, O and H. When 1.0 g of urea is dissolved in 1.0 L of water an osmotic pressure of 0.408 atm is measured at 25 °C. What is the molar mass of urea?

A 60 $^{g}/_{mol}$
B 65 $^{g}/_{mol}$
C 70 $^{g}/_{mol}$
D 75 $^{g}/_{mol}$
E 80 $^{g}/_{mol}$

Exercise 15:

Hydrogen peroxide (H_2O_2) can decompose into H_2O and O_2:

H_2O_2 (l) → H_2O (l) + ½ O_2 (g)

What is the equilibrium constant for the reaction at 25 °C?

Hint: ΔG° can be calculated from values given in the appendix in the book

A $5.5 \cdot 10^{20}$
B $6.0 \cdot 10^{20}$
C $6.5 \cdot 10^{20}$
D $7.0 \cdot 10^{20}$
E $7.5 \cdot 10^{20}$

Exercise 16:

For the decomposition of hydrogen peroxide (H_2O_2) the concentration as a function of time has been determined. Assuming a 1st order reaction calculate the half-life using the following data:

Time (hours)	0	60	120	500
[H_2O_2]	1.00 M	0.61 M	0.36 M	0.02 M

A 87.0 hours
B 88.0 hours
C 89.0 hours
D 90.0 hours

E	91.0 hours

Exercise 17:

The decomposition of hydrogen peroxide (H_2O_2) can be catalysed by iodide (I^-). Given the following dataset calculate the activation energy for the catalysed reaction

Temperature (ºC)	0	25	50	100
Rate constant, k, (M^{-1})	0.05	0.5	5	100

A	50 $^{kJ}/_{mol}$
B	55 $^{kJ}/_{mol}$
C	60 $^{kJ}/_{mol}$
D	65 $^{kJ}/_{mol}$
E	70 $^{kJ}/_{mol}$

Exercise 18:

2) Why is the boiling point of PH_3 (T_B = –88 °C) lower than that of NH_3 (T_B = –33 °C)?

A	The dispersion forces are stronger between NH_3 molecules than between PH_3 molecules.
B	There are stronger dipol-dipol interactions between NH_3 molecules than between PH_3 molecules.
C	Both the dispersion forces and the dipol-dipol interactions are stronger between NH_3 molecules than between PH_3 molecules.
D	The hydrogen bonds are much stronger between NH_3 molecules.
E	The hydrogen bonds are much stronger between PH_3 molecules

Exercise 19:

A reaction of the type 2 A $\rightarrow$ B + 2 C is 2nd order regarding [A]. The rate constant is determined to $k = 3 \cdot 10^{-3}\ M^{-1}\ s^{-1}$ at 298 K. How long time does it take to half the amount of A if the solution was prepared by diluting 0.5 mole of A to 1 L?

A	≈ 923 s
B	≈ 667 s
C	≈ 442 s
D	≈ 231 s
E	≈ 114 s

Exercise 20:

In coordination complexes a strong field ligand results in low spin configuration while weak field ligands results in a high configuration. Use the spectrochemical series of ligands to mark the right d-orbital diagrams for $[Fe(CN)_6]^{3-}$ and $[Mn(Br)_6]^{4-}$. Perfect octahedral geometry can be assumed in both cases.

	$[Fe(CN)_6]^{3-}$	$[Mn(Br)_6]^{4-}$
A	— — ↑↓ ↑↓ ↑	— — ↑↓ ↑↓ ↑
B	— — ↑↓ ↑↓ ↑↓	— — ↑↓ ↑↓ ↑
C	— — ↑↓ ↑↓ ↑	↑ ↑ ↑ ↑ ↑
D	↑ ↑ ↑ ↑ ↑	— — ↑↓ ↑↓ ↑
E	↑ ↑ ↑↓ ↑ ↑	↑ ↑ ↑ ↑ ↑

<u>Exercise 21</u>:

Even though formic acid is a relatively weak acid ($K_a=1.7\cdot10^{-4}$ M), it is still very effective when ants spray it upon their enemies during combat. What is the pH of a 1 M solution of formic acid, HCOOH?

A	1.9
B	2.3
C	2.7
D	3.1
E	3.5

<u>Exercise 22</u>:

The reaction $2\,A + B \rightleftharpoons 2C + D$ takes places in 1 L of water and has a $\Delta G° = -10\ ^{kJ}/_{mol}$ at 298 K. After reaching equilibrium there is 0.1 mole of A, 0.15 mole of B and 0.65 mole of C but how much of D is there?

A	≈ 0.1 mol
B	≈ 0.2 mol

C	≈ 0.4 mol
D	≈ 0.6 mol
E	≈ 1.0 mol

Exercise 23:

For the following two reactions the reaction enthalpies are known:

1)	$2\ Ba\ (s) + O_2\ (g) \rightarrow 2\ BaO\ (s)$	$\Delta H° = -1107$ kJ
2)	$Ba\ (s) + CO_2\ (g) + ½\ O_2\ (g) \rightarrow BaCO_3\ (s)$	$\Delta H° = -823$ kJ

Use these numbers to calculate the reaction enthalpy for the following reaction:

$BaCO_3\ (s) \rightarrow BaO\ (s) + CO_2\ (g)$

A	269 kJ
B	-376 kJ
C	-285 kJ
D	369 kJ
E	537 kJ

Exercise 24:

In which case are all the mentioned acids weak acids?

A	NaOH, HCl, HF
B	HCl, HF, HNO_3
C	HF, CH_3COOH, HCl
D	HF, CH_3COOH, HNO_2
E	CH_3COOH, HNO_2, HNO_3

Exercise 25:

What are the electron configurations for Na, B and O^{2-}?

	Na	B	O^{2-}
A	$1s^2\ 2s^2\ 2p^6$	$1s^2\ 2s^2\ 2p^1$	$1s^2\ 2s^2\ 2p^4$
B	$1s^2\ 2s^2\ 2p^6\ 3s^1$	$1s^2\ 2s^2\ 2p^1$	$1s^2\ 2s^2\ 2p^6$
C	$1s^2\ 2s^2\ 2p^6\ 3s^1$	$1s^2\ 2s^2\ 2p^1$	$1s^2\ 2s^2\ 2p^4$
D	$1s^2\ 2s^2\ 2p^6\ 3s^1$	$1s^2\ 2s^2$	$1s^2\ 2s^2\ 2p^5$
E	$1s^2\ 2s^2\ 2p^6\ 3s^1$	$1s^2\ 2s^2$	$1s^2\ 2s^2\ 2p^6$

Exercise 26:

For the equilibrium 3 A ⇌ 2 B + 1 C calculate the equilibrium constant assuming that [A] =0.1 M, [B] = 0.2 M and [C] =0.4 M.

A	0.8
B	0.16
C	8
D	16
E	80

Exercise 27:

The following redox reaction is taking place at acidic conditions:

Br^- (aq) + $Cr_2O_7^{2-}$ (aq) → Cr^{3+} (aq) + Br_2 (l)

Mark the right combination of coefficients below:

	Br^-	$Cr_2O_7^{2-}$	Cr^{3+}	Br_2
A	6	1	2	3
B	2	1	2	1
C	4	1	2	2
D	6	2	2	6
E	4	2	1	4

Exercise 28:

A frog uses glucose ($C_6H_{12}O_6$) as an anti-freeze agent during the winter. Calculate the freezing point depression assuming that the cells contain 60 g of glucose pr. 200 mL water and that the density of water is 1 g /mL. K_f = 1.86 °C kg/mol

A	≈ 1 °C
B	≈ 2 °C
C	≈ 3 °C
D	≈ 4 °C
E	≈ 5 °C

Exercise 29:

How many grams of ammonia (NH_3) can be produced from 77.3 grams of nitrogen (N_2) and 14.2 g of oxygen (O_2)?

A 91.5 g
B 79.7 g
C 47.0 g
D 120.0 g
E 13.3 g

Exercise 30:

What are the molecular geometries of ammonia, sulphur dioxide and formaldehyde? The geometries must be stated in the same order.

A Trigonal pyramidal, bent, trigonal planar
B Trigonal pyramidal, linear, trigonal planar
C Trigonal pyramidal, bent, trigonal pyramidal
D Trigonal planar, bent, trigonal pyramidal
E Trigonal planar, linear, trigonal pyramidal

Exercise 31:

Which Lewis-structure fits CN^- and O_2 best?

A 

B 

C 

D 

E 

Exercise 32:

1.2 g of NaOH is dissolved in 3 L of water. What is the resulting pH?

A ≈ 14
B ≈ 12
C ≈ 10
D ≈ 8
E ≈ 6

Exercise 33:

A catalyst works by?

A	Reducing the reaction barrier and making the reaction more exothermic
B	Increasing the reaction barrier and stabilising the product
C	Increasing the reaction barrier without stabilising the product
D	Reducing the reaction barrier without stabilising the product
E	Reducing the reaction barrier and stabilising the product

Exercise 34:

In which case are all 3 salts soluble in water?

A	$NaCl$, KNO_3, Li_2S
B	$NaCl$, $PbBr_2$, Li_2S
C	KNO_3, Li_2S, $AgCl$
D	$NaCl$, BaS, $PbBr_2$
E	$NaCl$, $AgCl$, $PbBr_2$

Exercise 35:

Balance the below redox-reaction and mark the correct combination of stoichiometric coefficients:

$Cr_2O_7^{2-}$ (aq) + H_3PO_2 (aq) → Cr^{3+} (aq) + H_3PO_4 (aq)

	$Cr_2O_7^{2-}$	H_3PO_2	H^+	Cr^{3+}	H_3PO_4	H_2O
A	1	2	8	2	2	4
B	1	3	8	2	3	4
C	2	3	16	4	3	8
D	2	4	8	4	4	4
E	3	4	16	6	4	8

Exercise 36:

The degradation of a drug, A, is 2nd order regarding [A].

Rate = $^{-d[A]}/_{A} = k \cdot [A]^2$

The rate constant, k, has a value of 0.01 M^{-1} s^{-1} at 37˚C.

How long time does it take to half the amount of the drug assuming that 0.2 mole is spread evenly in the entire blood volume (5 L)?

A	500 s
B	1500 s
C	2500 s
D	3500 s
E	4500 s

Exercise 37:

Which of the following reactions are expected to have a positive reaction entropy, ΔS°?

A	$2\ SO_2\ (g) + O_2\ (g) \rightarrow 2\ SO_3\ (g)$
B	$N_2\ (g) + 3H_2\ (g) \rightarrow 2\ NH_3\ (g)$
C	$BF_3(g) + NH_3(g) \rightarrow F_3BNH_3\ (s)$
D	$H_2O\ (g) \rightarrow H_2O\ (l)$
E	$2\ NH_4NO_3\ (s) \rightarrow 2\ N_2\ (g) + 4\ H_2O\ (g) + O_2\ (g)$

Exercise 38:

A reaction has the form 2 A + 1 B $\rightleftharpoons$ 1 C + 3 D. At equilibrium the concentration of the 4 compounds are [A] = 0.3 M, [B] = 0.2 M, [C] = 0.3 M and [D] = 0.2 M. Calculate ΔG° for the reaction at 25 °C.

A	$\approx -10\ ^{kJ}/_{mol}$
B	$\approx -5\ ^{kJ}/_{mol}$
C	$\approx 0\ ^{kJ}/_{mol}$
D	$\approx 5\ ^{kJ}/_{mol}$
E	$\approx 10\ ^{kJ}/_{mol}$

Exercise 39:

The standard redox potentials are known for 3 half reactions:

$Pb^{2+}\ (aq) + 2\ e^- \rightarrow Pb\ (s)$	$E^\circ = -0.13$ V
$Ni^{2+}\ (aq) + 2\ e^- \rightarrow Ni\ (s)$	$E^\circ = -0.25$ V
$Cd^{2+}\ (aq) + 2\ e^- \rightarrow Cd\ (s)$	$E^\circ = -0.40$ V

Which of the following species is most likely to take up electrons at standard conditions?

A	Pb^{2+} (aq)

B Cd^{2+} (aq)
C Ni (s)
D Cd (s)
E Pb (s)

Exercise 40:

Which of the following molecules is both containing a ketone and aldehyde functional group?

A $CH_3COCHOHCH_3$
B $CH_3COOCH_2CH_3$
C $CH_2OHCH_2CH_2CHO$
D CH_3COCH_2COOH
E CH_3COCH_2CHO

Exercise 41:

What is formed when an alkene and a ketone is reacting with hydrogen respectively?

A An alkyne and an aldehyde
B An alkyne and an alcohol
C An alkane and an aldehyde
D An alkane and an alcohol
E An aldehyde and a carboxylic acid

Exercise 42:

What is formed when reacting 3-methyl-1-pentene with Br_2.

A 2-bromo-3-methyl-1-pentene
B 3,4-dibromo-3-methyl-1-pentene
C 1,3-dibromo-3-methyl-pentane
D 1,2-dibromo-3-methyl-pentane
E 3-bromo-3-methyl-pentane

Exercise 43:

Wine makers often burn yellow sulfur to produce SO_2 to be used as a preservative in wine. What is the maximum amount of SO_2 which can be produced from 5 g of sulfur (S) and 12.5 g of oxygen (O_2)?

$S\ (s) + O_2\ (g) \rightarrow SO_2\ (g)$

A 7.5 g
B 10 g
C 12.5 g
D 15 g
E 17.5 g

Exercise 44:

Kerosene ($C_{12}H_{26}$) is a popular rocket engine fuel in combination with oxygen (O_2) as an oxidant. The reaction is:

$$2\ C_{12}H_{26}\ (l) + 37\ O_2\ (g) \rightarrow 24\ CO_2\ (g) + 26\ H_2O\ (g)$$

Calculate the volume of the gas produced in the rocket engine when burning 1 kg of kerosene at 1 atm assuming that the gas has a temperature of 3000 °C and that all gases are ideal.

A $2.85 \cdot 10^3$ L
B $3.95 \cdot 10^4$ L
C $8.22 \cdot 10^4$ L
D $1.05 \cdot 10^5$ L
E $3.25 \cdot 10^5$ L

Exercise 45:

Diamonds are essentially pure carbon which can burn at high temperatures. The largest known uncut diamond had a mass of 621 g when it was found in 1905. How many kg of CO_2 could be produced from complete combustion of this giant diamond?

A 1.28 kg
B 2.28 kg
C 2.44 kg
D 2.80kg
E 3.28 kg

Solutions to exam exercises adapted from course 26027

Exercise 1:

Thinking in terms of Le Chatelier's principle the solution with the most base should have the largest resistance to the addition of acid as it will have the smallest relative change. As the composition in answer A has the smallest amount of the conjugate base $NaCH_3COO$ this should cause the largest change in pH.

Alternatively use the Henderson-Hasselbalch equation to calculate the difference in pH for each solution:

$$pH = pK_a + log\frac{[Conjugate\ base]}{[Acid]}$$

$$pK_a(before) = pK_a(after)$$

$$\Delta pH = log\frac{[Conjugate\ base]}{[Acid]}_{After} - log\frac{[Conjugate\ base]}{[Acid]}_{Before}$$

The concentrations before and after are calculated easily as the volume is 1 L:

Before		**After**		**ΔpH**
[CH_3COOH]	**[$NaCH_3COO$]**	**[CH_3COOH]**	**[$NaCH_3COO$]**	
1.00	0.25	1.20	0.05	-0.77815
0.50	0.50	0.70	0.30	-0.36798
0.75	0.50	0.95	0.30	-0.32451
0.25	0.50	0.45	0.30	-0.47712
0.25	1.00	0.45	0.80	-0.35218

Again the pH will change the most for combination A.

Exercise 2:

Molarity is a measure of amount of solute (moles) pr. volume of solution. As the volume increases by 0.5 % pr. 25 K or 1 % for the total of 50 K a decrease in molarity of 1 % is expected.
Molality is a measure of amount of solute (moles) pr. mass of solvent. As mass is not changing with temperature this property is constant.

Combined E is the right answer.

Exercise 3:

The enthalpy is lower at the end of the reaction (to the right) than at the start. The change in enthalpy is thus negative and the reaction is classified as exothermic.
The “hill” on Path B is smaller than on path A and the activation energy has thus be decreased which is the action of a catalyst

Combined D is the right answer

Exercise 4:

First balance the equation:

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$$

Find the amount of the first reactant (glucose) used:

$$M(C) = 12.01\frac{g}{mol}, M(O) = 16.00\frac{g}{mol} \text{ and } M(H) = 1.008\frac{g}{mol}$$

$$M(C_6H_{12}O_6) = 6 \cdot M(C) + 12 \cdot M(H) + 6 \cdot M(O) = 180.56\frac{g}{mol}$$

$$m = n \cdot M \Rightarrow n(C_6H_{12}O_6) = \frac{m(C_6H_{12}O_6)}{M(C_6H_{12}O_6)} = \frac{10g}{180.56\frac{g}{mol}} = 0.0556mol$$

Find the amount of the second reactant (oxygen) used:

$$p \cdot V = n \cdot R \cdot T \Rightarrow n(O_2) = \frac{P \cdot V}{R \cdot T} = \frac{1atm \cdot 4L}{0.0821\frac{L \cdot atm}{mol \cdot K}298K} = 0.163mol$$

As $n(O_2) < 6 \cdot n(C_6H_{12}O_6)$ oxygen is a limiting reactant as seen from the reaction stoichiometry.

B is thus the right answer.

Exercise 5:

Convert the molar concentration into a mass concentration (p, unit g/L) in order to compare with the solubility at 25 °C (s_{25}) and at 100 °C (s_{100}):

$$M(Ag) = 107.9\frac{g}{mol}, M(Cl) = 35.45\frac{g}{mol}$$

$$M(AgCl) = M(Ag) + M(Cl) = 143.35\frac{g}{mol}$$

$$p(AgCl) = [AgCl] \cdot M(AgCl) = 0.0014\frac{g}{L}$$

As $p < s_{25} < s_{100}$ more AgCl can be added to both solutions before precipitation will start

Both NaCl and $AgNO_3$ are very soluble in water and will not precipitate at low concentrations. In general alkali metal ions, halides and nitrates are very soluble with only a few exceptions.

A is thus the right answer.

<u>Exercise 6</u>:

The atomic radius increases when either going to the left (effective nuclear charge increase when going right which means tighter bonding) or downwards in the periodic table (orbital size increases with increasing principal quantum number).

We stay in the 3^{rd} period so only the first rule is needed in this case. Cl thus has the smallest radius as it is located furthest to the right.

E is the right answer.

<u>Exercise 7</u>:

Lewis structure:

3 lone pairs are seen on I (the pairs to the left and right of I are bonding pairs) and the molecule is of the class AB_2E_3 with a trigonal bipyramidal arrangement of the electron pairs and a linear geometry as seen in D.

<u>Exercise 8</u>:

Ag^+ would precipitate with Cl^- so it cannot be $AgNO_3$ as nothing happens when adding NaCl. Both Ag^+ and Ba^{2+} will precipitate with SO_4^{2-} but $AgNO_3$ has already been eliminated so only $Al_2(SO_4)_3$ is a viable choice.

The right answer is thus D

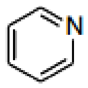
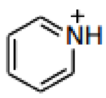
<u>Exercise 9</u>:

Using Le Chatelier's principle the reaction will move right when adding oxygen (minimising the change in oxygen by converting oxygen to ozone), right when increasing pressure (minimising the pressure change by combining 3 gas atoms into 2) and right when increasing the temperature (the reaction uses energy and converting oxygen into ozone thus reduces the temperature increase).

B is thus the right answer.

<u>Exercise 10</u>:

Describe conditions at equilibrium:

 + H_2O ⇄  + OH^-

	(Py)	(PyH$^+$)	
Initial (M)	1.0	0	0
Change (M)	-X	+X	+X
Equilibrium (M)	1.0-X	X	X

Write the equilibrium constant and solve for concentration:

$K_b = \frac{[PyH^+]\cdot[OH^-]}{[Py]} = \frac{X^2}{1.0-X} \approx X^2$ as K_b is very small and 1.0-X≈1.0 (weak base)

$$x = [OH^-] = \sqrt{K_b} = 4.12 \cdot 10^{-5} M$$

Find pOH:

$$pOH = -log[OH^-] = 4.38$$

Convert to pH:

$$pH = 14 - pOH = 9.62$$

B is the right answer

<u>Exercise 11</u>:

According to the reaction scheme in exercise 10 $[PyH^+] = [OH^-] = 4.12 \cdot 10^{-5}$ M.
According to Le Chatelier's principle the reaction will shift to the right upon adding water and the fraction of PyH^+ will increase.

Both answers are consistent with D

Exercise 12:

Use Hess rule: ΔH is additive

By looking at the reaction schemes:

Ammonia needs to be formed (Reversed reaction 3 is the only option)
Note: If ammonia were formed in more than one of the available reactions this would not be a good starting point. The starting point should be simple which ammonia is in this case. In other case it might by easier to start with a reactant and not a product!!!

Water is not in the final reaction (Add 6 times reaction 2 to eliminate water from reaction 3)
NO is not in the final reaction (Add 2 times reaction 1 to eliminate NO from reaction 3)
Formula is now double that which is needed (Divide the above with 2)

In an equation:

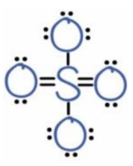
$$\Delta H_{Total} = \frac{6 \cdot \Delta H_2 + 2 \cdot \Delta H_1 - \Delta H_3}{2} = -92.6 \frac{kJ}{mol}$$

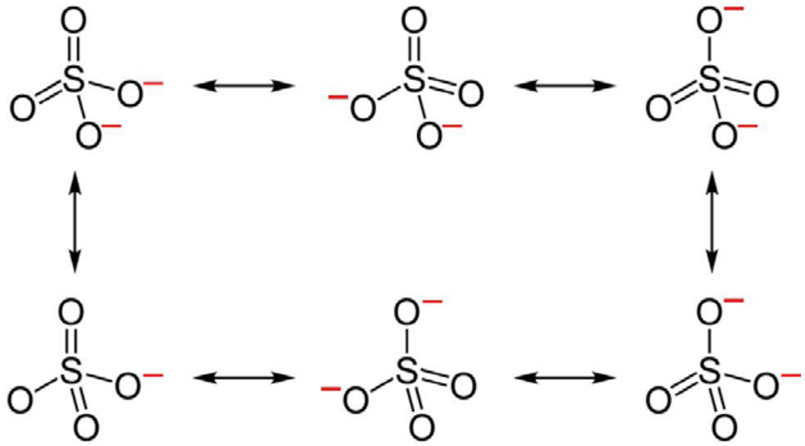
Alternatively use $\Delta H_R^o = \sum_n \Delta H_f^o(products) - \Delta H_f^o(reactants)$
In this case ΔH_f^o can be found for all the products and reactants in appendix 2 in the book. This approach is faster and often possible if the compounds look simple. This shortcut can often save time at the exam!!!

Answer is closest to possibility A

Exercise 13:

A resonance structure of AB_4 type with tetrahedral geometry:





B tetrahedral is the right answer.

Exercise 14:

$\pi = c \cdot R \cdot T$, note that c is here used for molarity (M is used in the book) but we need M for the molar mass later. Molarity is anyway just a different word for concentration. To add to the confusion the unit for c is $M = \frac{mol}{L}$.

$c = \frac{n}{V}, n = \frac{m}{M}$

$$\pi = c \cdot R \cdot T = \frac{n \cdot R \cdot T}{V} = \frac{m \cdot R \cdot T}{V \cdot M} \Rightarrow M = \frac{m \cdot R \cdot T}{V \cdot \pi} = \frac{1.0g \cdot 0.0821\frac{L \cdot atm}{mol \cdot K} \cdot 298.15K}{1L \cdot 0.408atm} = 60.0\frac{g}{mol}$$

A is the correct answer

Exercise 15:

$$\Delta G_f^o\left(H_2O_2(l)\right) = -118.1\frac{kJ}{mol}, \Delta G_f^o\left(H_2O(l)\right) = -237.2\frac{kJ}{mol} \text{ and } \Delta G_f^o\left(O_2(g)\right) = 0\frac{kJ}{mol}$$

$$\Delta G_R^o = \sum\nolimits_n \Delta G_f^o(products) - \Delta G_f^o(reactants) = -237.2\frac{kJ}{mol} - \left(-118.1\frac{kJ}{mol}\right) = -119.1\frac{kJ}{mol}$$

$$\Delta G_R^o = -R \cdot T \cdot ln(K) \Rightarrow K = e^{\frac{-\Delta G_R^o}{R \cdot T}} = e^{\frac{-(-119.1 \cdot 10^3)\frac{J}{mol}}{8.31451\frac{J}{mol \cdot K} \cdot 298.15K}} = 7.33 \cdot 10^{20}$$

Answer is closest to E

Exercise 16:

For a first order reaction the rate constant k can be found by plotting ln[H_2O_2] as a function of t and take the negative value of the slope:

$$k = 0.0078h^{-1}$$

For a 1st order reaction:

$$t_{\frac{1}{2}} = \frac{1}{k} \cdot ln2 = 88.9h$$

Option C is the right answer.

Note that even though the dataset has a very good correlation coefficient ($R^2 = 0.9995$) it is insufficient to just pick two points for finding k. A graphical method including all the data is needed. Using e.g. the last two points gives $k = 0.0076h^{-1}$ while using the 2nd and 3rd point gives $k = 0.0088h^{-1}$. This spread is well beyond the range of the spread in the answers! Also note that the data points are not equally spaced meaning that the relative weight is different for each data point. Using points that are the furthest apart is in general the best strategy if only selecting two points.

<u>Exercise 17</u>:

Plot ln k as a function of 1/T (Known as an Arrhenius plot). The slope α then equals $-E_a/R$

Remember to convert T to Kelvin first.

$$\alpha = -7813.2K \Rightarrow E_a = 64.96\frac{kJ}{mol}$$

D is the right answer

Note that even though the dataset has a very good correlation coefficient ($R^2 = 0.9986$) it is insufficient to just pick two points for finding the activation energy. A graphical method including all the data is needed. Using e.g. the last two points gives $\alpha = -7224.7K$ while using the 2nd and 3rd point gives $\alpha = -8873.9K$. This spread is well beyond the range of the spread in the answers! Also note that the data points are not equally spaced meaning that the relative weight is different for each data point. Using points that are the furthest apart is in general the best strategy if only selecting two points.

<u>Exercise 18</u>:

Nitrogen is significantly more electronegative than phosphorus (3 vs. 2.1) and NH_3 can therefore, like water, form strong hydrogen bonds.

D is the right answer

Exercise 19:

For a 2nd order reaction:

$$t_{\frac{1}{2}} = \frac{1}{k \cdot [A_0]} = 667s$$

B is the right answer

Exercise 20:

From the spectrochemical series it can be seen that CN^- is a strong field ligand while Br^- is a weak field ligand. That is CN^- increases the energy distance between the d-orbitals more than Br^-. As a consequence CN^- will to a higher extent double occupy the low energy orbitals creating a low spin configuration while Br^- will single occupy the orbitals creating a high spin configuration.

C is thus the right answer

Exercise 21:

Describe conditions at equilibrium:

	HCOOH	+	H_2O	⇄	$HCOO^-$	+	H_3O^+
Initial (M)	1.0				0		0
Change (M)	-X				+X		+X
Equilibrium (M)	1.0-X				X		X

Write the equilibrium constant and solve for concentration:

$K_a = \frac{[HCOO^-]\cdot[H_3O^{+-}]}{[HCOOH]} = \frac{X^2}{1.0-X} \approx X^2$ as K_a is very small and 1.0-X≈1.0 (weak base)

$$x = [H_3O^+] = \sqrt{K_a} = 0.013M$$

Find pH:

$$pH = -log[H_3O^+] = 1.88$$

A is the right answer

Exercise 22:

Write the general expression for the equilibrium constant:

$$K = \frac{[C]^2 \cdot [D]}{[A]^2 \cdot [B]}$$

Combine it with the relation between K and Gibbs free energy and solve for [D]

$$\Delta G_R^o = -R \cdot T \cdot ln(K) \Rightarrow K = e^{\frac{-\Delta G_R^o}{R \cdot T}}$$

$$[D] = e^{\frac{-\Delta G_R^o}{R \cdot T}} \cdot \frac{[A]^2 \cdot [B]}{[C]^2} = e^{\frac{10 \cdot 10^3 \frac{kJ}{mol}}{8.314 \frac{J}{mol \cdot K} \cdot 298K}} \cdot \frac{(0.1\ M)^2 \cdot 0.15M}{(0.65\ M)^2} = 0.200M \Longrightarrow 0.2mol$$

B is the right answer

<u>Exercise 23</u>:

Use Hess rule: ΔH is additive

By looking at reaction schemes:

BaO needs to be formed (Use reaction 1 as is)
Ba and O is not in the final reaction (Subtract 2 times reaction 2 to eliminate Ba and O from reaction 1)
Formula is now double that which is needed (Divide the above with 2)

In an equation:

$$\Delta H_{Total} = \frac{\Delta H_1 - 2 \cdot \Delta H_2}{2} = 269.5 \frac{kJ}{mol}$$

A is the right answer

<u>Exercise 24</u>:

From tables in book:

NaOH: strong base
HF: weak acid
HCl: strong acid
CH_3COOH: weak acid
HNO_2: weak acid
HNO_3: strong acid

Only D contains 3 weak acids

Exercise 25:

Ground state electron configurations can be found in a table in the book
As oxygen is charged two extra electrons are added to the 2p shell

B is the right answer

Exercise 26:

$$K = \frac{[B]^2 \cdot [C]}{[A]^3} = 16$$

D is the right answer

Exercise 27:

Assign oxidation states:

-I		+VI		+III		0
Br^- (aq)	+	$Cr_2O_7^{2-}$ (aq)	→	Cr^{3+} (aq)	+	Br_2 (l)

Cr↓3
Br↑1

For every Cr 3 Br are needed:

$6\ Br^-\ (aq) + Cr_2O_7^{2-}\ (aq) \rightarrow 2\ Cr^{3+}\ (aq) + 3\ Br_2\ (l)$

8 negative charges on left side, 6 positive on right side. As environment is acidic balance charges with H^+ on left side:

$14\ H^+(aq) + 6\ Br^-\ (aq) + Cr_2O_7^{2-}\ (aq) \rightarrow 2\ Cr^{3+}\ (aq) + 3\ Br_2\ (l)$

Balance H by adding water on right side:

$14\ H^+(aq) + 6\ Br^-\ (aq) + Cr_2O_7^{2-}\ (aq) \rightarrow 2\ Cr^{3+}\ (aq) + 3\ Br_2\ (l) + 7\ H_2O\ (l)$

Verify that the amount on oxygen is equal on both sides

A is the correct answer

Exercise 28:

$$\Delta T_f = i \cdot K_f \cdot m$$

$i = 1$ for glucose (i.e. no dissociation into ions)

$$M(C) = 12.01\frac{g}{mol}, M(O) = 16.00\frac{g}{mol} \text{ and } M(H) = 1.008\frac{g}{mol}$$
$$M(C_6H_{12}O_6) = 6 \cdot M(C) + 12 \cdot M(H) + 6 \cdot M(O) = 180.56\frac{g}{mol}$$

$$m = \frac{[C_6H_{12}O_6]}{\delta \cdot M(C_6H_{12}O_6)} = \frac{\frac{60g}{0.2L}}{1\frac{kg}{L} \cdot 180.56\frac{g}{mol}} = 1.66\frac{mol}{kg}$$

$$\Delta T_f = K_f \cdot m = 3.09K$$

C is the right answer

Exercise 29:

N_2 (g) + 3 H_2 (g) → 2 NH_3 (g)

$$n(N_2) = \frac{m(N_2)}{M(N_2)} = \frac{77.3g}{28.02\frac{g}{mol}} = 2.76mol$$
$$n(H_2) = \frac{m(H_2)}{M(H_2)} = \frac{14.2g}{2.016\frac{g}{mol}} = 7.04mol$$

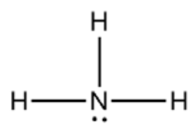
H_2 is the limiting factor as $n(H_2) < 3 \cdot n(N_2)$

$$n(NH_3) = \frac{2}{3}n(H_2) = 4.69mol$$
$$m(NH_3) = n(NH_3) \cdot M(NH_3) = 4.69mol \cdot 17.034\frac{g}{mol} = 79.89mol$$

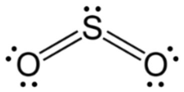
B is the right answer

Exercise 30:

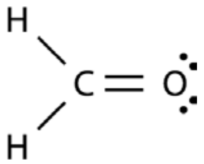
Lewis structures (each bond equals two bonding electrons):



AB_3E type with a tetrahedral arrangement of the electron pairs and a trigonal pyramidal geometry



AB_2E type with a trigonal planar arrangement of the electron pairs and a bent geometry



AB_3 type with a trigonal planar geometry – slightly distorted as atoms are not identical

A is the right answer

<u>Exercise 31</u>:

The structure should satisfy the octet rule for all atoms
Structures with formal charges should be avoided
Structures with small formal charges are preferred over structures with large formal charges

Applying the above rules leads to B as the right answer.

<u>Exercise 32</u>:

$$M(Na) = 22.99\tfrac{g}{mol},\ M(O) = 16.00\tfrac{g}{mol} \text{ and } M(H) = 1.008\tfrac{g}{mol}$$
$$M(NaOH) = M(Na) + M(H) + M(O) = 39.998\tfrac{g}{mol}$$

$$c(NaOH) = \frac{n}{V} = \frac{m}{V \cdot M} = 0.010M$$

NaOH is a strong base and is fully dissociated

$$pH = 14 - pOH = 14 + log[OH^-] = 12$$

B is the right answer

<u>Exercise 33</u>:

A catalyst by definition works by lowering the reaction barrier. As the products and the reactants are the same no stabilisation or change of thermodynamic properties has taken place.

D is the right answer

Exercise 34:

Compounds containing alkali metal ions (e.g. K^+, Na^+ and Li^+) are generally soluble
Compounds containing halides (e.g. Cl^- and Br^-) are generally soluble unless they also contain Ag^+, Hg_2^{2+} or Pb^{2+}
Compounds containing S^{2-} are generally not soluble unless they also contain alkali metal ions (e.g. K^+, Na^+ and Li^+) or the ammonium ion (NH_4^+)

$NaCl$: soluble
KNO_3: soluble
Li_2S: soluble
$AgCl$: not soluble
$PbBr_2$: not soluble
BaS: not soluble

A is the right answer

Exercise 35:

Assign oxidation states:

+I		+VI		+III		+V
H_3PO_2 (aq)	+	$Cr_2O_7^{2-}$ (aq)	→	Cr^{3+} (aq)	+	H_3PO_4 (l)

Cr↓3
P↑4

For every 4 Cr 3 P are needed:

$$2\ Cr_2O_7^{2-}(aq) + 3\ H_3PO_2(aq) \rightarrow 4\ Cr^{3+}(aq) + 3\ H_3PO_4(aq)$$

4 negative charges on left side, 12 positive on right side. As environment is acidic balance charges with H^+ on left side:

$$16\ H^+(aq) + 2\ Cr_2O_7^{2-}(aq) + 3\ H_3PO_2(aq) \rightarrow 4\ Cr^{3+}(aq) + 3\ H_3PO_4(aq)$$

Balance H by adding water on right side:

$$16\ H^+(aq) + 2\ Cr_2O_7^{2-}(aq) + 3\ H_3PO_2(aq) \rightarrow 4\ Cr^{3+}(aq) + 3\ H_3PO_4(aq) + 8\ H_2O(l)$$

Verify that the amount on oxygen is equal on both sides

C is the correct answer

Exercise 36:

For a 2nd order reaction:

$$t_{\frac{1}{2}} = \frac{1}{k \cdot [A_0]} = 2500s$$

C is the right answer

Exercise 37:

Look at the change in the amount of gas molecules on each side. If gas is produced the entropy increases. Other effects like type of gas, change in amount of solids and liquids are insignificant.

For A 1 gas molecule is lost
For B 2 gas molecules are lost
For C 2 gas molecules are lost
For D 1 gas molecule is lost
For E 7 gas molecules are produced

E is the right answer

Exercise 38:

$$K = \frac{[C] \cdot [D]^3}{[A]^2 \cdot [B]} = 0.133M$$

$$\Delta G_R^o = -R \cdot T \cdot ln(K) = 4.99\frac{kJ}{mol}$$

Remember temperature in Kelvin and R with units of $^{J}/_{mol\ K}$

D is the right answer

Exercise 39:

A positive E° indicates a favourable reaction.

Reaction A has the least negative value and Pb^{2+} is the most likely to take up electrons at standard conditions

Exercise 40:

A: Contains a ketone and an alcohol
B: Contains an ester
C: Contains an alcohol and an aldehyde
D: Contains a ketone and an acid
E: Contains a ketone and an aldehyde

E is the right answer

Exercise 41:

For an alkene hydrogen breaks one of the bonds in the double bond forming an alkane
For a ketone an alcohol is formed

D is the right answer

Exercise 42:

A Br atom is added in both position 1 and 2 while breaking the double bond. 1,2-dibromo-3-methyl-pentane is formed

D is the right answer

Exercise 43:

$$n(S) = \frac{m(S)}{M(S)} = \frac{5g}{32.07\frac{g}{mol}} = 0.1559mol$$
$$n(O_2) = \frac{m(O_2)}{M(O_2)} = \frac{12.5g}{32.00\frac{g}{mol}} = 0.3908mol$$

S is the limiting factor as $n(S) < n(O_2)$

$$n(SO_2) = n(S) = 0.15559mol$$
$$m(SO_2) = n(SO_2) \cdot M(SO_2) = 0.1559mol \cdot 64.07\frac{g}{mol} = 9.98g$$

B is the right answer

Exercise 44:

$$n(C_{12}H_{26}) = \frac{m(C_{12}H_{26})}{M(C_{12}H_{26})} = \frac{1000g}{170.328\frac{g}{mol}} = 5.871mol$$
$$n(CO_2) + n(H_2O) = 25 \cdot n(C_{12}H_{26}) = 146.78mol$$
$$p \cdot V = n \cdot R \cdot T \Rightarrow V = \frac{n \cdot R \cdot T}{P} = \frac{146.78mol \cdot 0.0821\frac{L \cdot atm}{mol \cdot K} \cdot 3273.15K}{1atm} = 3.94 \cdot 10^4 L$$

Answer B is correct

Exercise 45:

C (s) + O_2 (g) → CO_2 (g)

$$n(CO_2) = n(C)$$
$$\frac{m(CO_2)}{M(CO_2)} = \frac{m(C)}{M(C)} \Rightarrow m(CO_2) = \frac{m(C) \cdot M(CO_2)}{M(C)} = \frac{621g \cdot 44.01\frac{g}{mol}}{12.01\frac{g}{mol}} = 2.28\ kg$$

B is the right answer

List of correct answers for exercises

Exercise	E16	F17	E17	F18	Previous exam exercises
1	E	C	C	D	A
2	D	D	B	A	E
3	E	A	A	A	D
4	A	C	E	C	B
5	B	A	B	B	A
6	D	C	B	A	E
7	B	A	B	D	D
8	E	A	C	A	D
9	B	D	A	A	B
10	E	A	C	C	B
11	D	B	A	B	D
12	B	E	E	D	A
13	A	E	A	B	B
14	C	D	A	D	A
15	B	D	A	B	E
16	B	A	C	C	C
17	C	A	B	E	D
18	D	C	B	B	D
19	B	C	B	D	B
20	B	D	B	B	C
21			A	D	A
22			C	A	B
23			D	C	A
24			B	B	D
25			C	B	B
26			B	A	D
27			A	C	A
28			B	C	C
29			D	A	B
30			D	C	A
31					B
32					B
33					D
34					A
35					C
36					C
37					E
38					D
39					A
40					E
41					D
42					D
43					B
44					B
45					B